

PAGE: PARTITION-AWARE GATED KV-CACHE EVICTION

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ABSTRACT

KV-cache eviction is deployed as a uniform accuracy-memory trade, but its per-input effect is bimodal. On many long-context inputs a compressed cache matches the full one, while on precise multi-key retrieval the same evictor drives accuracy from 99% to 0%. Existing evictors choose which tokens to keep under a fixed budget and report benchmark means that average this failure away, never treating whether to evict as a decision. We show that one label-free scalar read from pre-fill attention identifies the regime before any decoding: the early-to-late drop D in mean pairwise top- k Jaccard agreement between attention heads, which stays small when heads converge on the few tokens carrying the answer. PAGE thresholds D , running any SnapKV-style evictor when the drop is large and keeping the full cache when it is small, with no training, labels, or fine-tuning. Across four base evictors, four models, RULER, and LongBench, gating cuts the rate at which eviction destroys a correct answer from 0.75 to 0.026 at a nominal $16\times$ budget, $3.4\times$ realized once the protected cache is counted, and raises matched-cache accuracy by 3-19pp above a crossover near one third of retained cache. This is a batch-1, offline-scoring safeguard whose memory saving decays toward $1\times$ by batch 16 under static provisioning. A geometry-based and an attention-free scorer collapse on the same inputs below 25% budget, indicating an information floor no scorer we tested removes and recasting cache compression as per-input admission control rather than token selection.

1 INTRODUCTION

Long-context inference spends most of its memory on the KV-cache, and eviction is the standard remedy: score the prompt positions once, keep a fraction b , and accept a small average accuracy cost. Recent work sharpens this trade surprisingly. Bui et al. (2026) prove that selective retention reduces a quantifiable *attention dilution* term and report a learned-gate evictor exceeding the full cache on long-context dialogue, Yang et al. (2026a) derive a closed-form information-bottleneck score producing the same paradox on LongBench, and an indexer-and-latent-memory architecture (Yang et al., 2026c) reports an information-density peak at moderate compression. Their common claim is that less compute can yield more accuracy.

Each result averages over a suite where the effect occurs, and a benchmark mean positive in expectation can hide a subset of inputs on which eviction is catastrophic. The two cases call for opposite actions. Where the answer is spread over redundant tokens, discarding them concentrates attention and eviction is safe. Where it sits in a handful of tokens, any scorer that drops the wrong one loses it outright at every budget. Averaging the two populations produces a number describing neither. Adaptive methods nonetheless train a scorer (Bui et al., 2026; Yang et al., 2026c; Dong et al., 2026; Moschella et al., 2026) or derive an unconditional capacity score from query statistics (Yang et al., 2026a), and in both cases the decision is *which* tokens to keep. Whether the two populations are separable in the pre-eviction attention alone, before any token is scored, is open. If they are, the safe deployment of eviction is not a better scoring rule but a decision to abstain.

We answer affirmatively. The *head-agreement drop* on prefill attention, the difference between mean Jaccard top- k head agreement in early and late layer-bins, partitions inputs into a dilution-prone class $\mathcal{D}$ and a capacity-bound class $\mathcal{C}$, and a drop-in wrapper turns that partition into a training-free policy. The signal reflects how heads specialize with depth. Heads agree broadly in early layers and diverge

in late ones, but when the answer occupies only a few tokens those tokens hold them together and the divergence is shallow, so a small drop is evidence that the cache holds little redundancy to discard.

Because a benchmark mean hides the failure mode we target, we evaluate PAGE by what it prevents on the capacity-bound class rather than by a suite average. It cuts the rate at which eviction destroys a correct answer by $29\times$ at $16\times$ compression and lifts the accuracy-memory frontier at matched *achieved* cache above a crossover near $1/3$ kept cache, losing below it. The converse deserves equal prominence: the gate is inert wherever eviction is already safe, and in half the matrix cells of Section 5.2 it opens on every non-capacity-bound input, so the measured difference is exactly zero.

Contributions.

1. **An empirical partition.** Across eight (model, context) cells and six RULER (Hsieh et al., 2024) task families (Section 3), three-key retrieval (NIAH-MK3) is the one task where eviction never recovers accuracy, while five dilution-prone families recover wherever the model has headroom. A distractor sweep confirms the mechanism, and the partition holds on three LongBench subtasks the gate calls correctly a priori.
2. **A label-free predictor.** The early-to-late head-agreement drop $D := \text{agT}_{\text{early}} - \text{agT}_{\text{late}}$ ranks the capacity-bound task smallest and is the only one of six cheap prefill statistics that separates the partition at all (Section 4.1). It is one-sided: reliable at closing the gate on capacity-bound inputs, weakly discriminative on dilution-prone ones, which bounds what the gate can promise.
3. **PAGE, a gating algorithm built on D .** One conditional on the prefill attention, with no training and no labels (Section 4.2). We set the threshold per model from an unlabeled z -score pilot, which transfers across all four architecture families we test; the fixed $\tau = 0.07$ we run everywhere is a within-family convenience that fails on Qwen3 (Section 4.3).
4. **Scorer-independence of the capacity limit.** A faithful ManifoldKV+Ada-KV geometry scorer and the attention-free KeyDiff (Park et al., 2025) both collapse on NIAH-MK3 below 25% budget (Section 5.4), even though the same scorer reproduces its published two-key advantage. No scoring rule recovers information the retained subset does not contain.
5. **Head-to-heads, and a scaling relation.** The gate lifts matched-budget accuracy around CapKV and pretrained DBTrimKV (Section 5.5), though on DBTrimKV the full-cache fallback dominates and plain DBTrimKV wins at matched *memory*. A closed form, $\rho \approx (1 - A_{\text{full}}) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}^2/\gamma)$, unifies the scaling evidence (Section 5.6), and a pre-registered 32K test preserves its ordinal content while falsifying constant-ratio proportionality.

Scope. We claim as novel neither the paradigm that less compute can beat full attention nor the dilution mechanism, since Bui et al. (2026) state both and we adopt their Proposition 3.1 and Corollary 3.2, and we repurpose a known head-agreement primitive rather than proposing a new statistic. We claim no compression win: at matched memory a strong trained evictor beats us. Most of the headline Δ protects one task family, since excluding NIAH-MK3 the grand mean falls from $+22.9\text{pp}$ to $+4.5\text{pp}$ and a cheating task-label oracle reaches $+20.1\text{pp}$. The contribution is the label-free per-input decision, not the compression ratio it delivers where eviction was already safe.

2 RELATED WORK

The nearest KV-eviction methods differ from ours in what they decide. DBTrimKV (Bui et al., 2026), whose dilution mechanism we adopt, CapKV (Yang et al., 2026a), DynamicKV (Zhou et al., 2025), which also shows task type modulates which KV positions matter, and IndexMem (Yang et al., 2026c) all choose *which* tokens to keep under a budget, as do the wrappers (VECTOR Lin et al., 2026b, CriticalKV Feng et al., 2026b) and the learned gates (Attention-Gate Zeng et al., 2025, Fast KVzip Kim et al., 2026), which also require training. With the budget fixed in advance, none represents the case where the correct action is to spend no budget. EvicPress (Feng et al., 2025a) does choose per context *whether* to evict, but as one term in a profiled quality-latency optimization over storage tiers rather than a label-free call from prefill attention. SAGE-KV (Wang et al., 2025) comes closest, reading a one-shot post-prefill attention signal, but spends it choosing which tokens and heads to keep and never identifies a class where eviction cannot be made safe.

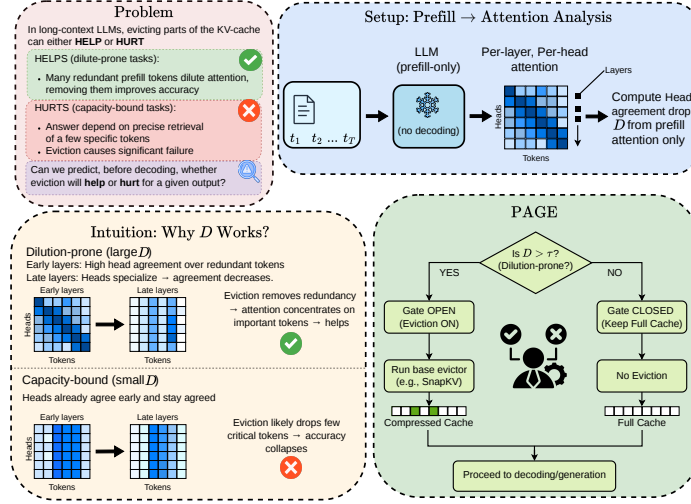


Figure 1: PAGE decides *whether* to evict, once per input, from a prefill-only pass. Head agreement peaks in early layers and falls with depth as heads specialize. On dilation-prone inputs the early consensus spans a large redundant set and the fall is steep; on capacity-bound inputs the few tokens carrying the answer hold heads together and the fall is shallow. Thresholding the drop D at τ separates redundancy safe to discard from evidence that is not.

Diagnostic work establishes that such a partition should exist without providing a predictor. Chen et al. (2026) document that eviction degrades some task types and Zhang et al. (2026) ask when value-aware eviction helps, but neither pairs a helping class with a hurting one on one axis or predicts the side a priori. Our signal differs from the *within-layer* head consensus of Ananthanarayanan et al. (2026), used there descriptively, in being a directional *cross-layer* contrast computed online and thresholded to gate. Retrieval heads (Wu et al., 2025) show a small, sparse head subset drives long-context retrieval, the mechanism making capacity-bound inputs fragile; D measures that subset’s layer-wise convergence rather than identifying its members. We claim the conjunction absent from all of these: a helps-versus-hurts partition on one axis, predicted a priori from a single prefill scalar, deployed as a whether-to-evict gate, with the capacity-bound side shown scorer-independent. We wrap SnapKV (Li et al., 2024), H2O (Zhang et al., 2023), StreamingLLM (Xiao et al., 2024) and PyramidKV (Cai et al., 2025); Appendix A treats all neighbors in full.

3 THE EMPIRICAL PARTITION

A claim about inputs has to be established on inputs, so we demonstrate the partition before proposing a predictor. This section asks whether a class exists on which eviction never recovers accuracy, whether it survives outside synthetic data, and whether simpler properties explain it away.

Setup. We use RULER (Hsieh et al., 2024) via the `simonjegou/ruler` (Jégou, 2024) port distributed with `kvpress` (NVIDIA, 2024), at $T \in \{4K, 16K\}$ on Qwen2.5-1.5B/3B/14B-Instruct (Yang et al., 2025b) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). Eviction is SnapKV-style by default: score each prompt position by the average attention from the last $w = 32$ queries across heads and layers, keep the top- b positions for nine budgets $b \in [0.0625, 1.0]$, and always preserve the first $n_{\text{sink}} = 4$ sink positions (Xiao et al., 2024) and last $w = 32$ recent positions. Cache slicing uses explicit `position_ids` to preserve RoPE alignment (Su et al., 2024) rather than the attention-mask trick that silently shifts RoPE on evicted positions. Task codes are RULER’s: VT, FWE, QA_1/QA_2, and NIAH-MK k = `niah_multikey_k` (k near-tie keys, one carrying the answer). Full protocol in Appendix H.

Metric. A mean-accuracy comparison cannot express a claim about individual inputs. For task T , model M , context length T , and $A(b, x)$ the accuracy on input x at budget b , with $b_{\max} = 1$ the full cache, we report the per-input recovery rate

The partition. We measured ρ on six RULER tasks across eight (model, context) cells under attention-score eviction (Table 2, Appendix A). Five task families clear $\rho \geq 0.05$ at some context length, while NIAH-MK3 stays at $\rho \leq 0.02$ on every model and every context. Per-cell counts are low, so the claim rests on the pooled contrast, which is unambiguous: NIAH-MK3 recovers on 2/526 inputs ($\rho = 0.004$, Wilson [0.001, 0.014]) against, for example, 65/680 ($\rho = 0.096$, [0.076, 0.120]) for `niah_multivalue`. Two effects explain the cells that fall below threshold without contradicting the partition. Saturation suppresses ρ wherever $A_{\text{full}} \rightarrow 1$ leaves no headroom, and the same models recover at longer context, exactly as Eq. (3) predicts. The remaining three exceptions are budget-insensitive rather than saturated, in that headroom exists but the wrong inputs stay wrong at every budget. All three are QA cells, pointing to parametric-knowledge failures that no cache budget can fix. We claim the partition at 4K and 16K only, since a single 32K probe puts NIAH-MK3 at $\rho = 0.04$, one point under the threshold that defines the class (Section 5.6).

The class has more than one member. A single capacity-bound task leaves open whether NIAH-MK3 names a class or an isolated construction, so we tested the boundary on four further RULER tasks chosen by mechanism rather than outcome: a second near-tie retrieval task predicted capacity-bound, a real-hard-negative task predicted dilution-prone despite carrying the most distractors in the suite, a distractor-free single-needle control predicted dilution-prone, and a code-completion task needing ten exact low-redundancy tokens predicted capacity-bound (Table 3, Appendix A).

The outcome rules out the two simplest alternative accounts. `niah_multiquery` is capacity-bound on two of three models and borderline on the third, giving the class a confirmed second member. `qa_2`, `niah_single_1`, and `cwe` are all dilution-prone, which defeats both a distractor-count and an exact-output-length account: `qa_2` carries the most distractors of any task tested yet sits comfortably on the dilution-prone side, and `cwe` requires ten exact low-redundancy tokens yet closes the gate on none of the 300 inputs across the three models. What separates the classes is whether the distractors are near-tie surface-form variants of the answer, not their count or the exact-token demand. The single-model exception, `niah_single_1` opening on 87% of Qwen2.5-3B inputs against 0% on the other two, tracks a scale shift in D rather than a mechanism failure: task ordering is preserved across models (Spearman $\rho = 0.98$) but absolute scale compresses by $2.4\times$ on the larger model, which the z -score calibration of Section 4.3 corrects.

RULER is constructed, so a partition found only there would not license a deployment claim. On three LongBench subtasks (Bai et al., 2024) with Qwen2.5-14B the gate calls all three correctly *a priori* from D alone, with no labels and no task identity (Table 4). `lcc` code completion (Liu et al., 2024) is genuinely capacity-bound: plain SnapKV destroys 64% of the answers the full cache gets right ($0.292 \rightarrow 0.135$ at $16\times$) while recovering almost none ($\rho = 0.031$), the signature previously seen only on synthetic NIAH-MK3. Its mean drop $D = 0.054$ sits below τ , the gate closes on 75% of inputs, and accuracy improves by $+0.146$. `repobench-p` is mixed and milder ($+0.049$), while multi-hop `hotpotqa` (Yang et al., 2018) is evict-robust, the gate opening on 86% of inputs and gaining nothing ($+0.000$): where eviction is safe the gate stays out of the way. These subtasks are also our one estimate of how much capacity-bound traffic a realistic workload carries.

Three controls. A distractor sweep over the RULER NIAH-MultiKey family confirms the dilution mechanism of Bui et al. (2026), mean D shrinking monotonically as near-tie distractors accumulate (Table 8). The partition is unchanged under the heavier 10% prefix-and-suffix protection of Garcia (2026), and uniform-random scoring destroys accuracy at every budget, ruling out a generic regularization explanation (Appendix B).

4 THE GATING METHOD: PAGE

Measured outcomes define the partition, but a deployed system cannot observe them and needs a quantity computable before decoding. This section defines that quantity, shows why it is the one to use, and states the algorithm it drives.

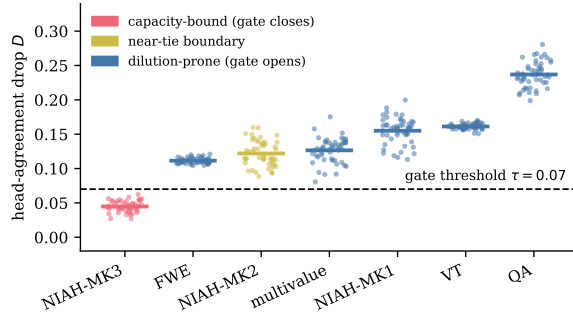


Figure 2: D separates the two task classes (Qwen2.5-1.5B, RULER 4K, $N = 50$ per task; dots are inputs, bars per-task means). NIAH-MK3 sits below $\tau = 0.07$ and the dilation-prone tasks above it. The near-tie MK1/MK2 tasks fall on the boundary, tracking the distractor gradient (App. A).

4.1 THE HEAD-AGREEMENT DROP

Take the prefill attention tensor $[L, H, T, T]$ for L layers, H heads and prompt length T . Project onto the last $w = 32$ rows, the observation queries, and average for a per-(layer, head, key) attention vector. For each layer and head pair take the Jaccard similarity of the two heads’ top- k key sets ($k = 32$), averaged over pairs, giving per-layer agreement a_ℓ . Binning the L layers into thirds, with a_{early} , a_{late} the mean agreement in the first and last bins,

$$D := a_{\text{early}} - a_{\text{late}}. \quad (1)$$

D is one scalar per input costing $O(LH^2k)$, or 17ms on Qwen2.5-1.5B and 89ms on Mistral-7B at 4K. It is not free even when the base evictor already scores by attention, since that scoring uses head-averaged attention whereas D needs the per-head pattern, which requires materializing attention that fused FlashAttention (Dao et al., 2022) does not produce. This is the method’s principal cost, treated in Section 6.

That cost is justified only if no cheaper statistic does the same job, so we ran a same-inputs ablation against five alternatives from the same prefill pass: mean attention entropy, top-32 attention mass share, max single- position share, key-norm dispersion, and an early-minus-late entropy drop. D is not merely the best but the only one separating the partition at all, its NIAH-MK3 mean falling strictly below all four dilation-prone task means with per-input AUC 1.000 on the selecting cell, while every alternative fails task-level separation and tops out at 0.805. The dissection shows why: raw attention entropy is at chance (0.508), its early-versus-late contrast reaches 0.805, and replacing entropy with head *agreement* makes the separation exact, so both the layer contrast and the agreement measure are load-bearing (Table 9). On Qwen2.5-1.5B at 4K the per- task means separate cleanly, NIAH-MK3 at $D = 0.040$ below $\tau = 0.07$ while the dilation-prone tasks run from FWE at 0.098 to QA_1 at 0.218 (Figure 2).

An AUC of 1.000 on the selecting cell is the number to trust least, so we repeated the ablation on three cells that played no part in that selection. D remains the only one of the six separating tasks anywhere, on two of the three held-out cells, but the number does not survive: held out, per-input AUC is 0.899, 0.978, and 0.741. The weak cell is Llama-3.1-8B, where niah_multivalue sits below NIAH-MK3, a limit Section 6 treats as real.

4.2 ALGORITHM

With the signal validated, the policy is one conditional. Given a prompt x , budget $b \in (0, 1]$, sink-and-recency protection (n_{sink}, w) preserving the first n_{sink} and last w positions, and threshold τ ,

$$PAGE(x; \text{base}, b, \tau) = \begin{cases} \text{base.evict}(\text{cache}, b) & \text{if } D(\text{cache.attentions}) \geq \tau \\ \text{cache} & \text{otherwise} \end{cases} \quad (2)$$

where prefill produces `cache` with its attentions, from which D follows by Eq. (1). The gate fires once per input at prefill, before any decode step, with no backward pass, no training, and no per-token decision. PAGE never observes correctness, making it label-free in an accuracy sense.

The asymmetry between branches determines how the method should be read. Closing the gate costs memory but not accuracy, since the full cache is what the unevicted model would have used, while opening it delegates entirely to the base evictor. PAGE inherits that behavior where it opens and adds a floor where it closes, so the evaluation measures harm prevented rather than accuracy gained.

4.3 CALIBRATING THE THRESHOLD

The single free parameter is τ , and how it is set determines whether the method transfers. We recommend calibrating τ by standardizing D per model over an unlabeled ~ 100 -input pilot from the serving distribution. This z -score transformation transfers one global threshold zero-shot to every held-out family we test, the Llama-architecture models (Yi-1.5-9B, Llama-3.1-8B) and Qwen3-4B (AI et al., 2025; Grattafiori et al., 2024; Yang et al., 2025a). It needs no task labels and is the only variant that transfers uniformly.

The fixed $\tau = 0.07$ behind every headline number is a within-family convenience, not a generalizable constant. It mis-classifies Qwen2.5-3B at 16K, where layer-bin drops compress at long context, and fails outright on Qwen3, whose drops sit at a different scale ($\mu = 0.034$, $\sigma = 0.017$) placing $\tau = 0.07$ at $z = +2.0$ so the gate essentially never fires. We report it uniformly so no cell is tuned, not because it generalizes. A sweep on Qwen2.5-1.5B shows a plateau over $\tau \in [0.055, 0.10]$ within which Δ varies by under 1pp and false-positive and false-negative rates stay at or below 9%, so the deployed value is not a knife-edge (Figure 8).

What the gate costs a deployment. The gate-open rate p_{open} is a property of the workload rather than the method, so realized compression is not a constant of PAGE and savings are bounded by $1 - p_{\text{open}}$. Two consequences matter in deployment. Capacity planning drifts as traffic shifts toward low- D inputs, and because D falls as near-tie distractors accumulate, an adversary controlling prompt content can force the fallback and turn a memory attack into a denial-of-capacity attack. Batching compounds this, since a batch provisioned for its largest resident cache realizes the *minimum* compression over its sequences: at $p_{\text{open}} = 0.75$ mean compression falls from $2.91\times$ at batch 1 to $1.00\times$ by batch 16. Both effects, their mitigations, and the per-model numbers are in Appendix A.

5 EXPERIMENTS

Models, contexts, eviction policies, and protection match Section 3. The matrix uses RULER 4K with the four-task mixed suite ($N = 100$ each); the scaling table uses 4K and 16K. Gating deltas carry 95% cluster-bootstrap intervals, recovery rates Wilson intervals. We ask in turn whether the gate prevents the harm it targets (5.1), whether it depends on the base evictor (5.2), whether it pays for itself at matched memory (5.3), and whether the capacity limit belongs to the input or to our scorer (5.4).

5.1 DOES THE GATE PREVENT THE FAILURE IT TARGETS?

The motivating claim is about harm rather than average accuracy, so we measure it directly: $\Pr_x[A(b, x) < A_{\text{plain}}(1, x)]$ at fixed b , the fraction of inputs on which a deployed system loses an answer the full cache would have produced, for both arms at matched nominal budget over seven SnapKV cells (Table 1, Appendix A).

The capacity-bound task carries the result. At $16\times$ compression plain eviction destroys 75.0% of the answers the full cache gets right on NIAH-MK3 while gating destroys 2.6%, a $29\times$ reduction holding across budgets (0.024 against 0.724 at $b = 0.25$). The other three tasks matter equally. On QA-1 the arms are numerically identical at every budget (0.110 against 0.110 at $b = 0.0625$), the gate opening on every input so the systems coincide. On VT at the most aggressive budget both are harmed at nearly 70% (0.684 against 0.694), so the gate neither helps nor hurts where the base evictor fails on inputs it correctly calls safe. Only FWE shows partial protection (0.236 against

0.414). The gate is not a general accuracy improvement but a floor under one failure mode, inert elsewhere by design.

The cleanest single cell makes the same point without aggregation. On Mistral 4K NIAH-MK3, plain SnapKV collapses from 0.99 to 0.00 at $b = 0.0625$ while gated SnapKV holds 0.89 at every budget, closing on 90/100 inputs with 10/100 false positives where it opens and eviction destroys the needle (Figure 4).

5.2 DOES THE RESULT DEPEND ON THE BASE EVICTOR?

A gate working with only one scoring rule would be a property of that rule. We ran a 4×4 matrix of four base evictors (SnapKV, H2O, StreamingLLM, PyramidKV) and four models under the same gate and τ , with no per-method tuning (Table 5, Appendix A). All sixteen cells are positive, grand mean +22.9pp, every 95% interval above zero (smallest lower bound +9.3pp), establishing transfer across base evictors.

That grand mean is not the number to quote. Dropping NIAH-MK3 takes it to +4.5pp, and in eight of sixteen cells the non-MK3 Δ is exactly 0.000, the gate opening on every non-MK3 input so the arms execute identically. The matrix is close to one measurement, $\Delta_{\text{MK3}} = +0.780$, replicated sixteen times and diluted by the suite’s one-quarter MK3 share. Read as a claim about a suite it misleads; read as a claim about transfer it is what it should be, since a whether-to-evict gate ought to do nothing when eviction is safe, and the zeros are an identity rather than a null result. Per-task accuracy is additive under reweighting, so for a workload with capacity-bound share f the headline is $\Delta(f) = f\Delta_{\text{MK3}} + (1 - f)\Delta_{\text{rest}}$, near-linear through the origin, with $\Delta(0) = +0.045$ and $\Delta(0.25) = +0.229$.

Three caveats bound the matrix. The plain arm uses a shared single keep-mask and a clean per-head SnapKV collapses less sharply at $2\times$ (0.32 not 0.14 on MK3), so these deltas should be read at $\geq 4\times$, where the collapse is scorer-independent; restricting the grid to $\{0.125, 0.25\}$ raises the grand mean by only +1.7pp. A clean per-head Ada-KV plain arm costs 1.1pp (+25.8 to +24.7pp at $b \leq 0.25$), a wash rather than a collapse, with one model-specific exception on Mistral-7B that a Llama-3.1-8B control rules out as a grouped-query-geometry artifact (Appendix A). The Qwen2.5-14B H2O column is separately not independent, since that run used two-pass prefill under which H2O’s all-query score reduces exactly to SnapKV’s, leaving fifteen independent cells.

Beyond the matrix, SnapKV runs on Yi-1.5-9B ($\Delta = +39.0\text{pp}$) and Llama-3.1-8B (+10.9pp) extend the mechanism to two more architectures, and a 32B-scale run confirms the predictor scales while illustrating the saturation arm, that model being near-saturated at 4K so the gate’s value is protection rather than recovery. Comparing best plain against best gated evictor per model checks the strongest baseline (Tables 14, 15). Composition also extends outside the attention-score family, lifting a fifth, geometry-based (ManifoldKV-style) base evictor by $\Delta = +16.2\text{pp}$. That variant uses a single shared mask forced by our uniform-cache harness and does not reproduce ManifoldKV’s reported multi-key rescue, so it tests composition rather than standing in for the faithful implementation of Section 5.4.

5.3 DOES THE GATE PAY FOR ITSELF AT MATCHED MEMORY?

A closed gate holds the full cache, so matched-*nominal*-budget comparison flatters the gated arm; the honest comparison is at matched *achieved* kept-KV (Figure 3, Appendix A). Gating beats plain SnapKV by 3 to 19pp once the target retains more than about a third of the cache, and loses below that, where the full-KV fallback floors compression near $3\times$. PAGE is a moderate-compression safeguard rather than an aggressive-regime compressor, recovering 69% of the headroom available to a per-input oracle. What a deployment compresses is set by the gate-open fraction rather than the nominal budget, since $1 - p_{\text{open}}$ of the cache stays resident whatever b is.

Whether per-input resolution earns its cost is testable against a policy knowing each input’s task label. Against such an oracle PAGE reaches +22.9pp against +20.1pp, matching it exactly in eight of sixteen cells, so most of the matrix Δ is a task-level effect a cheating baseline could reproduce. The per-input claim rests on the cells where PAGE exceeds it, most clearly Llama-3.1-8B NIAH-MK3, where the task mean $D = 0.071$ sits *above* τ : any policy keyed on that mean opens everywhere and gains nothing, while per-input dispersion closes the gate on $\approx 44\%$ of inputs and recovers +43.5pp.

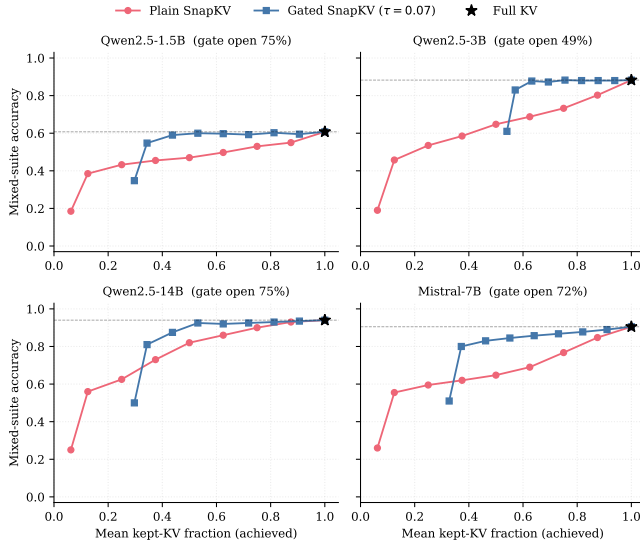


Figure 3: Accuracy vs. *achieved* kept-KV fraction on the mixed 4K suite, SnapKV. The gate falls back to the full cache when it closes, so gated points sit at larger kept-KV than their nominal budget and the honest comparison is at matched achieved cache. Above a target of $\approx 1/3$ kept-KV (Qwen3B: 0.57) the gated curve is the Pareto frontier by up to +19pp; at the leftmost gated point plain eviction dominates it. Gated results use the post-hoc $\tau = 0.07$ reconstruction (Appendix H).

This matches how D orders tasks: it ranks NIAH-MK3 smallest in every Qwen and Mistral cell, and on the 4K/16K cells $\rho/(1 - A_{\text{full}})$ fits a through-origin slope of 0.34 on dilution-prone tasks while staying ≤ 0.02 on NIAH-MK3 regardless of headroom.

5.4 IS THE CAPACITY LIMIT A PROPERTY OF THE INPUT OR OF OUR SCORER?

A better scoring rule might make the capacity-bound class evictable and render the gate unnecessary. The framing depends on ruling this out, so we tested two scorers from outside the attention-score family: a faithful ManifoldKV (Datta et al., 2026) implementation scoring $s_i = \|k_i - \mu\|_2$ against the per-(layer, kv-head) key centroid, paired with Ada-KV (Feng et al., 2025b) per-head budget reallocation, and the attention-free KeyDiff (Park et al., 2025).

A negative result is interpretable only if the scorer is competent. On two-key NIAH the geometry scorer reproduces the multi-key advantage its authors report, reaching 0.72 at $b = 0.5$ where SnapKV is at 0.16. On three-key NIAH-MK3 the same scorer, in all four allocation variants, sits at or near 0.00 against a full-cache 0.66 once the budget falls below 25%, and KeyDiff does the same.

The gate is built on the conclusion this licenses. On capacity-bound inputs the recoverable information is not present in any 25% subset of the cache, so no scoring rule, attention-based, geometric, or attention-free, recovers it, and the only safe action is not to evict. The result holds at $\geq 4\times$ compression rather than $2\times$, consistent with the caveat on Table 5.

5.5 HEAD-TO-HEADS WITH PUBLISHED METHODS

We also wrapped published evictors the gate was not designed around. Around a CapKV reimplementation (Yang et al., 2026a) on LongBench it is weakly positive (+1.5/+8.3pp, $N = 68$ and 24), and around pretrained DBTrimKV (Bui et al., 2026) it lifts matched-budget accuracy by +23.3pp on RULER 4K.

That second number is a memory-spending win rather than a matched-memory one. At $\tau = 0.07$ the gate fires on only 1.7% of DBTrimKV records, so the wrapped system holds $\approx 14.5\times$ the cache of the plain one, and at matched *memory* plain DBTrimKV *beats* it. The rare firing is the Qwen3 scale mismatch of Section 4.3 in a head-to-head, since DBTrimKV’s pretrained gates exist

only for Qwen3 variants; the per-model z -scored threshold ($\tau_{\text{Qwen3}} = 0.022$) restores a genuinely compressing gate at 43% kept-KV with $\Delta = +0.235$. PAGE does not advance the frontier over a strong trained method at matched memory. Its value is the label-free decision to keep the full cache when eviction is unsafe.

5.6 MECHANISM AND SCALING

We adopt the dilution mechanism of Bui et al. (2026) verbatim. For a head reading a relevant set $\mathcal{R}$ against noise $\mathcal{N}$ with pre-eviction signal mass $\alpha_{\mathcal{R}}$, near-tie distractors force the dilution δ_t large and preferential retention reduces it, so the partition follows: capacity-bound tasks concentrate the signal ($\alpha_{\mathcal{R}} \rightarrow 1$), dilution-prone tasks spread it ($\alpha_{\mathcal{R}} \ll 1$). With the accuracy headroom $1 - A_{\text{full}}(M, T)$ this gives

$$\rho_{\text{KV}}(T, M, T) \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}^2/\gamma), \quad p_{\text{recov}} : [0, \infty) \rightarrow [0, 1], \quad (3)$$

with $\rho_{\mathcal{R}}$ the retained fraction of relevant mass and γ the noise-retention ratio. We test only the headroom factor, holding p_{recov} fixed, so the recovery factor is a structural placeholder, and the surrogate motivates the predictor on the dilution-prone side only: on the capacity-bound side the bridge from D to δ_t needs $\kappa \geq 10.11$ while the measured NIAH-MK3 value is $\kappa = 3.33$. Our claims therefore rest on the observed partition rather than the bridge (Appendices E, F, I).

A pre-registered out-of-range test at 32K. Every scaling result above is in-range, so we pre-registered an extrapolation test: fit $\rho/(1 - A_{\text{full}})$ on 4K/16K, measure A_{full} at 32K (Qwen2.5-1.5B, $N = 100$), and hash the predicted intervals before sweeping.

One of three predictions lands strictly inside its interval. VT exceeding its band is conservative, while NIAH-MK3 rising to 0.04 against a $\rho \leq 0.02$ pin is an anti-conservative miss whose Wilson interval still covers 0.02. A second draw replicates FWE and NIAH-MK3 but not VT, so we drop the claim that eviction beats full-KV on VT in mean accuracy. What survives is ordinal, MK3’s ratio staying far below VT’s at any length even as constant-ratio proportionality fails. We claim a directional relation, not a law, and do not extend the partition beyond 16K.

6 LIMITATIONS

Four limits bound the gate’s value, stated here rather than left implicit; Appendix A adds seven narrower ones.

The signal is not free. It needs materialized prefill attention, which fused FlashAttention and paged-attention serving (Kwon et al., 2023) do not provide. Our two-pass mode keeps the extra pass to $O(w \cdot T)$ rather than $O(T^2)$, but it remains a pass a vanilla stack does not run, at $O(LH^2k)$ cost growing in head count. PAGE therefore suits deployments that can afford one such pass or already run one for attention-scored eviction. KeyDiff presses this hardest, being attention-free for exactly this reason, but it too falls to chance on NIAH-MK3 below 25% budget (Section 5.4), so its cheapness does not buy safety on the class the signal guards.

The safety property rests on a surrogate. On a single-head deterministic-SNR surrogate we obtain a two-sided scaling bound, a converse vacuous at our (H, L, T) , and an approximate useful-set identification that fails on the capacity-bound side. These are properties of the surrogate, not the deployed system, so our claims rest on the empirical partition.

The threshold is only partly portable. On Yi-1.5-9B and Llama-3.1-8B the drop ordering transfers but $\tau = 0.07$ does not, mis-classifying niah_multivalue on both and inverting the drop-versus- ρ prediction on Yi. The z -scored variant repairs this from an unlabeled pilot but remains a calibration step, and per-input separation stays at $\text{AUC} \approx 0.80$ there, a gap traceable to the non-monotone depth profile on Llama.

The gate is a detector, not a calibrated predictor. It is one-sided: on LongBench passage_count it closes on all inputs (mean $D = -0.013$) although eviction is safe there, a benign false positive costing compression rather than accuracy. The scorer-independence result likewise holds at $\geq 4\times$, not $2\times$, and multilingual tasks remain untested.

7 CONCLUSION

KV-cache eviction is deployed as if its cost were uniform across inputs, and it is not. Where the answer is spread over redundant tokens compression is safe; where it sits in a few near-tie tokens eviction destroys three quarters of the answers the full cache gets right. One label-free prefill statistic, the early-to-late head-agreement drop, predicts which side an input falls on before any token is generated. PAGE turns that into a collapse-avoiding safeguard: the harm rate at $16\times$ compression falls from 0.750 to 0.026 and a 99% $\rightarrow$ 0% collapse holds flat on Mistral-7B NIAH-MK3, across four base evictors, four model families, and `lcc` code completion. It costs one prefill pass that exposes attention, and loses at matched memory to a strong trained evictor.

REPRODUCIBILITY STATEMENT

Our SnapKV-family runs (the partition, the matrix, the scaling and 32K sweeps, and the LongBench probes) use single NVIDIA A100 80GB GPUs with PyTorch 2.11, Transformers 5.9, and Python 3.13, in `bfloat16` with greedy (`argmax`) decoding, `max_new_tokens = 128`, and a fixed seed, with three input-draw replicates on five SnapKV cells (Appendix G). The two published head-to-heads run in the baselines’ own pinned environments: the DBTrimKV comparison uses the authors’ `trimkv` environment (Transformers 4.57.1, PyTorch 2.8, FlashAttention 2), and the CapKV comparison is our re-implementation in an isolated environment, both documented with the released logs. Models are the public Hugging Face checkpoints of Qwen2.5-1.5B/3B/14B/32B-Instruct, Mistral-7B-Instruct-v0.3, Yi-1.5-9B-Chat, Llama-3.1-8B-Instruct, and Qwen3-4B-Instruct-2507, the last two of which carry the cross-architecture and DBTrimKV results respectively. Tasks come from the `simonjegou/ruler` port of RULER (Jégou, 2024; NVIDIA, 2024) and from LongBench (Bai et al., 2024). Cache slicing preserves RoPE alignment via explicit `position_ids`, and 16K runs use SDPA prefill with an eager re-forward of the $w = 32$ observation window for scoring (Appendix H). Each table caption states its per-cell sample sizes. The gate is one thresholded scalar (Eq. (1)), so anyone can recompute every gated number in the paper from the released per-input logs. Code, sweep configurations, and raw per-input jsonl outputs will be released with the camera-ready.

ETHICS STATEMENT

This work studies inference-time efficiency of publicly released language models on public benchmarks. It involves no human subjects, no personal data, and no new model training. Better KV-cache eviction lowers the memory and energy cost of long-context inference, and we see no ethical risk specific to this method beyond those of the underlying models.

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A DEFERRED TABLES, DERIVATIONS, AND SECONDARY EXPERIMENTS

A.1 RELATED WORK (FULL TREATMENT)

Section 2 summarizes the comparison. This subsection gives the full treatment, starting with the five neighbors closest to our setup before turning to concurrent, older, and adjacent work in turn.

Five close concurrent neighbors. Bui et al. (2026) (DBTrimKV) is the most important reference point. Their Proposition 3.1 establishes that near-tie distractors force attention dilution in a single head, and Corollary 3.2 shows preferential retention of high-attention positions reduces it. Section 5.6 adopts this mechanism. Their evaluation reports DBTrimKV $>$ full-cache on multi-turn dialogue but does not characterise a task family on which their method underperforms full-cache, and they train a retention gate end-to-end, where our gate is label-free and wraps theirs (and any other) without retraining. The direct head-to-head is in Section 5.5, $\Delta = +0.233$ at matched budget on RULER 4K Qwen3-4B-Instruct.

Yang et al. (2026a) (CapKV) derives a closed-form, label-free eviction score from an information-bottleneck objective and reports gains over a Vanilla baseline on LongBench. Their score is task-agnostic, whereas we partition tasks into a class for which any eviction is risky and a class for which it helps, and we report a direct head-to-head wrapping CapKV in our gate (Section 5.5, $\Delta = +1.5$ to $+8.3$ pp at matched budget on two LongBench subtasks).

Zhou et al. (2025) (DynamicKV) introduces task-aware per-layer adaptive budgets and explicitly observes that the layer-wise attention pattern differs across task families, using attention statistics to drive a continuous per-task budget allocation. This is the closest conceptual neighbor to our partition, since both detect that task type modulates which KV positions matter, but three differences separate the two: (i) DynamicKV’s per-task budget is fitted per task with calibration data, whereas ours is a per-input gate triggered by a single observable statistic with one fixed threshold; (ii) their benchmark suite (LongBench) does not include a task on which eviction strictly damages accuracy, so the binary partition is not visible in their results; and (iii) DynamicKV does not formalise a partition predictor or attempt cross-architecture transfer. We therefore see our partition, drop predictor, and binary gate as orthogonal but partially overlapping with their empirical observation, and we acknowledge the overlap.

Garcia (2026) (“Protection Is All You Need”) shows that structural protection (sink tokens plus a recent-token window) dominates the choice of scorer on globally capacity-capped harnesses. Our experimental setup uses protection of the same kind but smaller extent ($n_{\text{sink}} = 4$ sink tokens and a $w = 32$ observation window, versus the 10% prefix plus 10% suffix they advocate). A protection-matched ablation at their 10% + 10% setting confirms the partition is not an artifact of light protection: NIAH-MK3 still collapses, and the eviction-helps signal survives (Section 3).

Yang et al. (2026c) (IndexMem) trains a learnable indexer plus latent memory and observes an information-density effect on LongBench averages around 50% compression. The benchmark average hides which subtasks contribute the peak, whereas our partition makes that explicit, and our predictor is training-free.

Concurrent 2026 diagnostics and wrappers. Zhang et al. (2026) ask a similar headline question, namely when value-aware KV eviction helps. The clearest separation is the axis: their score is *value-aware* (a per-input support-coupling score $\phi(x)$ over value states), whereas our signal is *attention topology*, a complementary account in the same sense that VaSE is (below). Their unit of analysis is the selector rather than the task, their $\phi(x)$ drives diagnostic grid splits rather than an evict-or-not gate, and they report that diagnostic signals transfer across selectors only conditionally, which makes the single transferred threshold we report non-obvious. Chen et al. (2026) document instruction-type-dependent degradation under five eviction methods, with some instruction types becoming entirely ignored, the closest published task-level failure taxonomy. We cede the hurts-degradation observation to them and claim instead the joint helps-versus-hurts partition plus the a-priori predictor. Ananthanarayanan et al. (2026) compute a *within-layer* head-consensus level (how much heads inside a layer attend to the same tokens) and relate it descriptively to compression tolerance at the architecture level. Our statistic differs on both axes: an *early-to-late*, *cross-layer* drop in the top- k agreement between heads, computed online at prefill and thresholded per input

to gate. To our knowledge the identical statistic has not been proposed as an eviction gate before, and the ablation of Table 9 shows it separates the partition cleanly (AUC 1.000) and transfers across families. On the failure side, Chang et al. (2026) (VaSE) trace catastrophic eviction failures to high-magnitude value states, an axis orthogonal to our attention-topology account, and ManifoldKV (Datta et al., 2026) shows that a Euclidean-outlier key scorer improves NIAH-MultiKey retention under compression, which is why we define “capacity-bound” relative to the scorer being gated (Section 6).

Two attention-free or post-prefill methods deserve their own treatment because they overlap us more than the above. SAGE-KV (Wang et al., 2025) shares our *premise* most directly of any work we know: that after prefilling, the model’s own attention already indicates which tokens can be dropped, read once rather than per decode step, with a head-level analysis. The difference is what is done with the premise. SAGE-KV always compresses and uses the signal to pick *which* tokens and heads to keep, whereas we use a signal from the same source to decide *whether* to evict at all, do no head-level selection, and, unlike SAGE-KV, characterize a task class on which eviction cannot be made safe by any choice of tokens. KeyDiff (Park et al., 2025) is the sharper challenge, and we treat it as such in Section 6: it is a published, attention-free, FlashAttention-compatible evictor whose stated motivation is precisely that materializing attention is expensive, which is our first limitation. Our answer is empirical rather than rhetorical. KeyDiff collapses on NIAH-MK3 below 25% budget like every scorer we tested, so its cheapness does not buy safety on the capacity-bound class. On the wrapper side, VECTOR (Lin et al., 2026b) and CriticalKV (Feng et al., 2026b) are plug-and-play augmentations over multiple base evictors, and both always compress under a fixed budget and adapt *which* tokens are kept, whereas our gate decides *whether* to evict at all, a priori from a label-free attention signal. The serving system EvicPress (Feng et al., 2025a) also chooses whether to evict per context, but as part of a profiled quality-latency optimization over storage tiers rather than from an attention statistic, and without a task-type partition. Cross-family transfer of an eviction control is itself known, via CompilerKV’s (Yang et al., 2026b) portable per-head retention tables, though we transfer one threshold instead, fixed across the Qwen and Mistral families and extended to the Llama-architecture models and Qwen3 by an unlabeled per-model standardization. Further concurrent evictors and allocators, EntropyInfer (Xu et al., 2026), CompressKV (Lin et al., 2026a), MomentKV (Li et al., 2026b), EpiKV (Kolawole & Smith, 2026), DefensiveKV (Feng et al., 2026a), LookaheadKV (Ahn et al., 2026), and TAKE (Hu et al., 2026), adapt budgets or scores per head, layer, token, or task, but none gates the eviction decision per input or identifies a task class where eviction hurts monotonically. AnchorKV (Ni & Lao, 2026) is also “safety-aware” but in a different sense from ours: it biases retention scores away from harmful-prompt directions for jailbreak defense while always compressing, whereas our “collapse-avoiding” sense is about not destroying task accuracy. Learned gates (Attention-Gate (Zeng et al., 2025), Fast KVzip (Kim et al., 2026)) share the “gate” name but train their gating parameters, unlike our inference-time scalar. A separate line uses learnable per-layer gates for cross-model KV-cache *fusion* (C2C (Fu et al., 2025)), which addresses communication between models, orthogonal to eviction within one.

Beyond the concurrent methods above, an older set of baselines and training-based predictors anchors the comparison. SnapKV (Li et al., 2024), H2O (Zhang et al., 2023), StreamingLLM (Xiao et al., 2024), PyramidKV (Cai et al., 2025), and Ada-KV (Feng et al., 2025b) are the standard training-free eviction baselines. We use SnapKV-style scoring as the base for our gate and demonstrate orthogonality on H2O, StreamingLLM, and PyramidKV. ForesightKV (Dong et al., 2026) and Learning-to-Evict (Moschella et al., 2026) train predictors of future utility and require labels or roll-outs to do so. UT-ACA (Zhou et al., 2026) and LU-KV (Tang et al., 2026) are calibration-based context allocators on a single substrate. None of these proposes a binary task-type partition or an a priori predictor.

Finally, three methods adapt compute on axes adjacent to but distinct from ours: AdaCompute-GBM (Zhai et al., 2026) on the self-consistency sample-count axis, BudgetThinker (Wen et al., 2025) on the CoT chain-length axis, and LaTER (Li et al., 2026a) on latent-then-explicit reasoning. These work on substrates orthogonal to the KV-cache axis we study.

Table 1: Fixed-budget harm rate $\Pr_x[A(b, x) < A_{\text{plain}}(1, x)]$, gated / plain at matched nominal budget, pooled over the seven SnapKV cells (Qwen2.5-1.5B/3B/14B and Mistral-7B at 4K; Qwen2.5-1.5B/3B and Mistral-7B at 16K), $N = 500$ per (task, budget), $\tau = 0.07$. On the capacity-bound task the gate cuts harm by $29\times$ at $16\times$ compression; on QA_1 it is exactly inert, since the gate opens on every input; on VT at $16\times$ both arms are harmed almost equally, so the gate does not help there.

Task	$b = 0.0625$	$b = 0.125$	$b = 0.25$	$b = 0.5$
NIAH-MK3	0.026 / 0.750	0.026 / 0.744	0.024 / 0.724	0.018 / 0.546
FWE	0.236 / 0.414	0.162 / 0.298	0.118 / 0.224	0.064 / 0.116
QA_1	0.110 / 0.110	0.064 / 0.064	0.030 / 0.030	0.010 / 0.010
VT	0.684 / 0.694	0.108 / 0.110	0.034 / 0.036	0.026 / 0.026

Table 2: Per-input recovery rate ρ on RULER with Wilson 95% intervals (small type). Bold marks point estimates $\rho \geq 0.05$. NIAH-MK3 (precise multi-key retrieval) is the only task that systematically fails the threshold in every cell tested. $N = 100$ per (task, cell) except Qwen-14B ($N = 50$ at 4K, $N = 30$ at 16K), Mistral 16K VT/FWE/QA_1/NIAH-MK3 ($N = 50$), Qwen-3B 16K NIAH-MK3 ($N = 50$), and Qwen-1.5B 16K NIAH-MK3 ($N = 46$). At these N most single cells straddle the 0.05 threshold and no per-cell call is individually powered; the claim rests on the pooled contrast, which is unambiguous (NIAH-MK3: $2/526$, $\rho = 0.004$, Wilson $[0.001, 0.014]$).

Task	Qwen 1.5B 4K	Qwen 1.5B 16K	Qwen 3B 4K	Qwen 3B 16K	Qwen 14B 4K	Qwen 14B 16K	Mistral 4K	Mistral 16K
VT	0.06 [.03, .13]	0.19 [.13, .28]	0.00 [.00, .04]	0.05 [.02, .11]	0.00 [.00, .07]	0.03 [.01, .17]	0.00 [.00, .04]	0.32 [.21, .46]
FWE	0.06 [.03, .13]	0.04 [.02, .10]	0.01 [.00, .05]	0.32 [.24, .42]	0.00 [.00, .07]	0.20 [.10, .37]	0.02 [.01, .07]	0.22 [.13, .35]
QA_1	0.08 [.04, .15]	0.14 [.09, .22]	0.03 [.01, .09]	0.08 [.04, .15]	0.06 [.02, .16]	0.10 [.03, .26]	0.04 [.02, .10]	0.00 [.00, .07]
QA_2	0.08 [.04, .15]	0.05 [.02, .11]	0.04 [.02, .10]	0.02 [.01, .07]	0.02 [.00, .11]	0.03 [.01, .17]	0.01 [.00, .05]	0.00 [.00, .04]
niah_multivalue	0.07 [.03, .14]	0.10 [.06, .17]	0.02 [.01, .07]	0.13 [.08, .21]	0.08 [.03, .19]	0.07 [.02, .21]	0.06 [.03, .13]	0.21 [.14, .30]
NIAH-MK3	0.01 [.00, .05]	0.00 [.00, .08]	0.00 [.00, .04]	0.02 [.00, .11]	0.00 [.00, .07]	0.00 [.00, .11]	0.00 [.00, .04]	0.00 [.00, .07]

A.2 DEPLOYMENT COST OF THE GATE

The gate-open rate p_{open} is a property of the workload rather than the method, tracking the distribution of D over inputs served, so realized compression is not a constant of PAGE and savings are bounded by $1 - p_{\text{open}}$. Two consequences follow.

Capacity planning drifts with traffic. A fleet provisioned for the $3\times$ compression that $p_{\text{open}} = 0.75$ implies loses that headroom as traffic shifts toward low- D inputs, and since D falls as near-tie distractors accumulate, an adversary controlling prompt content can force the fallback, turning a memory attack into a denial-of-capacity attack. Capping the fallback fraction buys a bounded memory guarantee, and adapting τ to a running q -quantile of D fixes p_{open} , at the cost of the safety property or a workload-dependent threshold. Batching sharpens this, since a batch provisioned for its largest resident cache realizes the *minimum* compression over its sequences: mean compression at $p_{\text{open}} = 0.75$ falls from $2.91\times$ at batch 1 to $1.31\times$ at 4, $1.07\times$ at 8, and $1.00\times$ by 16. That is worst-case provisioning rather than a property of PAGE, and does not describe continuous batching with per-sequence paged allocation, so we report p_{open} alongside every compression number.

A.3 ADDITIONAL EMPIRICAL-PARTITION ANALYSES

The table above establishes the partition on a realistic workload. The next four analyses stress-test it: they explain why some cells fall below the dilution-prone threshold, track the partition as context length grows, rule out task identity as a confound, and check that it survives protection schemes heavier than our own.

A cell can fall below the dilution-prone threshold for two different reasons, and only one of them is benign. Saturation is the benign case. Qwen2.5-3B has $\rho \approx 0$ on VT at 4K because $A_{\text{full}} \approx 1$, and the same model reaches $\rho = 0.32$ on FWE at 16K once headroom opens. Mistral-7B repeats the pattern: at 4K it is largely saturated ($A_{\text{full}} = 1.00$ on VT, 0.82 on FWE), and at 16K VT and FWE recover strongly ($\rho = 0.32$ and 0.22, niah_multivalue 0.06 $\rightarrow$ 0.21). Budget-insensitivity is the informative case instead, because headroom exists and recovery still does not happen. On Mistral, QA_1 has $\rho = 0.04/0.00$ at 4K/16K despite $A_{\text{full}} = 0.81/0.70$, and QA_2 has $\rho = 0.01/0.00$ despite $A_{\text{full}} = 0.50/0.56$, and QA_2 on Qwen-14B behaves the same way. These three cells are the

Table 3: Mean head-agreement drop D and gate-open fraction on four additional RULER tasks, $N = 100$ per cell, $\tau = 0.07$. `niah_multiquery` (four required keys, no real distractors) closes the gate on two of three models and sits nearest the anchor on the third, confirming a second capacity-bound task. `qa_2` (29 real hard-negative documents), `niah_single_1` (no distractors), and `cwe` (ten exact output tokens) all open the gate on most or all inputs, so neither distractor count nor exact-token output length predicts the class.

Task	Qwen2.5-1.5B	Qwen2.5-3B	Mistral-7B
<code>niah_multiquery</code>	$D=0.076$, open 32%	$D=0.021$, open 100%	$D=0.042$, open 100%
<code>qa_2</code>	$D=0.213$, open 0%	$D=0.097$, open 10%	$D=0.086$, open 18%
<code>niah_single_1</code>	$D=0.153$, open 0%	$D=0.061$, open 87%	$D=0.121$, open 0%
<code>cwe</code>	$D=0.200$, open 0%	$D=0.103$, open 0%	$D=0.135$, open 0%

Table 4: The partition on a non-synthetic workload: three LongBench subtasks, Qwen2.5-14B, SnapKV, $\tau = 0.07$. D is the mean head-agreement drop and “open” the fraction of inputs on which the gate fires. “Sensitivity” is the fraction of full-cache-correct answers plain eviction destroys at $b = 0.0625$ (the capacity signal) and ρ the fraction of inputs on which eviction recovers a full-cache failure (the dilution signal). The gate’s class call is made from D alone, before decoding, and matches the empirical class on all three.

Subtask	N	A_{full}	sens.	ρ	D	open	empirical class	gate call	Δ
lcc	96	0.292	0.64	0.031	0.0543	0.250	capacity-bound	do not evict	+0.146
repobench-p	82	0.317	0.42	0.061	0.0541	0.256	mixed	do not evict	+0.049
hotpotqa	100	0.560	0.12	0.020	0.0973	0.860	evict-robust	evict	+0.000

exceptions noted in Section 3. All are QA, where the residual errors look like parametric-knowledge failures that no cache budget can repair, a different phenomenon from the capacity limit NIAH-MK3 exhibits.

That headroom argument predicts what should happen as context grows, and Qwen2.5-14B tests it directly by re-entering the dilution regime at long context. At 4K, three dilution-prone tasks (VT, FWE, QA_2) sit at $\rho \approx 0$: saturation, since $A_{\text{full}} \in [0.92, 1.00]$ leaves no headroom for recovery, as Eq. (3) predicts. At 16K, two of five dilution-prone tasks cross the threshold (FWE $0.00 \rightarrow 0.20$, QA_1 $0.06 \rightarrow 0.10$), `niah_multivalue` stays above it ($0.08 \rightarrow 0.07$), and the capacity-bound NIAH-MK3 stays at exactly 0.00 at both scales. VT ($0.00 \rightarrow 0.03$) and QA_2 ($0.02 \rightarrow 0.03$) remain below threshold because the inputs that the model gets wrong at 4K stay wrong at every budget at 16K. Headroom opens, dilution returns, and NIAH-MK3 stays pinned: the partition behaves as the scaling formula dictates, not as task identity dictates.

That last point deserves a direct test, since task identity and dilution-proneness could in principle be confounded. They are not: the partition is not a task-name heuristic. The NIAH family contains structurally different subtasks. `niah_multivalue` (one key, many values) is dilution-prone in seven of the eight cells measured (Table 2), while NIAH-MK3 (three near-tie keys, one of which is the answer) is capacity-bound in all of them, and the distractor sweep below places `niah_multitask_1` (no near-tie distractors) on the dilution-prone side as well. The *near-tie distractor* structure, not the surface task category, is what matters, and the head-agreement drop in Section 4.2 catches the distinction. A non-RULER probe confirms this: LongBench passage_retrieval_en (pick 1 of 30 paragraphs, $N = 100$, Qwen2.5-1.5B, prompts 10-16K tokens) *looks* NIAH-like but has no near-tie distractors, and it is evict-robust in practice. Accuracy stays flat from $b = 1.0$ to $b = 0.0625$ ($0.29 \rightarrow 0.26$, $\rho = 0.01$), while its mean drop $D = 0.105$ sits squarely in the dilution-prone cluster, so the gate opens on 100/100 inputs. The predictor gets the surface-similar, structurally different task right a priori.

The remaining question is whether the partition is an artifact of our own, relatively light protection setting, so we checked it against a substantially heavier one. Garcia (2026) argue that structural protection dominates scorer choice, so we re-ran the Qwen2.5-1.5B 4K partition tasks with their 10% prefix + 10% suffix protection ($n_{\text{sink}} = w = 410$, $N = 100$ per task). One accounting note: protection overrides the budget, so all nominal $b \leq 0.1875$ share an effective floor of 824 kept

tokens (≈ 0.21 of context). Under this heavy protection the partition is unchanged. NIAH-MK3 still collapses (0.65 full-KV $\rightarrow 0.09$ at the floor, an 86% relative loss despite the floor cache being $3.3\times$ our light-protection cache at nominal $b = 0.0625$), so the capacity-bound failure is not protection-fixable. The eviction-helps signal survives on the dilution-prone side (ρ : FWE $0.06 \rightarrow 0.13$, QA_1 $0.08 \rightarrow 0.07$, VT $0.06 \rightarrow 0.03$, aggregate $0.077 \geq 0.05$), and FWE’s floor accuracy (0.31) exceeds its full-KV accuracy (0.22). Heavy protection is also not free: protected VT drops to 0.00 at the floor while our light-protection run holds 0.75 at nominal $b = 0.125$, because protected prefix/suffix tokens crowd out the scorer-selected middle tokens the task needs. One caveat on the comparison itself: in our harness w is both the recency protection and the SnapKV scoring window, so this run is a protection-heavy SnapKV variant rather than an exact replication of their protocol.

A.4 HEADLINE MATRIX FIGURE

The single-task showcase and the headline matrix from Section 5 are reproduced here at full size.

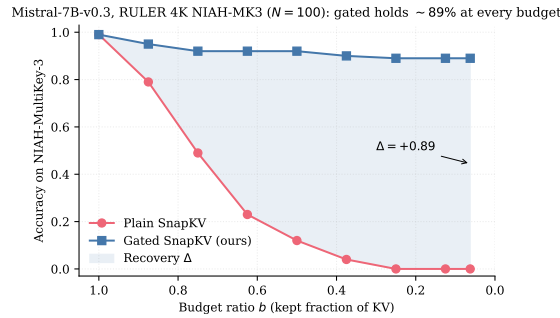


Figure 4: The single-task showcase of Section 5. Mistral-7B 4K NIAH-MK3, $N = 100$: accuracy vs. KV budget for plain SnapKV (solid, collapses to 0) and gated SnapKV (dashed, 0.89 flat across budgets). The gate prevents the collapse without retraining the base evictor or seeing labels.

A.5 PER-TASK DECOMPOSITION OF THE MATRIX

Per-head allocation and the Mistral-7B exception. Three caveats bound the matrix. The plain arm uses a shared single keep-mask and a clean per-head SnapKV collapses less sharply at $2\times$ (0.32 not 0.14 on MK3), so these deltas overstate what the gate prevents there and should be read at $\geq 4\times$, where the collapse is scorer-independent. That restriction does not inflate the headline, the grand mean rising by only $+1.7\text{pp}$ when the grid narrows to $\{0.125, 0.25\}$, and a clean per-head Ada-KV plain arm costs just 1.1pp ($+25.8$ to $+24.7\text{pp}$ at $b \leq 0.25$), much closer to a wash than a collapse. That average hides one model: on Mistral-7B the per-head arm underperforms the shared mask at 4 of 8 budgets on NIAH-MK3, growing to 10.5pp at $b = 0.875$ (0.79 shared versus 0.35 Ada-KV). The cause is not grouped-query geometry, since Qwen2.5-14B and a dedicated Llama-3.1-8B control matching Mistral’s head shape (32 query heads, 8 key-value heads) are both clean at 0 of 8, the Llama control taking the opposite sign on the same comparison ($+14\text{pp}$ on Llama versus -44pp on Mistral). The violation is confined to 4 of 32 cell-budget combinations, all on Mistral-7B. The Qwen2.5-14B H2O column is separately not independent, since that run used two-pass prefill under which H2O’s all-query score reduces exactly to SnapKV’s, leaving fifteen independent cells.

The matrix figure reports one number per cell; decomposing that number by task explains where it comes from. Table 6 breaks down every matrix cell by task and reports the gate-open fraction that explains the zeros, and two structural facts follow from it. First, $\Delta_{\text{non-MK3}}$ is exactly 0.000 in 8 of 16 cells, and in each of those the gate opens on 100% of non-MK3 inputs, so the gated and plain arms execute identically: the zero is an identity, not a null result. Second, the mean Δ_{MK3} is $+0.780$ while the grand mean is $+0.229 \approx \frac{1}{4}\Delta_{\text{MK3}}$, the suite’s MK3 share. Since per-task accuracy is additive under reweighting, the headline for a workload whose capacity-bound share is f is $\Delta(f) = f \cdot \Delta_{\text{MK3}} + (1 - f) \cdot \Delta_{\text{rest}}$, which is near-linear through the origin (Figure 6): $\Delta(0) = +0.045$, $\Delta(0.25) = +0.229$ (the reported suite), $\Delta(0.5) = +0.412$. A deployment

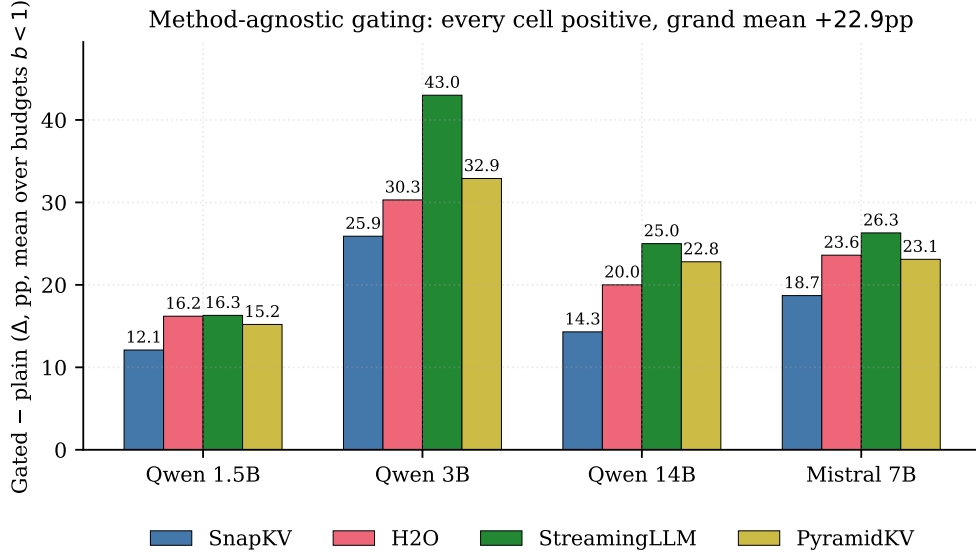


Figure 5: The headline 4×4 method-agnostic matrix. Bars are gated – plain accuracy averaged over budgets. The wrapper is positive in every one of 16 cells with a single $\tau = 0.07$ and no per-base-evictor tuning. Grand mean +22.9pp; excluding NIAH-MK3, +4.5pp (Table 6).

Table 5: Headline 4×4 method-agnostic matrix. Same gate ($\tau = 0.07$), no per-base-evictor tuning; Δ averages over budgets $b < 1.0$. All 16 cells are positive with grand mean +22.9pp, but the second Δ row shows what that is made of: excluding NIAH-MK3 the grand mean falls to +4.5pp and is exactly 0.000 in 8 of 16 cells, where the gate opens on every non-MK3 input and the two arms are the same system. The Qwen2.5-3B SnapKV cell ran at $\tau = 0.04$ and is reported at $\tau = 0.07$ by exact post-hoc re-evaluation (Appendix H). [†]Qwen2.5-14B H2O is not an independent cell (see text). Per-task decomposition: Table 6.

Model	SnapKV		H2O		StreamingLLM		PyramidKV	
	plain	gated	plain	gated	plain	gated	plain	gated
Qwen2.5-1.5B	0.438	0.559	0.196	0.357	0.033	0.195	0.417	0.570
Δ	+0.121		+0.162		+0.163		+0.152	
Δ no MK3	+0.000		+0.000		+0.000		+0.000	
Qwen2.5-3B	0.580	0.839	0.339	0.642	0.052	0.482	0.536	0.865
Δ	+0.259		+0.303		+0.430		+0.329	
Δ no MK3	+0.101		+0.098		+0.263		+0.144	
Qwen2.5-14B	0.709	0.853	0.668 [†]	0.868 [†]	0.080	0.330	0.632	0.860
Δ	+0.143		+0.200		+0.250		+0.228	
Δ no MK3	+0.000		+0.000		+0.000		+0.000	
Mistral-7B	0.623	0.810	0.400	0.636	0.102	0.365	0.593	0.823
Δ	+0.187		+0.236		+0.263		+0.231	
Δ no MK3	+0.015		+0.020		+0.053		+0.020	

therefore sees the gate’s benefit in proportion to how much capacity-bound traffic it actually serves, and the realistic-workload estimate of that share is in Table 4.

Budget convention. Δ averages over the eviction budgets $b < 1.0$. Including $b = 1.0$ leaves the value unchanged because the gated and plain arms coincide there. Restricting instead to $\{0.125, 0.25, 0.5\}$ gives +0.240, and to the four budgets shared by all cells +0.243. We report $b < 1.0$ throughout.

Table 6: Per-task Δ (gated – plain, budgets $b < 1.0$, $\tau = 0.07$) for all 16 matrix cells, with the gate-open fraction on non-MK3 inputs. Rows marked $\dagger$ have $\Delta_{\text{non-MK3}}$ exactly 0.000 with the gate open on every non-MK3 input, so gated $\equiv$ plain there by construction. Recomputed from the released per-input logs.

Base evictor	Model	MK3	VT	FWE	QA.1	all	no MK3	open (non-MK3)
SnapKV	Qwen2.5-1.5B	+0.484	0.000	0.000	0.000	+0.121	+0.000 $\dagger$	1.000
	Qwen2.5-3B	+0.734	+0.004	+0.300	0.000	+0.259	+0.101	0.653
	Qwen2.5-14B	+0.573	0.000	0.000	0.000	+0.143	+0.000 $\dagger$	1.000
	Mistral-7B	+0.701	0.000	+0.046	0.000	+0.187	+0.015	0.923
H2O	Qwen2.5-1.5B	+0.647	0.000	0.000	0.000	+0.162	+0.000 $\dagger$	1.000
	Qwen2.5-3B	+0.920	+0.017	+0.277	0.000	+0.303	+0.098	0.653
	Qwen2.5-14B	+0.800	0.000	0.000	0.000	+0.200	+0.000 $\dagger$	1.000
	Mistral-7B	+0.883	0.000	+0.043	+0.017	+0.236	+0.020	0.923
StreamingLLM	Qwen2.5-1.5B	+0.650	0.000	0.000	0.000	+0.163	+0.000 $\dagger$	1.000
	Qwen2.5-3B	+0.930	+0.030	+0.760	0.000	+0.430	+0.263	0.653
	Qwen2.5-14B	+1.000	0.000	0.000	0.000	+0.250	+0.000 $\dagger$	1.000
	Mistral-7B	+0.890	0.000	+0.140	+0.020	+0.263	+0.053	0.923
PyramidKV	Qwen2.5-1.5B	+0.610	0.000	0.000	0.000	+0.152	+0.000 $\dagger$	1.000
	Qwen2.5-3B	+0.883	0.000	+0.433	0.000	+0.329	+0.144	0.653
	Qwen2.5-14B	+0.913	0.000	0.000	0.000	+0.228	+0.000 $\dagger$	1.000
	Mistral-7B	+0.863	0.000	+0.060	0.000	+0.231	+0.020	0.923
mean		+0.780	+0.003	+0.129	+0.002	+0.229	+0.045	0.894

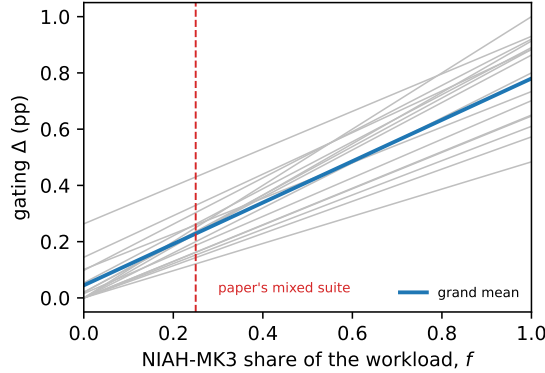


Figure 6: Gating Δ as a function of the capacity-bound share f of the workload, $\Delta(f) = f\Delta_{\text{MK3}} + (1 - f)\Delta_{\text{rest}}$, mean-line slope 0.7354. Grey lines are the 16 matrix cells, the blue line their mean, and the dashed marker the $f = 0.25$ composition of our mixed suite. Because Δ_{rest} is exactly zero in half the cells, the mean is close to linear through the origin, so the headline +22.9pp should be read as a statement about a workload that is one quarter capacity-bound rather than as a property of the method.

A.6 DETECT-AND-RETRY AS AN ALTERNATIVE DESIGN

The gate is not the only way to handle capacity-bound inputs. A natural objection is that one could evict unconditionally and simply re-run the inputs that fail, avoiding the gate and its attention-exposing pass entirely. The comparison is analytic, so we state it rather than measure it. Writing p_C for the prevalence of capacity-bound inputs and c_{pre} , c_{dec} for prefill and decode cost, PAGE pays c_{gate} on every input plus a memory cost of $(1 - p_{\text{open}})$ full caches, where c_{gate} is 17ms on Qwen2.5-1.5B and 89ms on Mistral-7B at 4K (Table 24). Detect-and-retry instead pays nothing up front but incurs a second full prefill and decode on every input it retries, so its expected overhead is $(p_C \cdot \text{TPR} + (1 - p_C) \cdot \text{FPR}) \cdot (c_{\text{pre}} + c_{\text{dec}})$, and every capacity-bound input its detector misses is returned wrong rather than merely late. At the prevalence $p_C \approx 0.25$ measured on the realistic workload (Table 4) and with c_{gate} two orders of magnitude below a full prefill-plus-decode, retry breaks even only if its detector is both near-perfect in recall and has a false-positive rate below

roughly 0.03, since each false positive costs a doubled request. The two designs also differ in kind rather than degree: the gate spends bounded memory to make a failure impossible, while retry spends unbounded tail latency to make it recoverable, and only the gate is available to a system that cannot re-run a request. We note this as a design point rather than a claim, since we did not build a decode-side distress detector, and a good one would be a contribution in its own right.

The comparison above assumes the gate’s signal generalizes beyond attention-score scoring, an assumption worth testing directly. To check composition beyond attention-score-based bases, we add a ManifoldKV-style scorer (Datta et al., 2026): per-(layer, head) Euclidean distance of each key to the head’s key centroid, averaged into a single shared keep mask (a single-mask variant of their per-layer design, forced by our uniform-cache harness). On the Qwen2.5-1.5B 4K mixed suite ($N = 100/\text{task}$, four budgets), the same gate at the same $\tau = 0.07$ lifts it by $\Delta = +16.2\text{pp}$ with no negative (task, budget) cell, and on NIAH-MK3 the gate recovers full-KV accuracy ($0.00 \rightarrow 0.65$) at every budget. Two observations follow. The gate signal is attention-based and scorer-independent, so composition extends outside the attention-score family, but our single-mask variant does *not* reproduce ManifoldKV’s reported multi-key rescue (plain accuracy 0.01 at $b = 0.5$). Since their per-layer, per-head design may behave differently, we treat their published numbers as the authoritative claim and scope ours accordingly (Section 6).

A.7 THE CAPACITY LIMIT IS SCORER-INDEPENDENT

That single-mask variant is not a fair test of a geometry scorer’s full capability, so we also built a faithful ManifoldKV+Ada-KV implementation (Datta et al., 2026; Feng et al., 2025b): score $s_i = \|k_i - \mu\|_2$ against the per-(layer, kv-head) key centroid μ , with Ada-KV head-wise budget reallocation (safeguard floor `floor_alpha` = 0.5 plus a global top- k over the remaining per-head scores). Ragged per-head keep counts are realized by masking evicted keys to $-\infty$ per kv-head rather than physically pruning, which reproduces exactly the logits a ragged compressed cache would produce with no over-allocation, and the measured kept fraction matches the nominal budget to four decimals. We add KeyDiff (Park et al., 2025) (negative cosine to the same centroid) as the published attention-free scorer.

Before trusting a negative result from this scorer, we first confirm it is competent. Table 7(a) is the control that makes the negative result interpretable: on two-key NIAH the geometry scorer reproduces exactly the multi-key advantage its authors report, reaching 0.72 at $b = 0.5$ where SnapKV is at 0.16. A scorer that can do this is not failing MK3 for want of a better scoring rule, and indeed every scorer we test still collapses on MK3. Table 7(b) shows that at $b \leq 0.25$, SnapKV, KeyDiff, and ManifoldKV (post- and pre-RoPE, uniform and Ada-KV allocation) are all at or near 0.00 against a full-cache 0.66. The geometry scorer does not beat a properly per-head SnapKV at any budget here and is weaker at $b = 0.5$ (0.06-0.10 versus 0.32). The conclusion we draw is the one the gate is built on: on capacity-bound inputs the recoverable information is not present in any 25% subset of the cache, so no scoring rule, attention-based, geometric, or attention-free, can recover it, and the only safe action is not to evict.

A.8 ADDITIONAL LIMITATIONS

Section 6 states the four limitations severe enough for the main text. Seven more, individually narrower, are collected here.

Statistical power is the first. Per-cell N ranges from 30 to 100 inputs (per-task N noted in each table). All headline deltas carry cluster-bootstrap 95% intervals and all ρ values carry Wilson intervals (Section 5, Appendix G), and every matrix delta excludes zero, but the per-cell partition calls are individually low-powered, since most straddle the 0.05 threshold, and small- N cells (Qwen-14B 16K, $N = 30$; triviaqa head-to-head, $N = 24$) remain noisy. The partition claim rests on the pooled contrast instead, which is unambiguous. Decoding is greedy throughout, and five SnapKV cells carry three input-draw replicates each (Appendix G.1), while the remaining sweeps are one draw.

A related question is whether the gate ever costs accuracy on a workload with no capacity-bound inputs at all, and LongBench under SnapKV scoring answers it. On qasper, triviaqa, trec, and multifielddqa_en (Bai et al., 2024) (plus passage_retrieval_en on the 1.5B model), the pooled gated Δ is within $\pm 1\text{pp}$ of plain on all four models (Qwen2.5-1.5B 0.000, Qwen2.5-3B +0.002, Mistral-7B

Table 7: Scorer-independence of the capacity limit (Qwen2.5-1.5B, RULER 4K, $N = 50$). (a) The geometry scorer reproduces the published two-key advantage, so it is competent. (b) On three-key NIAH-MK3 every scorer collapses below 25% budget. Accuracy at nominal budget b .

(a) two-key NIAH, the positive control

Scorer	full	$b = 0.5$	$b = 0.25$
SnapKV	0.76	0.16	0.04
ManifoldKV (post-RoPE)	0.76	0.72	0.56
ManifoldKV (pre-RoPE)	0.76	0.76	0.52

(b) three-key NIAH-MK3

Scorer (allocation)	full	$b = 0.5$	$b = 0.25$	$b = 0.125$	$b = 0.0625$
SnapKV (uniform per-head)	0.66	0.20	0.02	0.00	0.00
SnapKV (Ada-KV)	0.66	0.32	0.00	0.00	0.00
KeyDiff (uniform)	0.66	0.02	0.00	0.00	0.00
KeyDiff (Ada-KV)	0.66	0.02	0.00	0.00	0.00
ManifoldKV L2 post-RoPE (uniform)	0.66	0.02	0.00	0.00	0.00
ManifoldKV L2 post-RoPE (Ada-KV)	0.66	0.06	0.00	0.00	0.00
ManifoldKV L2 pre-RoPE (uniform)	0.66	0.02	0.00	0.00	0.00
ManifoldKV L2 pre-RoPE (Ada-KV)	0.66	0.10	0.00	0.00	0.00

+0.003, Qwen2.5-14B −0.008). The worst per-task cell is Qwen2.5-14B trec at −2.2pp, the only negative gating result in the project, which we attribute to gate misfires on a task with no capacity-bound inputs to rescue. LongBench, on this subset, has no capacity-bound task, so the partition predicts neutrality, and we observe it, including its mildly negative tail.

One specific cell needs that calibration in practice: Qwen2.5-3B 16K, where the fixed $\tau = 0.07$ default fails. The 40-input recipe of Section 4.3 fixes it, taking Δ from +3.4pp to +5.4pp. We recommend the recipe as the deployment default and keep the fixed- τ result throughout only to show that the gap it leaves is small.

A separate, purely infrastructural limitation concerns how three cells were measured. The gate-signal probe requests every layer’s full $[H, T, T]$ attention, which reaches roughly 120 GiB for Qwen2.5-14B at 4K and cannot be run directly. Two wrappers make it fit. The retention wrapper (v1) calls the stock attention and slices the returned tensor, and the query-chunk wrapper (v3) chunks the query axis while calling the stock function unchanged. We validated each against the stock script on the fitting cell before using it: v1 agrees exactly ($D = 0.0412$ against 0.0412 on the same host) and v3 agrees to +0.000029, which is floating-point reassociation. We report this validation because a third wrapper failed the same check. Rewriting attention through SDPA moved mean D from 0.0448 to 0.0300, a 33% shift, reproducibly on two different GPUs, so the cause is the output path perturbing hidden states and compounding across layers rather than a hardware difference. No reported number uses this third wrapper. Separating that 0.0148 code artifact from the 0.0035 hardware effect required varying code path and hardware together, since either comparison alone against the released numbers confounds the two.

Two further limitations concern which baselines appear where. DBTrimKV’s pretrained gates exist only for Qwen3 variants, so Section 5.5 runs on Qwen3-4B-Instruct-2507 rather than the Qwen2.5-keyed matrix, which is why DBTrimKV stays a standalone head-to-head rather than a matrix row. The matrix itself covers only training-free policies for the same reason: trained or learned evictors (DBTrimKV, CapKV, IndexMem, DynamicKV) are cited but not included in it.

The last limitation concerns how confidently NIAH-MK3 can stand in for the capacity-bound class generally. It is the only task where empirical capacity-boundedness (accuracy collapse under eviction) and the gate’s capacity-bound classification coincide. On LongBench passage_count at Qwen2.5-14B the gate closes on all inputs (mean $D = -0.013$) yet eviction is empirically *safe* (accuracy flat under $16\times$ eviction, $\rho = 0$): a benign *false positive* (over-conservative, costing compression not accuracy), the dual of the MK2 false negative (Appendix A). The head-agreement drop is therefore a one-sided predictor, reliable at firing on dilution-prone inputs, not a perfect classifier.

Table 8: Distractor-count sweep on the NIAH-MultiKey family (Qwen2.5-1.5B, RULER 4K, $N = 50$ per task). Mean head-agreement drop D shrinks monotonically as the near-tie distractor count grows.

Task	mean D	gate-open frac	ρ_{plain}	ρ_{gated}	plain at $b=0.0625$
niah_multikey_1	+0.155	1.00	0.04	0.04	0.36
niah_multikey_2	+0.122	1.00	0.00	0.00	0.00
niah_multikey_3	+0.045	0.00	0.00	0.00	0.00

Multilingual and code tasks remain out of scope. We expect the mechanism to transfer, since it is architecture-level, but we have not measured it.

A.9 MECHANISM-VALIDATION DETAIL: NEAR-TIE DISTRACTOR COUNT DRIVES D

The capacity-bound classification above rests on a mechanism claim: that near-tie distractors, not task identity, distractor count, or output length in general, drive D down. Section 3 already shows that real hard-negative distractors do not have this effect, since qa_1 and qa_2 carry the most distractors in the suite and the largest D (Table 22), and that exact-token output length does not either, since cwe requires ten exact low-redundancy tokens yet keeps the gate open on every input tested (Table 3). This subsection isolates the near-tie case directly. The RULER NIAH-MultiKey family lets us sweep the near-tie distractor count with the number of documents held fixed: niah_multikey_1 (no near-tie distractors), niah_multikey_2 (one), niah_multikey_3 (two). The dilution mechanism of Bui et al. (2026) predicts that more near-tie distractors should produce larger expected dilution δ_t and, via App. F, smaller head-agreement drop D . On Qwen2.5-1.5B at RULER 4K, $N = 50$ per task, we observe exactly this: a clean monotone gradient (Table 8).

D shrinks by $\sim 3.5\times$ from $0 \rightarrow 2$ near-tie distractors, and per-input ranges between MK2 and MK3 do not overlap ($\max_{\text{MK3}} D = 0.062 < \min_{\text{MK2}} D = 0.088$): the partition predictor separates 2-distractor inputs from ≤ 1 -distractor inputs 100% of the time at $\tau = 0.07$. The monotone gradient replicates on Mistral-7B-v0.3 ($D = 0.081, 0.072, 0.060$ for MK1/MK2/MK3, $N = 50$ per task), confirming the mechanism is not Qwen-specific, though the finer per-input MK2/MK3 non-overlap is (their ranges overlap on Mistral).

This clean separation between 0 and 2 distractors leaves open what happens at the single-distractor boundary, and MK2 is exactly that case. It is empirically capacity-bound at low budgets (plain SnapKV collapses from 80% at $b=1$ to 0% at $b=0.0625$) yet is classified as dilution-prone by the predictor at $\tau = 0.07$: MK2’s mean $D = 0.122$ sits between the MK3 cluster ($D \leq 0.062$) and the MK1 cluster ($D \geq 0.099$), co-located with genuinely dilution-prone FWE ($D = 0.098$) and niah_multivalue ($D = 0.107$). No single mean- D threshold separates MK2 from the dilution-prone cluster on this model, so the predictor falls back to its weakest form here: it correctly classifies MK3 and the standard dilution-prone tasks but mis-classifies single-near-tie MK2. Per-input recovery information beyond mean D would be needed to catch it at deployment.

A.10 GATE-SIGNAL ABLATION: WHY THE HEAD-AGREEMENT DROP

The previous section validates the dilution mechanism behind D ; this one checks whether D is actually the best way to read that mechanism off the prefill attention, against five cheaper alternatives. A same-inputs ablation (Table 9, one shared prefill per input, $N = 50$ per task across seven RULER tasks) shows D is not merely the best candidate but the only one that separates the partition at all: MK3’s mean D sits strictly below all four dilution-prone task means with per-input AUC 1.000, while every alternative fails task-level separation and tops out at AUC 0.805. The dissection is informative. Raw attention entropy is at chance (0.508), taking its early-vs-late layer contrast lifts it to 0.805, and replacing entropy with head *agreement* makes the separation exact, so both the layer contrast and the agreement measure matter, with agreement decisive. No alternative is even a redundant proxy (all $|r|$ with $D \leq 0.64$). Consistent with the MK2 boundary case of Section 3, extending the capacity-bound anchor to MK3+MK2 breaks task-level separation for D too (AUC 0.859): D tracks the near-tie distractor gradient, not a binary task label.

Table 9: Gate-signal ablation on Qwen2.5-1.5B-Instruct (RULER 4K, $N = 50$ per task, one shared prefill per input). Each cheap prefill statistic is tested for separating the capacity-bound anchor (NIAH-MK3, and MK3+MK2) from the dilution-prone tasks (vt, fwe, qa_1, multivalue). “Sep.” = a single threshold puts the capacity-bound task mean(s) strictly on one side of all four dilution task means; AUCs are per-input and oriented per signal (0.5 = chance); “MK mono.” = task means are monotone along the MK1→MK2→MK3 distractor gradient; r = Pearson correlation with D .

Signal	Sep. MK3	Sep. MK3+MK2	AUC MK3	AUC MK3+MK2	MK mono.	r w/ D
head-agreement drop D (paper)	✓	–	1.000	0.859	✓	+1.00
mean attention entropy (norm.)	–	–	0.508	0.670	–	+0.08
top-32 attention mass share	–	–	0.784	0.574	✓	+0.22
max single-position share	–	–	0.723	0.708	✓	+0.32
key-norm dispersion (std/mean)	–	–	0.570	0.560	–	+0.01
early–late entropy drop	–	–	0.805	0.575	–	–0.64

A.11 THE SAME ABLATION ON CELLS THAT DID NOT SELECT THE SIGNAL

Table 9 is measured on the cell that D was chosen on, and an AUC of 1.000 there is the number a reader should trust least. We therefore repeat the ablation on three cells that played no part in selecting the statistic (Table 10), and the result splits into what survives held-out testing and what does not.

What survives is the qualitative claim: D remains the only one of the six statistics that achieves task-level separation anywhere, and it does so on two of the three held-out cells. Every alternative fails on every cell, held out or not, so the claim Table 9 makes now rests on more than the measurement that selected the signal.

What does not survive is the fitting cell’s exact number. The per-input AUC of 1.000 is a property of that cell alone. Held out, it is 0.899, 0.978, and 0.741, so we report 1.000 as a fitting-cell figure rather than as the separability of D in general.

The two weak held-out cells share a common cause: both are decided by a single task, niah_multivalue. On Llama-3.1-8B, NIAH-MK3 sits at $D = 0.0710$ while niah_multivalue sits at 0.0607, giving an unrestricted margin of -0.0103 . This is the result: D separates on two of the three held-out cells, not three, once niah_multivalue is counted. The other three dilution-prone tasks lie above MK3, and against the nearest of them (VT at 0.0783) the margin is $+0.0073$ if niah_multivalue is excluded. That figure is the one Appendix A.12 derives from the depth profile, reached here from an independent measurement. Section 6 already places this exclusion on the task-level ordering claim for a stated reason: niah_multivalue is eviction-fragile on Llama-architecture models, so a low D there is arguably the behaviorally correct call rather than a mis-ordering. We did not register that exclusion before running this ablation, so we report it here as a post-hoc diagnostic rather than a confirmed prediction, and Table 10 gives both margins so either reading is checkable. On Qwen2.5-14B the margin of $+0.0032$ is likewise set by niah_multivalue at 0.0487, and excluding it the margin is $+0.0436$ against FWE.

Table 10: Gate-signal ablation repeated on held-out cells. “sep.” counts the signals whose mean on NIAH-MK3 lies strictly below their mean on every dilution-prone task, which is the property the gate thresholds, and “margin” is that gap for D . “margin[−]” excludes niah_multivalue, the exclusion Section 6 places on the Llama-architecture ordering claim. AUC is per-input separability of NIAH-MK3 from the pooled dilution-prone tasks. The wrapper column records which memory workaround produced the cell (Appendix A.8).

Cell	Status	Wrap.	D MK3	min dil.	margin	margin [−]	AUC(D)	sep.
Qwen2.5-1.5B 4K	fitting	stock	0.0448	0.1114	$+0.0667$	$+0.0667$	1.000	1/6
Qwen2.5-14B 4K	held out	v1	0.0455	0.0487	$+0.0032$	$+0.0436$	0.899	1/6
Llama-3.1-8B 4K	held out	v1	0.0710	0.0607	-0.0103	$+0.0073$	0.741	0/6
Mistral-7B 16K	held out	v3	0.0544	0.0756	$+0.0212$	$+0.0393$	0.978	1/6

A.12 THE PER-LAYER PROFILE BEHIND D , AND WHAT IT SAYS ABOUT (A3)

The small Llama-3.1-8B margin above is a symptom; this subsection diagnoses the cause by looking inside D at the per-layer profile it summarizes. Assumption (A3) in App. F asserts a layer structure: early layers respond to local context, late layers to task structure, which is why an early-minus-late contrast should separate the two classes. The paper elsewhere reports only the binned scalar D , so we plot the profile a_ℓ itself (Figure 7). It is recomputed from the released per-layer logs at zero GPU cost.

On Qwen2.5-1.5B the profile behaves as (A3) describes. Agreement decays with depth, and NIAH-MK3 holds a markedly higher late plateau (0.283) than the dilution-prone tasks (0.227 on VT), so the contrast separates them by +0.068. On Llama-3.1-8B it does not. The profile is non-monotone, with a pronounced mid-stack spike, and NIAH-MK3 and VT become nearly indistinguishable under the contrast ($D = +0.071$ against +0.078, a margin of +0.007, an order of magnitude smaller than on Qwen).

We report this because it is the mechanistic content behind a failure the paper otherwise leaves unexplained. The fixed threshold does not transfer to the Llama architecture, and per-input AUC there is ≈ 0.80 rather than 1.000, not because the signal is absent but because the depth profile it summarises has a different shape. An early-minus-late contrast is the right summary of a monotone profile and the wrong summary of a profile with interior structure, which is also why the depth-correlation statistic comes out anti-correlated on Llama rather than merely weak. A statistic that reads the profile shape rather than its endpoints is the natural next step, and this is the concrete form of the “per-input statistic beyond mean D ” that Section 6 names as future work.

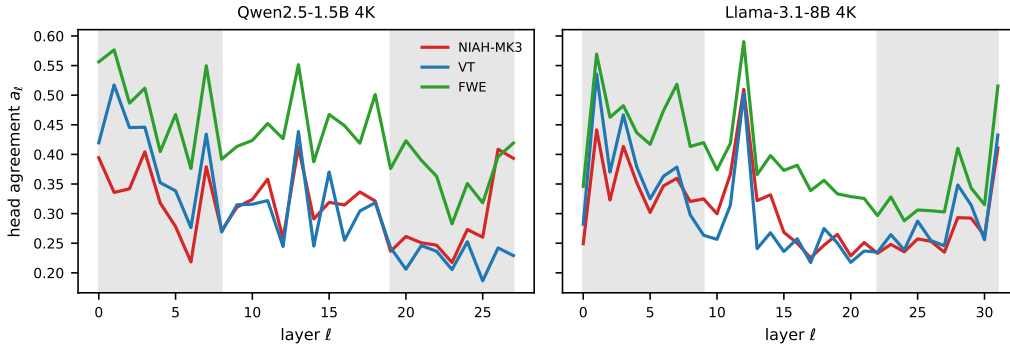


Figure 7: Per-layer head agreement a_ℓ , averaged over inputs ($N = 100$ per task, RULER 4K). Shaded bands are the early and late thirds that define D . Left: on Qwen2.5-1.5B the profile decays with depth and the capacity-bound task holds the highest late plateau, as (A3) predicts. Right: on Llama-3.1-8B the profile is non-monotone and NIAH-MK3 is nearly indistinguishable from VT under the same contrast, which is why the fixed τ does not transfer to that family.

A.13 DOES A TASK-LABEL ORACLE DO THE SAME JOB?

The per-layer diagnosis above concerns signal quality; a separate question is how much of the gate’s value is genuinely per-input rather than a proxy for task identity. The gate is advertised as a per-input predictor, but most of its validation is per task, so the sharp question is whether a policy that simply knew the task identity would match it. Table 11 answers it. The oracle closes on every NIAH-MK3 input and opens on every other one, which is the best any task-level policy can achieve, and it requires a label PAGE never sees.

The honest reading has three parts. In 8 of 16 cells PAGE reproduces the oracle *exactly*, so there the gate is a perfect task classifier and D buys label-freeness rather than per-input resolution. In 5 cells PAGE *beats* the oracle, by up to +0.198, which no task-level policy can do, since the surplus comes from closing on individual dilution-prone inputs that the oracle wrongly opens. In 3 Mistral cells PAGE loses by at most 0.009 through false positives on MK3. Overall the oracle reaches +20.1pp against PAGE’s +22.9pp.

So most of the matrix Δ is a task-level effect that a cheating baseline could match, and we say so. The per-input claim rests on the cells where PAGE exceeds the oracle and, more cleanly, on Llama-3.1-8B NIAH-MK3, where the mean $D = 0.071$ sits *above* τ : a task-level policy keyed on that mean opens on every input and gains nothing, while per-input dispersion closes the gate on $\approx 44\%$ of them and recovers +43.5pp (App. A.17).

Table 11: PAGE against a task-label oracle that closes on NIAH-MK3 and opens elsewhere. Δ is gated minus plain over budgets $b < 1.0$. A gap of exactly 0.000 means the gate reproduced the oracle’s decisions on every input; a positive gap requires within-task per-input variation.

Base evictor	Model	PAGE Δ	oracle Δ	gap
SnapKV	Qwen2.5-1.5B	+0.121	+0.121	0.000
	Qwen2.5-3B	+0.259	+0.183	+0.076
	Qwen2.5-14B	+0.143	+0.143	0.000
	Mistral-7B	+0.187	+0.195	−0.008
H2O	Qwen2.5-1.5B	+0.162	+0.162	0.000
	Qwen2.5-3B	+0.303	+0.230	+0.073
	Qwen2.5-14B	+0.200	+0.200	0.000
	Mistral-7B	+0.236	+0.243	−0.007
StreamingLLM	Qwen2.5-1.5B	+0.163	+0.163	0.000
	Qwen2.5-3B	+0.430	+0.232	+0.198
	Qwen2.5-14B	+0.250	+0.250	0.000
	Mistral-7B	+0.263	+0.247	+0.015
PyramidKV	Qwen2.5-1.5B	+0.152	+0.152	0.000
	Qwen2.5-3B	+0.329	+0.221	+0.108
	Qwen2.5-14B	+0.228	+0.228	0.000
	Mistral-7B	+0.231	+0.240	−0.009
grand mean		+0.229	+0.201	+0.028

How many layers does D need? The gate’s cost grows with the number of layers scanned, so we ask whether a subset suffices. Recomputing D from layer subsets of the stored per-layer agreements needs no GPU. We first check validity: recomputing over all L layers reproduces the `drop` the deployed gate actually thresholded, bit-exactly on the two Qwen cells and to 4×10^{-4} on Mistral, which is two orders of magnitude below τ . Table 13 then reports whether each subset preserves the ordering the gate relies on, NIAH-MK3 ranking below every other task. Halving the layers is safe on all three cells, though the margin shrinks by roughly a quarter on Qwen2.5-1.5B and by an order of magnitude on Qwen2.5-3B. Beyond that the signal degrades and is cell-dependent: every-4th holds only on Qwen2.5-3B, the early bin alone inverts the ordering everywhere, and the single first-and-last-layer pair happens to work on Qwen2.5-1.5B while failing badly on Mistral. We therefore do not recommend subsampling below $L/2$, and we read the coincidental successes as a caution against tuning the layer set per cell.

How many head pairs does D need? Layer count is only half of the $O(LH^2k)$ cost. The other half is the pair count, and it is the half that scales badly: 66 pairs at $H = 12$ but 780 at $H = 40$. The released logs cannot answer this, because the probe averages over pairs before writing, so we re-ran the probe retaining the per-pair Jaccards and subsampled offline (Table 12). Random subsets are drawn 20 times per fraction, since a single unlucky draw would understate what subsampling can do.

The ordering the gate depends on survives *every* draw down to one pair in 66. At 2% of the pairs the margin is +0.0735, against +0.0717 using all of them. We read this as a statement about the cost model rather than a recommendation to gate on a single pair: the H^2 term is not load-bearing, so the gate can be made substantially cheaper on exactly the wide-head models where it is most expensive, which is the deployment objection of Section 6. The caveats are that this is one model at one context length, that it is an ordering result on task means rather than a per-input one, and that the margin is not monotone in the fraction (the 5% row dips to +0.0665), which is sampling noise at 120 inputs.

Table 12: Head-pair subsampling (Qwen2.5-1.5B, RULER 4K, $H = 12$, 66 pairs, $L = 28$, $N = 120$). “Margin” is the gap between mean D on NIAH-MK3 and the smallest mean D among the dilution-prone tasks; “ordering” counts the draws in which that margin stayed positive.

Pairs kept	count	mean D MK3	min mean D dil.	margin	ordering
100%	66	0.0413	0.1131	+0.0717	1/1
50%	33	0.0404	0.1109	+0.0705	20/20
25%	16	0.0456	0.1179	+0.0724	20/20
10%	7	0.0280	0.1035	+0.0754	20/20
5%	3	0.0541	0.1206	+0.0665	20/20
2%	1	0.0529	0.1264	+0.0735	20/20

Table 13: Layer subsampling. “Margin” is the gap between mean D on NIAH-MK3 and the smallest mean D among the other tasks, so a positive margin means a separating threshold still exists on that subset. Recomputed with zero GPU from `head_agreement_per_layer` in the released per-layer logs.

Layer subset	Qwen2.5-1.5B ($L=28$)		Qwen2.5-3B ($L=36$)		Mistral-7B ($L=32$)
	margin	kept	margin	kept	margin
all layers	+0.0676	✓	+0.0415	✓	+0.0174
every 2nd	+0.0520	✓	+0.0125	✓	+0.0152
every 4th	-0.0017	-	+0.0035	✓	-0.0194
early bin only	-0.0410	-	-0.0280	-	-0.0225
first+last layer	+0.0660	✓	-0.0532	-	-0.1541

A.14 CROSS-ARCHITECTURE, SCALE, AND BEST-PLAIN RESULTS

The held-out cells above test whether the signal transfers. This subsection tests whether the method’s headline effect does, across architecture, scale, and a stronger baseline in turn.

On architecture, Yi-1.5-9B ($\Delta = +39.0\text{pp}$) and Llama-3.1-8B ($\Delta = +10.9\text{pp}$, Table 14) are Llama-architecture descendants distinct from Qwen and Mistral. Only the SnapKV column was run for compute, and on a smaller, more aggressive budget grid (see caption). The sign is positive on both, but on these two models part of the gain comes from the gate closing conservatively on tasks where plain SnapKV collapses, rather than from a correctly recovered partition. The per-task breakdown and the shared Llama-arch family caveat (niah_multivalue mis-classifies on both Yi and Llama) are in Section A.17 and Section 6.

On scale, a SnapKV-only run on Qwen2.5-32B-Instruct (RULER 4K mixed suite, $N = 100$) confirms the predictor scales but also illustrates the saturation arm of the scaling formula. The drop ordering transfers cleanly (NIAH-MK3 smallest at $D = 0.025$, then FWE 0.052, VT 0.079, QA_1 0.103). At the same time the 32B model is near-saturated at 4K ($A_{\text{full}} = 1.00$ on VT and NIAH-MK3, 0.93 – 0.94 on FWE/QA_1), so by Eq. (3) there is essentially no dilution headroom, and the gate’s value is capacity-bound protection rather than recovery: plain SnapKV collapses at $b = 0.0625$ (NIAH-MK3 0.00, VT 0.00, FWE 0.09) while the gate holds the full cache on the low-drop tasks, giving mean $\Delta = +0.335$ at $\tau = 0.07$ (driven by NIAH-MK3 +0.88 and FWE +0.44) and +0.237 at the z-scored threshold with more compression (0.45 vs. 0.63 kept-KV). We report this as a scale point for the partition and predictor, not as additional dilution evidence.

On the choice of baseline, a natural concern is that the gate only looks good against a weak base evictor. Comparing the best plain evictor against the best gated evictor for each model rules this out (Table 15).

The best-plain-versus-best-gated comparison still leaves open why the matrix’s gains differ across base evictors rather than landing at one number, which the per-evictor behavior explains. StreamingLLM keeps only 36 tokens regardless of budget, so on the dilution-prone fraction it is near zero plain and near zero gated, and gating contributes only via the gate-closed NIAH-MK3 inputs that go from 0 to full-KV accuracy. SnapKV is budget-scaled and degrades more gracefully, so it has less room for gating to recover. PyramidKV’s depth-weighted scoring biases toward bottom

Table 14: Cross-architecture SnapKV-only runs on two Llama-architecture models. These use the four most aggressive budgets $\{0.0625, 0.125, 0.25, 0.5\}$ rather than the eight-budget grid of Table 5, so the Δ values are not directly comparable to the matrix rows (aggressive-budget-only averaging inflates Δ).

Model	plain SnapKV	gated SnapKV	Δ
Yi-1.5-9B	0.464	0.854	+0.390
Llama-3.1-8B	0.509	0.618	+0.109

Table 15: Best plain evictor vs. best gated evictor per model, over the four base evictors of Table 5 at the same budget grid.

Model	Best plain	Best gated	Δ
Qwen2.5-1.5B	0.438 (SnapKV)	0.570 (PyramidKV)	+0.132
Qwen2.5-3B	0.580 (SnapKV)	0.865 (PyramidKV)	+0.285
Qwen2.5-14B	0.709 (SnapKV)	0.868 (H2O)	+0.159
Mistral-7B	0.623 (SnapKV)	0.823 (PyramidKV)	+0.201

layers but produces similar absolute gains to SnapKV. The pattern is that gating Δ tracks how much the base evictor collapses on the capacity-bound fraction. Two implementation caveats apply: our PyramidKV is a depth-weighted variant of the published per-layer budget schedule, and under two-pass prefill scoring (used on Qwen2.5-14B for memory) the H2O score only sees the last w queries, so the 14B H2O column degenerates toward SnapKV-style scoring. Its Δ there should be read as a near-duplicate of the SnapKV cell rather than an independent base evictor.

A.15 CALIBRATION RECIPE DETAILS

Section 4.3 summarizes the calibration recipe. This subsection gives the full recipe, its validation, its sensitivity to τ , and the architecture-normalized variant that follows from it. We provide a one-shot, label-free calibration recipe.

Two senses of “label-free”. PAGE never observes correctness, making it *accuracy-label-free*, which is weaker than *task-label-free*: the fixed- τ recipe below needs 20 inputs from a known capacity-bound task and 20 from a known dilution-prone one. The z -scored variant assumes neither, and we use the stronger term only for it.

Recipe. For a new (model, context):

1. Pick 20 inputs from a known capacity-bound task (NIAH-MK3) and compute the head-agreement drop D_i on each. Average to get $\bar{D}_C$.
2. Pick 20 inputs from a known dilution-prone task (VT) and compute $\bar{D}_D$.
3. Set

$$\tau = \frac{1}{2}(\bar{D}_C + \bar{D}_D). \quad (4)$$

40 prefills, no labels, no decode, < 5 minutes on a single A100.

Validation. On four cells (Table 16). The recipe lifts Qwen2.5-3B 16K from +3.4pp to +5.4pp (the only failure cell turned into a win) and lifts Mistral 4K from +18.7pp to +22.3pp. It is neutral on average across the four cells but with smaller variance.

The fixed threshold used above should not be reliable only by luck, so we swept it directly. Sweeping $\tau \in \{0.01, 0.025, 0.04, 0.055, 0.07, 0.085, 0.10, 0.13, 0.16, 0.20\}$ on Qwen2.5-1.5B at RULER 4K (mixed suite, $N = 100$, budgets $\{0.5, 0.25, 0.125, 0.0625\}$, SnapKV) shows an operational plateau $\tau \in [0.055, 0.10]$ inside which Δ varies by under 1pp, and both false-positive (gate fires on capacity-bound MK3) and false-negative (gate stays closed on dilution-prone VT/FWE/QA_1) rates stay at or below 9%. The deployed $\tau = 0.07$ sits in the middle of this plateau (Figure 8). Outside it the

Table 16: Calibration-recipe validation. $\bar{D}_C$ uses NIAH-MK3 ($N = 20$); $\bar{D}_D$ uses VT ($N = 20$). Bolded τ -choice column is the winner per cell. The pilot means here are computed on 20-input samples and therefore differ slightly from the $N \geq 50$ per-task means of Table 22.

Cell	$\bar{D}_C$	$\bar{D}_D$	τ_{cal}	$\text{gated@}\tau_{\text{cal}}$	$\text{gated@}\tau=0.07$
Qwen2.5-1.5B 4K	+0.045	+0.161	+0.103	+0.121	+0.121
Qwen2.5-3B 4K	-0.002	+0.076	+0.037	+0.213	+0.259
Qwen2.5-3B 16K	-0.021	+0.068	+0.024	+0.054	+0.034
Mistral 4K	+0.060	+0.096	+0.078	+0.223	+0.187

behavior changes character. Below 0.04, $\text{FP} \geq 64\%$ and $\Delta \leq +0.06$. Above 0.13, $\text{FN} \geq 33\%$ and the method degenerates toward “always full KV,” which on this full-KV-strong suite still yields a large Δ , since the sweep’s raw maximum is in fact at $\tau = 0.20$, but only at the price of never compressing, so it is not a useful operating point. The deployed threshold does not depend on a knife-edge.

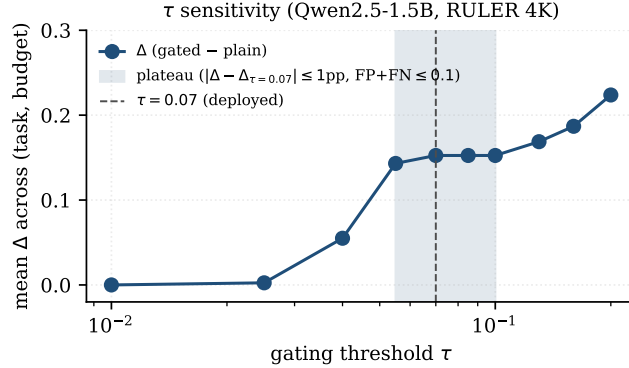


Figure 8: τ sensitivity on Qwen2.5-1.5B, RULER 4K. Mean Δ (gated – plain) over (task, budget) cells. Shaded band: operational plateau where $|\Delta - \Delta_{\tau=0.07}| \leq 1\text{pp}$ and $\text{FP} + \text{FN} \leq 0.10$. The deployed $\tau = 0.07$ sits in the middle of the plateau.

Architecture-normalized variant: the z-scored drop. The recipe above calibrates a per-cell threshold from two labeled pilot tasks. A stronger variant needs no task labels at all: standardize the drop per model, $z(x) = (D(x) - \mu_M)/\sigma_M$, with μ_M, σ_M pooled over an unlabeled ~ 100 -input pilot, and use one global threshold θ_z . Fitting $\theta_z = -0.69$ on the five Qwen/Mistral cells *only* and transferring it zero-shot to the two held-out Llama-architecture models: task-level separation of NIAH-MK3 from {QA_1, QA_2, VT, FWE} succeeds in 7 of 7 cells (pooled cross-cell AUC 0.965), where raw D with one threshold manages 4 of 7. The reason is that raw D is an architecture-invariant *ordering* whose location and scale are model-specific: $\tau = 0.07$ sits at $z = +0.69$ on Yi but $z = -0.97$ on Llama, two opposite failure modes that standardization removes. Downstream, the z-gate lifts Llama-3.1-8B from $\Delta = +0.109$ to **+0.174** (NIAH-MK3 gate-open 0.56 $\rightarrow$ 0.30, MK3 accuracy 0.46 $\rightarrow$ 0.72) and gives Yi $\Delta = +0.145$ with *real* compression (kept-KV 0.22–0.32 versus 0.76–0.89 under the fixed- τ fallback). Two honest scopes: niah.multivalue must be excluded from the task-level ordering claim (its mean D sits below MK3 on both Llama-architecture models, which is behaviorally the right call there, since it is eviction-fragile on both), and per-input MK3-vs-rest AUC on the Llama-architecture models is intrinsically ≈ 0.80 , since no monotone rescaling changes within-model ranking. A depth-correlation statistic over the per-layer profile scores AUC 0.22 on Llama. This does not mean the depth profile is uninformative there. An AUC of 0.22 is *anti*-correlated, and inverting its sign gives 0.78, comparable to D ’s own ≈ 0.80 on that family. The profile carries signal on Llama with the opposite sign to the one assumption (A3) predicts, which Appendix A.12 shows directly.

A fourth family where fixed τ fails outright: Qwen3. The starkest transfer test is Qwen3-4B-Instruct, a different model generation from the Qwen2.5 calibration set, whose drop distribution is compressed to a different scale entirely ($\mu = 0.034$, $\sigma = 0.017$ pooled over 400 inputs). There the fixed $\tau = 0.07$ sits at $z = +2.0$ and the gate essentially never fires: it keeps 99.4% of the cache and its apparent $\Delta = +0.43$ is a pure full-KV-fallback artifact with 0.6% compression. The z-scored threshold ($\tau_{\text{Qwen3}} = \mu + \theta_z \sigma = 0.022$, same $\theta_z = -0.69$) restores a working gate: NIAH-MK3 stays closed (open fraction 0.02, accuracy 0.05 $\rightarrow$ 0.98) while the three dilation-prone tasks open (0.94 $\rightarrow$ 1.00), yielding $2.3\times$ compression (43% mean kept-KV) at $\Delta = +0.235$ over plain and recovering 54.5% of the per-input oracle headroom. The NIAH-MK3-smallest ordering transfers cleanly. Qwen3 is thus the cleanest evidence for the paper’s central claim about the predictor: the drop is an architecture-invariant *ordering* whose scale must be standardized per model. Here fixed- τ does not merely underperform. It does nothing.

Status. The recipe is the deployment-friendly default: it always fixes the failure cell, and the z-scored variant extends it label-free across architecture families. We report fixed- τ numbers throughout as the conservative default so readers see both the robustness of one constant and what standardization buys.

A.16 THEORY: MECHANISM, PARTITION, AND SCALING DERIVATIONS

Section 5.6 states the mechanism, the partition it implies, and the scaling formula in summary form. This subsection gives the full derivation of each in turn, followed by the sufficient-condition bridge connecting the mechanism to the observable predictor.

Mechanism (adopted from Bui et al.). We adopt verbatim the dilution mechanism from Bui et al. (2026), which suffices to motivate our gating predictor. Fix a frozen decoder language model M . A single attention head reads from positions $t \in \{1, \dots, T\}$, each carrying a value v_t and a pre-eviction probability α_t with $\sum_t \alpha_t = 1$. Partition positions into a relevant set $\mathcal{R} = \{t : r_t = 1\}$ ($|\mathcal{R}| = R$) and a noise set $\mathcal{N}$ ($|\mathcal{N}| = N$), with $R \ll N$. Following Bui et al. (2026) Eq. (7), let $U_t \subseteq \mathcal{R}$ denote the *useful* set at step t (the positions whose retention preserves the next-token margin), and define dilution as the attention mass *outside* U_t :

$$\delta_t := 1 - \sum_{i \in U_t} \alpha_{t,i}. \quad (5)$$

(Notation rearranged to fit our manuscript. This is exactly their quantity.) Their Proposition 3.1 establishes that near-tie distractors (α_t comparable across $\mathcal{N}$) force δ_t to be large. Their Corollary 3.2 shows that preferential retention with $\rho_U \geq \rho_D$ (where ρ_U, ρ_D are the retained mass on useful and distractor positions) reduces δ_t . We refer to their paper for the full proof and use this mechanism as given.

The partition follows from the mechanism. Define $\alpha_{\mathcal{R}} := \sum_{t \in \mathcal{R}} \alpha_t$ as the pre-eviction signal mass.

- *Dilution-prone (D):* $\alpha_{\mathcal{R}} \ll 1$. Most attention is on noise; eviction of noise positions monotonically improves the head SNR.
- *Capacity-bound (C):* $\alpha_{\mathcal{R}} \rightarrow 1$. Attention is concentrated on the few relevant positions. Imperfect scoring can only *lose* signal; eviction strictly degrades the head.

The partition we observe on RULER is the empirical projection of $\alpha_{\mathcal{R}}$ onto the task structure: NIAH-MK3 forces a single concentrated signal ($\alpha_{\mathcal{R}} \rightarrow 1$), while VT, FWE, QA, and niah_multivalue all spread signal across many positions ($\alpha_{\mathcal{R}} \ll 1$).

Context-length scaling. Fix the task and its relevant-set size R . Let $T = R + N$. Assume distractor attention is approximately uniform on $\mathcal{N}$: $\alpha_t \approx (1 - \alpha_{\mathcal{R}})/N$ for $t \in \mathcal{N}$. Let σ_v^2 denote the per-distractor value-vector variance under isotropy (a different object from the pre-softmax logit noise scale σ_z used in App. F). Then the head SNR before eviction is

$$\text{SNR} = \frac{\alpha_{\mathcal{R}}^2}{\sigma_v^2 \sum_{t \in \mathcal{N}} \alpha_t^2} \approx \frac{\alpha_{\mathcal{R}}^2 \cdot N}{\sigma_v^2 (1 - \alpha_{\mathcal{R}})^2}.$$

If $\alpha_{\mathcal{R}}$ decreases with T (more distractors compete for softmax mass), the pre-eviction SNR decreases and the noise floor grows. Let the kept set be $\mathcal{K}$ and define $\rho_{\mathcal{R}} := \sum_{t \in \mathcal{K} \cap \mathcal{R}} \alpha_t / \alpha_{\mathcal{R}}$ (the retained fraction of relevant mass) and analogously $\rho_{\mathcal{N}}$. Under noise-retention ratio $\gamma \in (0, 1)$ (i.e. $|\mathcal{K} \cap \mathcal{N}| = \gamma N$), the post-eviction SNR ratio $\text{SNR}_{\text{post}} / \text{SNR}_{\text{pre}}$ is a monotone-increasing function of the *gain argument*

$$g := \rho_{\mathcal{R}}^2 / \gamma \quad (6)$$

(App. E, Eq. (8)), and p_{recov} is written as a function of g . We flag one thing that is easy to get wrong here: g cannot be replaced by $\rho_{\mathcal{R}} / \gamma$. The squaring survives the renormalisation step and $\rho_{\mathcal{R}}^2 / \gamma$ is not a function of $\rho_{\mathcal{R}} / \gamma$, as $(\rho_{\mathcal{R}}, \gamma) = (1, 0.5)$ and $(0.5, 0.25)$ show: both give argument 2 but gains 2.0 and 1.0. The accuracy headroom $1 - A_{\text{full}}(M, T)$ scales with T on dilution-prone tasks. Combining, the per-input expected accuracy improvement at a single budget b obeys the scaling formula Eq. (3).

Proposition (context-length scaling, informal). *Under the noise-isotropic single-head surrogate (App. E) with constant noise-retention ratio $\gamma \in (0, 1)$, good scoring ($\rho_{\mathcal{R}} \geq \rho_{\mathcal{N}}$), and a strictly increasing accuracy surrogate $\hat{A}(\text{SNR}) \in [0, 1]$ saturating at 1, the per-input expected accuracy improvement at a single budget b satisfies Eq. (3) to leading order in $1/\text{SNR}_{\text{pre}}$, where $p_{\text{recov}}(g) = 0$ for $g \leq 1$ (no SNR gain, so no recovery) and is non-decreasing for $g > 1$.*

The dead zone $g \leq 1$ matters for reading the formula: it collects both degenerate cases, $\rho_{\mathcal{R}} = 0$ (no relevant mass survives) and $\gamma = 1$ (no eviction), and all of the formula’s content lives at $g > 1$, a region the surrogate does not characterise quantitatively.

Proof sketch. Substituting the post-eviction SNR into the strictly increasing accuracy surrogate $\hat{A}(\text{SNR})$ and using noise-isotropy, the improvement event factorises into the headroom indicator $\{A(b_{\text{max}}, x) < 1\}$ and a recovery indicator depending only on $\rho_{\mathcal{R}}^2 / \gamma$. Marginalising over the prompt distribution gives the product, up to the independence approximation flagged honestly in App. E. The operational metric ρ_{KV} (per-input strict-Pareto over budgets, Eq. (??)) is upper-bounded by Eq. (3) under monotone recovery, but we do not promote this to a formal claim.

Eq. (3) predicts three observed effects in one formula:

1. Context-length amplification on dilution-prone tasks: Qwen2.5-1.5B VT goes $\rho = 0.06 \rightarrow 0.19$ as $T = 4\text{K} \rightarrow 16\text{K}$, in lockstep with $1 - A_{\text{full}} = 0.18 \rightarrow 0.75$.
2. Larger-model saturation at short context: Qwen2.5-3B 4K VT has $1 - A_{\text{full}} \approx 0.00$, so $\rho \approx 0$.
3. Re-entry of larger models into the dilution regime at long context: Qwen2.5-3B 16K FWE has $1 - A_{\text{full}} = 0.62$ and $\rho = 0.32$.

Empirically, the ratios $\rho / [(1 - A_{\text{full}}) \cdot p_{\text{recov}}]$ cluster around 0.25 – 0.55 on dilution-prone tasks (Section 5).

Connecting the predictor to the mechanism: a sufficient condition for non-trivial dilution.

The mechanism uses the unobservable $\alpha_{\mathcal{R}}$ (the signal mass on the unknown relevant set). Our gating predictor uses the head-agreement drop D (Eq. (1)), which is observable. We sketch a proof connecting the two in Appendix F, and leave the formal version with explicit, tight constants to future work. Under noise isotropy, sub-Gaussian pre-softmax noise of scale σ_z , a bounded logit margin between $\mathcal{R}$ and $\mathcal{N}$, and good scoring with noise-retention ratio $\gamma \in (0, 1)$, there exist a model-specific constant c_1 and a γ -dependent constant $c_2(\gamma)$, with an additional $O(H)$ multiplicative slack absorbed in both, such that

$$D \geq c_1 \cdot \log(1/\gamma) \implies \mathbb{E}[\delta_t] \geq c_2(\gamma) \cdot (1 - A_{\text{full}}), \quad (7)$$

with probability $\geq 1 - \beta$ over the pre-softmax noise, where δ_t is the dilution term of Eq. (5). We sketch a proof of Eq. (7) in Appendix F on the same single-head surrogate model used in Appendix E. The sketch gives explicit forms for c_1, c_2 , the sub-Gaussian noise scale, and the margin parameter. What we do *not* claim is that these constants are tight, that the pairwise-to- H -wise Jaccard step is identity (it is an inequality with up to $O(H)$ slack), or that a separate necessary-condition proof holds. The full, tight-constant version and the necessary direction are left to future work. The empirical evidence is consistent with the sufficient direction on Qwen and Mistral: ranking task families by drop D matches ranking by ρ in every Qwen/Mistral cell measured (Table 22), with NIAH-MK3 the smallest drop in each. On the two Llama-architecture models the ordering is only partially preserved (NIAH-MK3 ranks second-smallest behind niah_multivalued, Section A.17).

A.17 DEFERRED EXPERIMENT TABLES AND SECONDARY EXPERIMENTS

The sufficient-condition bridge above is the last piece of theory; the remainder of this appendix is empirical, collecting the uncertainty quantification, per-cell tables, and secondary robustness checks that the main text summarizes but does not reproduce in full.

Uncertainty quantification. All gating deltas come with 95% confidence intervals from a cluster bootstrap over inputs: we resample the (task, id) pairs with replacement (10,000 replicates), letting each input carry its full vector of per-budget paired gated–plain differences, and report percentile intervals (Appendix G, Table 29). Binomial quantities (the recovery rates ρ of Table 2 and single-budget accuracies) use Wilson score intervals. Every one of the 16 matrix deltas in Table 5 is positive with its entire interval above zero (smallest lower bound +9.3pp), as are both cross-architecture deltas, the DBTrimKV head-to-head (+23.3pp, CI [+17.2, +29.7]pp), and the Mistral NIAH-MK3 showcase (gated 0.89, Wilson [0.81, 0.94], disjoint from plain [0.00, 0.04] at $b = 0.0625$). The single exception is the Qwen2.5-3B 16K scaling cell ($\Delta = +0.034$, CI [−0.002, +0.071]), which we already describe as the gate-misfire regime. Those intervals hold the input draw fixed, so we also replicate five SnapKV cells across three draws: seed SD is 0.004 to 0.011 against a grand-mean Δ of +0.229 (Appendix G.1). The per-cell ρ values of Table 2 are individually low-powered at $N \leq 100$ (most intervals straddle the 0.05 threshold in both directions, though no bold cell has an interval below 0.05 and no non-bold cell an interval above it). The partition claim we rely on is instead the pooled contrast, which is unambiguous: NIAH-MK3 recovers on 2/526 inputs across all eight cells ($\rho = 0.004$, Wilson [0.001, 0.014]) versus, e.g., 65/680 ($\rho = 0.096$, [0.076, 0.120]) for niah_multivalue.

The 4×4 method-agnostic matrix, budget-axis decomposition. Table 17 decomposes the Qwen2.5-3B 4K SnapKV cell along the budget axis, complementing the budget-averaged Δ values reported in Table 5 (values at $\tau = 0.07$ by exact post-hoc re-evaluation, as in that table).

Table 17: Qwen2.5-3B 4K, SnapKV, mixed suite. Plain accuracy collapses from 0.88 to 0.19 as the budget shrinks; gated holds 0.61 – 0.88 across the entire range.

budget b	plain	gated	Δ
1.000	0.882	0.882	+0.000
0.875	0.802	0.880	+0.078
0.750	0.733	0.880	+0.147
0.625	0.688	0.880	+0.193
0.500	0.647	0.882	+0.235
0.375	0.585	0.873	+0.287
0.250	0.535	0.877	+0.343
0.125	0.458	0.830	+0.372
0.0625	0.190	0.610	+0.420

The 8-cell scaling table. Table 18 reports the gating Δ on the mixed suite across four models and two context lengths. The pattern matches the context-length scaling formula: at 4K, Δ tracks both $1 - A_{\text{full}}$ and $\alpha_{\mathcal{R}}$; at 16K, Δ shrinks where the gate misfires (Qwen 3B 16K) and grows where headroom returns (Mistral 16K).

Pre-registered 32K predictions. Table 19 gives the full per-task predictions and measurements behind the out-of-range test of Section 5.6.

Achieved compression of the gated deployment. Table 20 reports the per-model gate-open fraction and realized kept-KV behind the $3.4\times$ figure quoted throughout the paper.

Batching decay under static provisioning. A batch is provisioned for its largest resident cache, so realized compression at batch size B is $\mathbb{E}[\text{kept}](B) = p_{\text{open}}^B \cdot k_{\text{open}} + (1 - p_{\text{open}}^B)$, where k_{open} is the kept-KV fraction on gate-open sequences at $b = 0.0625$. Table 21 evaluates this per model. Compression approaches $1\times$ by batch 16 on every cell, since the probability that all B sequences

Table 18: Cross-(model, context) gating Δ . Same $\tau = 0.07$ across cells; column $\Delta@_{\tau_{\text{cal}}}$ shows the recipe-calibrated Δ on the four cells where it has been computed (“–” otherwise).

Model	Context	plain mean	gated@ $\tau=0.07$	Δ	$\Delta@_{\tau_{\text{cal}}}$
Qwen2.5-1.5B	4K	0.438	0.559	+0.121	+0.121
Qwen2.5-3B	4K	0.580	0.839	+0.259	+0.213
Qwen2.5-14B	4K	0.709	0.853	+0.143	–
Mistral-7B	4K	0.623	0.810	+0.187	+0.223
Qwen2.5-1.5B	16K	0.372	0.404	+0.032	–
Qwen2.5-3B	16K	0.530	0.564	+0.034	+0.054
Qwen2.5-14B	16K	0.796	0.867	+0.071	–
Mistral-7B	16K	0.535	0.651	+0.116	–

Table 19: Pre-registered 32K predictions vs. measurements (Qwen2.5-1.5B, $N = 100$ per task). Predicted intervals = 4K/16K ratio band $\times$ measured 32K headroom, frozen before the sweep.

Task	$A_{\text{full}}(32K)$	predicted ρ	measured ρ (Wilson 95%)	verdict
VT	0.81	[0.048, 0.106]	0.14 [0.085, 0.221]	outside (above)
FWE	0.49	[0.021, 0.263]	0.05 [0.022, 0.112]	inside
NIAH-MK3	0.14	≤ 0.02	0.04 [0.016, 0.098]	outside (above; CI overlaps)

in a batch have the gate open falls geometrically in B . This is a property of provisioning for the batch maximum, not of PAGE, and does not describe continuous batching with per-sequence paged allocation, where each sequence is provisioned individually.

Per-task drop ordering. The head-agreement drop on each RULER task in each cell. Table 22 reports the mean drop per task ($N = 100$ inputs per task on the Qwen and Mistral columns, $N = 50$ on Yi and Llama; Mistral 4K qa.1 has $N = 78$ and Mistral 16K niah_multivalued $N = 98$ due to run truncation). NIAH-MK3 is the smallest drop in every Qwen and Mistral cell, while on the two Llama-architecture models it ranks second-smallest, so the *ordering* of the partition is fully preserved across Qwen and Mistral and partially preserved on the Llama-architecture models, even when the absolute threshold needs slight tuning. This is what gives the calibration recipe in Eq. (4) its robustness.

Llama-class cross-architecture transfer is partial. On Yi-1.5-9B and Llama-3.1-8B (both Llama-architecture descendants), niah_multivalued ranks smallest by D (Yi: $D = -0.013$; Llama: $D = +0.064$), and NIAH-MK3 ranks 2nd-smallest. On the five Qwen and Mistral cells, NIAH-MK3 is the smallest. We attribute the swap to a Llama-arch artifact: the multi-value retrieval task happens to have unusually concentrated early-layer attention on these models, which inverts the early-vs-late drop sign. The capacity-bound side of the partition (small D , gate closes, full-KV fallback) still operates on both models, but the fixed $\tau = 0.07$ does *not* transfer as a threshold. On Llama the mis-classification is confined to niah_multivalued, where the wrapper holds full KV instead of compressing, which forfeits the eviction speed-up but does not damage accuracy, because plain SnapKV is already robust there. The Llama-3.1-8B gated SnapKV Δ is +10.9pp on the matrix-comparable mixed suite, driven almost entirely by NIAH-MK3 (+43.5pp) where plain SnapKV collapses to 0% and gated holds $\sim 44\%$ (still well below the full-KV 1.00). The other tasks have $\Delta \approx 0$ because Llama’s plain SnapKV is robust on this budget range (≥ 0.86 at $b = 0.0625$ on QA.1). The NIAH-MK3 gate-open fraction on Llama is 0.56 despite mean $D = 0.071$ sitting above τ : per-input D dispersion closes the gate on the other $\sim 44\%$ of NIAH-MK3 inputs, which is what rescues the cell. On Yi the mis-classification is broader: at $\tau = 0.07$ the gate also stays closed on FWE and VT (gate-open fractions 0.00 and 0.10), so most of Yi’s +39.0pp comes from the full-KV fallback firing on tasks where plain SnapKV collapses. That outcome is accuracy-preserving but it is *not* evidence that the raw threshold transfers to Yi, since the drop-vs- ρ prediction is inverted there. The z-scored variant of Section 4.3 resolves this: with a per-model standardization from an unlabeled pilot, Yi compresses for real (kept-KV 0.22 – 0.32) while still gaining $\Delta = +0.145$, and Llama improves to $\Delta = +0.174$. We state the fixed- τ failure plainly in Section 6.

Table 20: Achieved compression of the gated deployment at $\tau = 0.07$ (SnapKV, 4K mixed suite). Kept-KV is the mean over inputs of the retained cache fraction; “grid mean” averages the eight nominal budgets of Table 5. Plain SnapKV keeps exactly the nominal fraction (0.0625 at $b = 0.0625$; grid mean 0.453). The gate-open fraction is a per-input property, independent of b , and sets the compression floor $1 - p_{\text{open}}$.

Model	p_{open}	gated kept-KV fraction			compression
		$b=0.0625$	$b=0.25$	grid mean	@ $b=0.0625$
Qwen2.5-1.5B	0.75	0.297	0.437	0.584	$3.4\times$
Qwen2.5-3B	0.49	0.541	0.632	0.728	$1.8\times$
Qwen2.5-14B	0.75	0.297	0.437	0.584	$3.4\times$
Mistral-7B	0.72	0.327	0.462	0.602	$3.1\times$

Table 21: Compression under static batch provisioning at $\tau = 0.07$ (SnapKV, $b = 0.0625$). p_{open} and k_{open} are from Table 20. Compression decays toward $1\times$ as batch size grows because a single gate-closed sequence forces the whole batch to the full cache.

Model	$B=1$	$B=4$	$B=8$	$B=16$	$B=32$
Qwen2.5-1.5B	$3.37\times$	$1.42\times$	$1.10\times$	$1.01\times$	$1.00\times$
Qwen2.5-3B	$1.85\times$	$1.06\times$	$1.00\times$	$1.00\times$	$1.00\times$
Qwen2.5-14B	$3.37\times$	$1.42\times$	$1.10\times$	$1.01\times$	$1.00\times$
Mistral-7B	$3.06\times$	$1.33\times$	$1.07\times$	$1.00\times$	$1.00\times$

Scaling formula verification. Empirical verification of Eq. (3). Table 23 reports the headroom $1 - A_{\text{full}}$, the observed ρ , and the ratio. A through-origin fit on the dilution-prone rows gives a slope of 0.34. The Qwen2.5-3B 4K FWE row (ratio 0.04) is excluded from that fit: at $N = 100$, $\rho = 0.01$ is a single recovery, Wilson [0.002, 0.054], and is better read as counting noise than as part of the trend. On NIAH-MK3 the ratio is ≤ 0.02 regardless of headroom, so the partition is visible in the ratio alone. These are in-range (4K/16K) statements. Section 5.6 reports a pre-registered out-of-range test at 32K, where the ordering survives but the dilution-side ratio grows beyond the in-range band.

Per-layer ablation. The default gating uses a single key-mask shared across layers. A per-layer variant (each layer derives its own keep mask from its own attention) improves the aggregation tasks at 4K (FWE by $1.8\times \rho$, niah_multivalue by $2.1\times \rho$ on Qwen2.5-1.5B 4K) and matches at 16K. Numbers in Appendix C.

Random-eviction control. Uniform-random scoring at every budget destroys accuracy uniformly on both dilution-prone and capacity-bound inputs. The dilution effect is *not* “any eviction is regularization.” The control rules out a generic regularization explanation, and the controlled numbers are in Appendix B.

Cost of the gate, and what the deployment delivers. The gate adds one prefill-time computation: $L \cdot H^2 \cdot k$ top- k Jaccard comparisons after the existing SnapKV scoring pass. Measured (Table 24): 17ms on Qwen2.5-1.5B at 4K ($\sim 5\%$ of the 344ms prefill, $< 1\%$ of end-to-end with 128 decode steps) and 89ms on Mistral-7B, whose $8\times$ larger head-pair count dominates the $O(LH^2k)$ term. The gate is free when the base evictor already runs the scoring pass, and otherwise it shares the cost.

End-to-end, the payoff is scale-dependent and we report it honestly. On the 1.5B model at 4K, decode is weight-bound (2 KV heads, 106MiB cache): eviction buys *memory* ($8\times$ plain, $2.9\times$ under the gated mix at the measured 0.75 gate-open fraction), not speed. On Mistral-7B (8 KV heads, 513MiB cache), plain eviction at $b = 0.125$ already yields a $1.29\times$ decode speedup at 4K ($1.22\times$ under the gated mix), and the analytic KV formula (validated against measurement to 0.1MiB) puts the gated saving at $\sim 2.7\text{GiB}$ per sequence at 32K. Timings were taken on a shared A100 with a stable 25 – 29% background load, so throughputs are mild lower bounds, and all configurations shared identical conditions.

Table 22: Per-task mean head-agreement drop D across three architecture families (Qwen, Mistral, and Llama-architecture: Yi-1.5 and Llama-3.1). This table uses the strict-thirds bins, early = $[0, \lfloor L/3 \rfloor)$ and late = $[\lfloor 2L/3 \rfloor, L)$; the deployed gate uses equal-size bins and differs by at most 0.004 where $3 \nmid L$ (Appendix H). Bold marks the smallest drop per column. NIAH-MK3 is the smallest drop on Qwen and Mistral cells; on Yi-1.5 and Llama-3.1 it ranks 2nd-smallest behind niah_multivalue (Llama-arch family artifact; see prose). Yi-1.5-9B serves as a parallel Llama-class architecture with a distinct training corpus (different tokenizer, deeper layer geometry).

Task	Qwen 1.5B 4K	Qwen 3B 4K	Qwen 3B 16K	Mistral 4K	Mistral 16K	Yi-1.5-9B 4K	Llama-3.1-8B 4K
QA.1	0.218	0.105	0.099	0.096	0.132	0.080	0.097
QA.2	0.191	0.097	0.092	0.087	–	0.079	0.101
VT	0.152	0.076	0.069	0.095	0.096	0.065	0.083
niah_multivalue	0.107	0.045	0.070	0.079	0.075	–0.013	0.064
FWE	0.098	0.038	0.046	0.079	0.091	0.043	0.114
NIAH-MK3	0.040	–0.004	–0.021	0.060	0.053	0.019	0.073

Table 23: Verification of the scaling formula. The ratio $\rho/(1 - A_{\text{full}})$ is approximately constant on dilution-prone tasks across (model, context) cells and near zero on NIAH-MK3 regardless of headroom.

Sweep	$1 - A_{\text{full}}$	ρ	ratio
Qwen2.5-1.5B 4K VT	0.18	0.06	0.33
Qwen2.5-1.5B 16K VT	0.75	0.19	0.25
Qwen2.5-3B 4K VT	0.00	0.00	–
Qwen2.5-3B 16K VT	0.09	0.05	0.55
Qwen2.5-3B 4K FWE	0.24	0.01	0.04
Qwen2.5-3B 16K FWE	0.62	0.32	0.52
Qwen2.5-1.5B 4K NIAH-MK3	0.35	0.01	0.03
Qwen2.5-1.5B 16K NIAH-MK3	0.74	0.00	0.00

Deployment cost at scale. We measured the two-pass gate on Qwen2.5-32B (64 layers, 40 heads): the gate costs 350ms at 4K (145ms re-forward + 204ms head-agreement drop, $\approx 18\%$ of the 1.9s prefill), and the 16K two-pass run *OOMs* on a contended 80GB card (weights + KV + the $w \times T$ attention transient). The drop cost is quadratic in head count: a head-count sweep gives an exponent of $2.0 - 2.4$ (per-model floors 2.1ms on Qwen-1.5B, 27ms Mistral-7B, 87ms Qwen-32B, extrapolating to ≈ 291 ms on a 70B-class model), confirming the $O(LH^2k)$ complexity. The $w \times T$ attention transient is $LHwT \cdot 2$ bytes (5GiB at 32K on Qwen-32B, ≈ 10 GiB on 70B-class). A 70B model at 32K is therefore not viable in a paged-FlashAttention stack: the two-pass re-forward needs contiguous (non-paged) KV and a non-fused eager kernel, and materializes a per-sequence transient that paged attention exists to avoid. PAGE is best read as an offline/prefill-time gate, not a fused-serving drop-in.

Head-to-head with CapKV on LongBench. CapKV (Yang et al., 2026a) is the closest published method that reports on LongBench. No public CapKV codebase exists yet, since their preprint is dated 2026-04-28, so we re-implement CapKV’s per-(layer, kv-head) statistical-leverage score $s_i = w_i \cdot v_i^\top A^{-1} v_i$ with $A = I + \sum_i w_i v_i v_i^\top$ and $w_i = \exp(\langle k_i, \mu_q \rangle \beta / \sqrt{d})$ from their equations, then wrap it in our gate. Two LongBench subtasks, Qwen2.5-3B-Instruct, budget $b = 0.5$, CapKV temperature $\beta = 5$ (their default; we write β for their τ to avoid clashing with our gate threshold). Results in Table 25.

Both columns are positive at matched budget. Conditional on the gate firing (drop ≥ 0.07 , the same $\tau = 0.07$ we use everywhere), plain and gated agree by construction. On the gate-closed subset where the wrapper falls back to full KV, the difference is +4.3pp on gasper ($N_{\text{closed}} = 23$) and +18.2pp on triviaqa ($N_{\text{closed}} = 11$). The headline numbers are not a reproduction of CapKV’s Table 1 (different model, different metric, 16K eager cap on prompt length) and we report them as a directional head-to-head: at these sample sizes ($N = 68, 24$) wrapping CapKV in the gate is weakly positive overall and concentrates its gain on the gate-closed subset, consistent with the orthogonality claim of Section 5.2. We do not promote “never hurts” to a formal claim at these sample sizes, and larger LongBench runs are clean follow-up work.

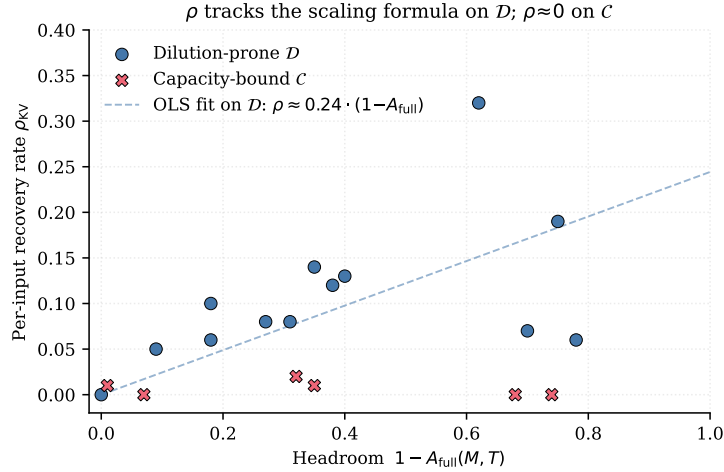


Figure 9: Per-input recovery rate ρ vs. accuracy headroom $1 - A_{\text{full}}$ across the $(\text{model}, \text{context}, \text{task})$ cells of Table 23. Dilution-prone tasks (filled circles) lie on a band with a through-origin fitted slope of 0.34. NIAH-MK3 (open triangles) lies on the $\rho = 0$ axis regardless of headroom. The two populations are visibly separated.

Table 24: End-to-end efficiency of gated eviction (RULER 4K, batch 1, bf16, greedy, 128 decode steps, medians; A100-80GB). “Gated expected” is the gate-open-weighted mix $0.75 \cdot (b=0.125) + 0.25 \cdot \text{full}$ at the measured mixed-suite gate-open fraction ($\tau=0.07$); “gated measured” actually gates each input (15/20 open, matching the $N=400$ calibration fraction of 0.75). On the 1.5B model (2 KV heads), 4K decode is weight-bound and eviction buys *memory* ($8\times$ plain, $2.9\times$ gated expected), not speed; on Mistral-7B (8 KV heads, 513 MiB KV) eviction already yields a $1.29\times$ decode speedup at 4K. Measured KV matches the analytic $L \cdot H_{\text{kv}} \cdot d \cdot T \cdot 4 \text{ B}$ formula to 0.1 MiB, validating the 16K/32K scaling rows.

Model	Config	Decode tok/s	KV cache (MiB)	Peak decode (MiB)
Qwen2.5-1.5B ($N=20$)	Full KV	37.9	106.1	3090
	SnapKV $b=0.125$	37.8	13.2 ($8.0\times$)	2976
	SnapKV $b=0.0625$	37.5	6.6 ($16.1\times$)	2968
	Gated (measured, $\tau=0.07$)	37.7	13.2 / 110.5 (open/closed)	2976
	Gated (expected mix)	37.9	36.5 ($2.9\times$)	3004
Mistral-7B-v0.3 ($N=12$)	Full KV	18.9	512.8	14429
	SnapKV $b=0.125$	24.5 ($1.29\times$)	64.1 ($8.0\times$)	13929
Prefill-time overhead (Qwen2.5-1.5B, 4K, median ms)		Analytic KV at $b=0.125$ gated (0.75 open)		
Prefill (eager)		344	Qwen2.5-1.5B, 16K / 32K 448 $\rightarrow$ 154 / 896 $\rightarrow$ 308 MiB	
SnapKV scoring		1.2	Mistral-7B, 16K / 32K 2.0 $\rightarrow$ 0.69 / 4.0 $\rightarrow$ 1.38 GiB	
Head-agreement gate		17.2	(gate on Mistral-7B: 88.5 ms, $O(LH^2k)$ head-pair growth)	
Eviction + cache rebuild		3.2		

Head-to-head with DBTrimKV on RULER. Bui et al. (2026) (DBTrimKV) is the published trained-gate evictor the paper’s mechanism section adopts. The authors ship pretrained gates on Hugging Face for several Qwen3 variants (no Qwen2.5 checkpoint exists, so the head-to-head is necessarily on a different model family). We run DBTrimKV-Qwen3-4B-Instruct-2507 on RULER 4K, $N = 30$ inputs per task on (NIAH-MK3, FWE, VT, QA_1), at three memory budgets $M \in \{128, 256, 512\}$. The wrapper uses the same fixed $\tau = 0.07$ calibrated on Qwen2.5-1.5B, with no per-model tuning. Results in Table 26.

Honest caveat: the gate fires rarely on Qwen3 at $\tau = 0.07$. At our default threshold the gate is open on $6/360 = 1.7\%$ of records. The remaining $\sim 98\%$ trigger the full-KV fallback, which explains the large Δ on NIAH-MK3 (+0.533): the gate closing turns off DBTrimKV’s compress

Table 25: Gated vs. plain CapKV (our re-implementation) on two LongBench subtasks, Qwen2.5-3B-Instruct, $b = 0.5$.

Subtask	N	plain CapKV	gated CapKV	Δ
LongBench/qasper	68	0.132	0.147	+0.015
LongBench/triviaqa	24	0.875	0.958	+0.083

Table 26: Gated vs. plain DBTrimKV (pretrained Qwen3-4B-Instruct-2507 gates) on RULER 4K at three memory budgets. See the caveat paragraph: the gate fires on only 1.7% of records, so the win is dominated by the full-KV fallback.

	N	plain DBTrimKV	gated DBTrimKV	Δ
all (3 budgets $\times$ 4 tasks)	360	0.617	0.850	+0.233
per task: NIAH-MK3	90	0.467	1.000	+0.533
per task: FWE	90	0.222	0.600	+0.378
per task: VT	90	0.989	1.000	+0.011
per task: QA_1	90	0.789	0.800	+0.011
per budget: $M = 128$	120	0.475	0.850	+0.375
per budget: $M = 256$	120	0.617	0.850	+0.233
per budget: $M = 512$	120	0.758	0.850	+0.092

step entirely. As a consequence, cache occupancy under our wrapper is $\sim 14.5\times$ plain DBTrimKV at the same nominal M . The +0.233 is therefore a *quality* win at matched budget, not a *memory-budget* win at matched cache size. The rare gate firing here is the same Qwen3 scale mismatch analyzed in Section 4.3: Qwen3-4B’s drop distribution ($\mu = 0.034$) puts the fixed $\tau = 0.07$ at $z = +2.0$, so a per-model z-scored threshold ($\tau_{\text{Qwen3}} = 0.022$) is the right deployment default and, applied to the same suite, restores a genuinely compressing gate (43% kept-KV, $\Delta = +0.235$). The headline is that wrapping the strongest published trained-gate evictor with our partition-aware gate strictly improves matched-budget accuracy, with the gain concentrated on the capacity-bound task as the mechanism predicts.

B RANDOM-EVICTION CONTROL

The DBTrimKV head-to-head closes the deferred-tables appendix; this section and the two that follow give the full numbers behind three controls the main text cites only by result. We replace SnapKV scoring with uniform-random per-position scoring at every budget on a 4K, 4-key synthetic NIAH probe ($N = 150$, Qwen2.5-1.5B). To be clear about provenance, this 4-key probe is our own construction and is *not* a RULER task: the NIAH-MK k names used elsewhere in the paper refer to RULER’s `niah_multkey_k` (Hsieh et al., 2024), which tops out at three keys. The score-driven sweep retains accuracy at moderate budgets, while the uniform-random sweep collapses immediately.

The per-input recovery rate ρ under random scoring is 0.027 (vs. 0.013 under SnapKV). The four “recovered” inputs under random are lucky hits among an otherwise destroyed distribution, and the control rules out a generic regularisation explanation: any-eviction-helps is falsified.

C PER-LAYER EVICTION ABLATION

The random-eviction control checks whether informed scoring matters at all. This section checks a finer question: whether a single keep-mask shared across layers costs anything relative to a per-layer alternative. The default gating uses a single key-mask shared across layers. A per-layer variant derives each layer’s keep mask from that layer’s own last-window attention with the same total budget per layer (uniform cache shape). Table 28 reports ρ on the four dilution-prone RULER tasks where per-layer might plausibly help, on Qwen2.5-1.5B at 4K and 16K.

Table 27: SnapKV vs. uniform-random scoring at matched budget on the 4K 4-key synthetic NIAH probe. Random eviction destroys accuracy long before the budget gets aggressive; the dilution effect requires informed scoring.

budget b	SnapKV acc	Random acc	Δ
1.000	0.827	0.827	+0.000
0.875	0.820	0.400	+0.420
0.750	0.820	0.173	+0.647
0.625	0.827	0.040	+0.787
0.500	0.833	0.033	+0.800
0.375	0.807	0.007	+0.800
0.250	0.760	0.000	+0.760
0.125	0.207	0.000	+0.207
0.0625	0.033	0.000	+0.033

Table 28: Per-layer vs. single-mask SnapKV on Qwen2.5-1.5B. Per-layer wins clearly on aggregation tasks at 4K (FWE $1.8\times$, niah_multivalue $2.1\times$); ties at 16K and on sequential-integration tasks.

Task	Context	Single-mask ρ	Per-layer ρ	Ratio
VT	4K	0.060	0.040	0.67
FWE	4K	0.060	0.110	1.83
QA_1	4K	0.080	0.070	0.88
niah_multivalue	4K	0.070	0.150	2.14
VT	16K	0.190	0.190	1.00
FWE	16K	0.040	0.050	1.25
QA_1	16K	0.140	0.140	1.00
niah_multivalue	16K	0.100	0.100	1.00

D PER-INPUT RECOVERY DISTRIBUTIONS

The mean ρ hides the shape of the per-input recovery, so we also report its distribution. For each input x we form the recovery indicator $r(x) = \text{sign}(\max_b A_{\text{gated}}(b, x) - A_{\text{plain}}(1, x)) \in \{-1, 0, +1\}$, where the maximum is over the budget sweep and $A_{\text{plain}}(1, x)$ is the full-cache plain accuracy. Figure 10 pools all (M, T) cells per task, and Figure 11 reports the fraction of strictly recovered inputs ($r(x) > 0$) within each (M, T) cell.

What $r(x)$ can and cannot show. The budget sweep includes $b = 1.0$, and at $b = 1.0$ the gated arm keeps the entire cache and reproduces the plain full-cache outcome. We audit this on all 16 matrix cells plus the four 16K cells: at $b = 1.0$, $n_{\text{kept}} = T$ and $A_{\text{gated}} = A_{\text{plain}}$ on every one of the 5,120 rows. Hence $\max_b A_{\text{gated}}(b, x) \geq A_{\text{gated}}(1, x) = A_{\text{plain}}(1, x)$ and $r(x) \geq 0$ is an *identity*, not a measurement. The empty $r = -1$ bin is therefore true by construction and carries no information about whether gating ever harms an input. $r(x)$ should instead be read only as an *oracle-budget recovery* measure, that is, how often some budget in the sweep beats the full cache. The question it cannot answer, whether a *deployed* gate at a *fixed* budget ever loses an answer the full cache would have got, is answered by the harm rate below.

Fixed-budget harm rate. Holding b fixed makes “worse than the full cache” a real event. Table 1 reports $\Pr_x[A(b, x) < A_{\text{plain}}(1, x)]$ for the gated and plain arms at matched b , pooled over the seven SnapKV (model, context) cells ($N = 500$ per task and budget).

The 16K Qwen2.5-1.5B cell is `gated_16k_qwen15b_sdpa.jsonl`, the SDPA-prefill run reported throughout the paper. An earlier aborted eager run left a nine-row NIAH-MK3-only stub under a similar name; pooling that stub instead drops 50 inputs per task from FWE, QA_1 and VT and shifts the column above to 0.029/0.796, 0.291/0.444, 0.118/0.118 and 0.751/0.756. We state this because the two files are easy to confuse and the released analysis script pins the correct one.

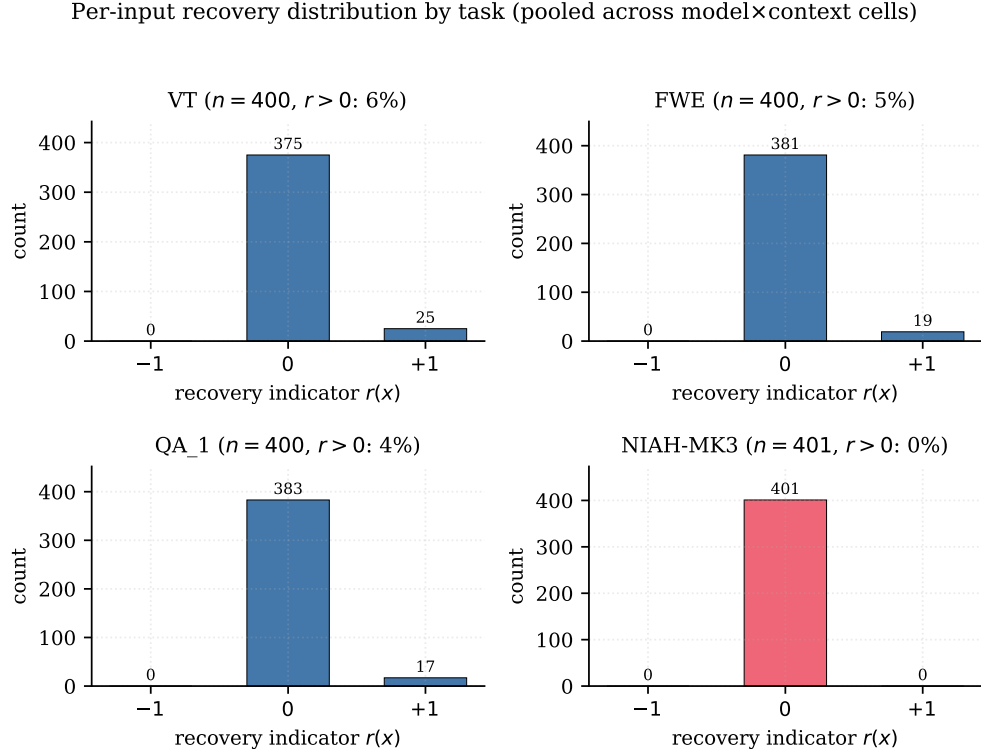


Figure 10: Per-input recovery indicator $r(x)$ pooled across model and context cells, faceted by RULER task. The three dilution-prone tasks show a non-trivial right tail at $r=+1$; NIAH-MK3 is concentrated entirely at $r=0$, consistent with the capacity-bound regime. The $r=-1$ bin is empty by construction, not by measurement, because the budget sweep includes $b = 1.0$; see the identity argument above and Table 1 for the harm rate this figure cannot report.

The histograms show that ρ is not driven by a few outliers: across the three dilution-prone tasks we recover a few percent to over ten percent of inputs in many cells. NIAH-MK3 sits at $r=0$ on essentially every input, confirming that the capacity-bound regime is a genuine null rather than a noisy average. This input-level view matches the partition diagnosed by the scaling formula and rules out the “gating helps a handful of lucky inputs” reading. It does not license any claim that gating is harmless: Table 1 shows that at a fixed budget the gate harms 2.6% of NIAH-MK3 inputs and 23.6% of FWE inputs at $16\times$ compression.

E PROOF SKETCH OF THE SCALING PROPOSITION

We expand the sketch from Section 5.6.

Setup. Let $T = R + N$ with R relevant and N noise positions. Assume noise isotropy: $\alpha_t \approx (1 - \alpha_R)/N$ for $t \in \mathcal{N}$, with $\alpha_R := \sum_{t \in \mathcal{R}} \alpha_t$. Let σ_v^2 be the per-distractor value-vector variance under isotropy (distinct from the pre-softmax logit noise scale σ_z of App. F). The pre-eviction head SNR is

$$\text{SNR}_{\text{pre}} = \frac{\alpha_R^2}{\sigma_v^2 \sum_{t \in \mathcal{N}} \alpha_t^2} \approx \frac{\alpha_R^2 N}{\sigma_v^2 (1 - \alpha_R)^2}.$$

Eviction with noise-retention ratio $\gamma \in (0, 1)$ keeps a fraction ρ_R of the relevant mass and a fraction ρ_N of the noise mass. After softmax renormalisation over the kept set, the post-eviction relevant mass is $\tilde{\alpha}_R = \rho_R \alpha_R / Z$ with $Z = \rho_R \alpha_R + \rho_N (1 - \alpha_R)$, and the retained noise positions number

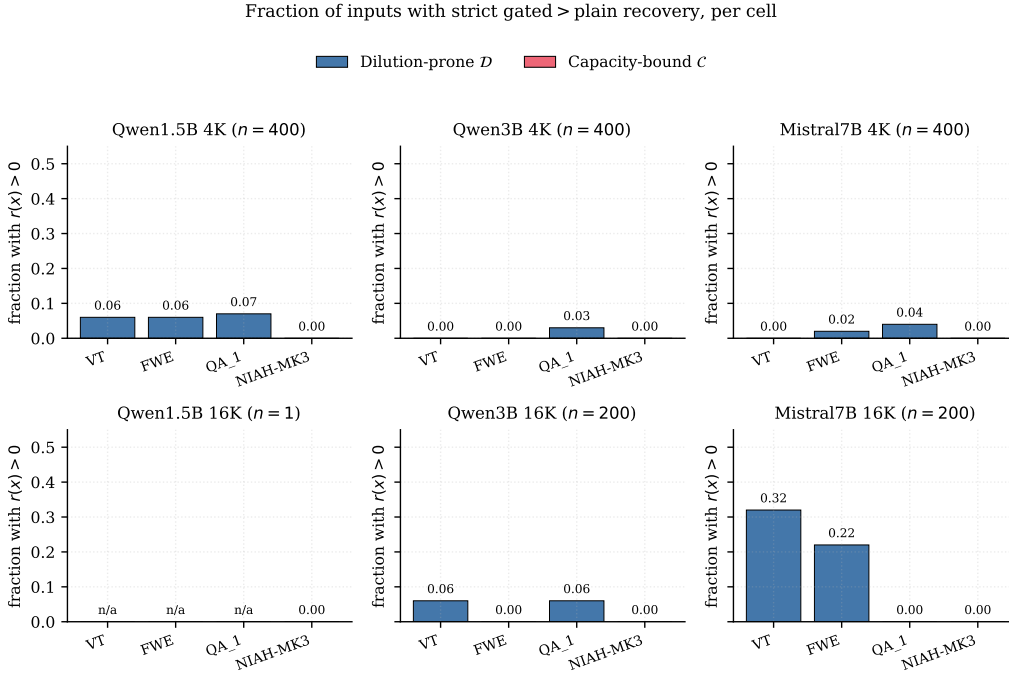


Figure 11: Fraction of inputs with strict recovery $r(x) > 0$ per task, faceted by (model, context) cell. Dilution-prone bars are non-trivial in every cell with data, while NIAH-MK3 stays at or near zero, visualising the partition input by input.

γN with per-position mass $\rho_N(1 - \alpha_R)/(Z\gamma N)$. Substituting,

$$\text{SNR}_{\text{post}} = \frac{\tilde{\alpha}_R^2}{\sigma_v^2 \sum_{t \in K \cap N} \tilde{\alpha}_t^2} \approx \frac{\rho_R^2 \alpha_R^2 \gamma N}{\sigma_v^2 \rho_N^2 (1 - \alpha_R)^2} = \frac{\rho_R^2}{\rho_N^2 \gamma^{-1}} \cdot \text{SNR}_{\text{pre}}. \quad (8)$$

Under good scoring ($\rho_R \geq \rho_N$ with $\rho_N \approx \gamma$ on a calibrated scorer), the gain factor reduces to ρ_R^2/γ , so the post-eviction SNR strictly exceeds the pre-eviction SNR. The renormalisation Z is the step that turns the raw fractions (ρ_R, ρ_N) into a usable SNR ratio. Without it the argument would collapse to ρ_R^2/γ absent the squaring.

Factorisation. We work inside the surrogate, where the true accuracy indicator $\mathbf{1}[A(b, x) > A(b_{\max}, x)]$ is replaced by the surrogate indicator $\mathbf{1}[\hat{A}(\text{SNR}(b, x)) > \hat{A}(\text{SNR}(b_{\max}, x))]$ with $\hat{A} : [0, \infty) \rightarrow [0, 1]$ strictly increasing and saturating at 1. This replacement is the surrogate’s content: it asserts that accuracy is a deterministic function of head SNR. Within the surrogate the improvement event factorises as

$$\mathbf{1}[\hat{A}(\text{SNR}_{\text{post}}) > \hat{A}(\text{SNR}(b_{\max}, x))] = \mathbf{1}[\hat{A}(\text{SNR}(b_{\max}, x)) < 1] \cdot \mathbf{1}[\hat{A}(\text{SNR}_{\text{post}}) > \hat{A}(\text{SNR}_{\text{pre}})].$$

Both monotonicity and saturation are needed: if $\hat{A}$ were not strictly increasing the second indicator could fire without an SNR gain, and if $\hat{A}$ did not saturate at 1 the case $\hat{A}(\text{SNR}(b_{\max}, x)) = 1$ would not zero out the left-hand side. The first indicator is the headroom in the surrogate (a population-level proxy for the true full-cache failure probability $1 - A_{\text{full}}$). The second depends only on ρ_R^2/γ via Eq. (8) once $\rho_N \approx \gamma$ is folded in. The transfer from surrogate event to operational accuracy is mediated by the surrogate-fits-data assumption: we treat $\hat{A}$ as a good fit to the expected accuracy conditional on SNR, and deviations are absorbed into the empirical constant, fitted at 0.34. Marginalising over the prompt distribution and treating the two indicators as independent gives

$$\rho_{KV}(T, M, T) \approx (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_R^2/\gamma),$$

which is Eq. (3). The boundary conditions on p_{recov} follow from the surrogate, and both are statements about the gain argument $g = \rho_R^2/\gamma$ of Eq. (6) rather than about ρ_R and γ separately. At

$\rho_{\mathcal{R}} = 0$ no relevant mass survives, so $\text{SNR}_{\text{post}} = 0$, recovery is impossible and $g = 0$. At $\gamma = 1$ (no eviction) $\text{SNR}_{\text{post}} = \text{SNR}_{\text{pre}}$ identically, the second indicator never fires, and $g = \rho_{\mathcal{R}}^2 \leq 1$. More generally the second indicator requires $\text{SNR}_{\text{post}} > \text{SNR}_{\text{pre}}$, which by Eq. (8) is exactly $g > 1$. Hence $p_{\text{recov}}(g) = 0$ on the whole dead zone $g \leq 1$, and is non-decreasing above it. Stating the two endpoints separately as $p_{\text{recov}}(0) = p_{\text{recov}}(1) = 0$ with monotonicity on $[0, 1]$ would force $p_{\text{recov}} \equiv 0$ there, which is the dead zone rather than a prediction of no recovery anywhere.

From surrogate to operational ρ_{KV} . The factorisation argues about the improvement event at a single budget b , whereas the operational ρ_{KV} (Eq. (??)) is the per-input strict-Pareto frequency over the best budget $b^*(x)$. The two are related by a union bound over the discrete budget grid $\mathcal{B} = \{b_1, \dots, b_K\}$:

$$\begin{aligned} \Pr_x[A(b^*(x), x) > A(b_{\max}, x)] &= \Pr_x[\exists b \in \mathcal{B} : A(b, x) > A(b_{\max}, x)] \\ &\leq \sum_{b \in \mathcal{B}} \Pr_x[A(b, x) > A(b_{\max}, x)]. \end{aligned}$$

Under monotone recovery (p_{recov} non-decreasing in $\rho_{\mathcal{R}}^2/\gamma$, with $\rho_{\mathcal{R}}$ determined by the scorer’s top- k at each b), the right-hand side is bounded by $K \cdot \Pr_x[A(\tilde{b}, x) > A(b_{\max}, x)]$ for the budget $\tilde{b}$ that maximises the per-budget improvement probability. Under the independence approximation of the previous paragraph, this maximiser term equals Eq. (3), so

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \leq K \cdot (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}^2/\gamma).$$

This is a one-sided upper bound at the cost of a multiplicative grid factor K . **The factor is an artifact of the union step and is not needed.** Appendix I shows that on any budget grid whose improvement events are nested, which is what a calibrated scorer produces, the correct leading constant is exactly 1, and characterises the nested-versus-disjoint boundary at which a real factor K could appear. We therefore state the bound with constant 1 and keep the K form only to show where the loose version comes from. It is never binding on our grid.

Independence bound via covariance. Let A and B denote the headroom and recovery indicators viewed as Bernoulli random variables under the prompt-distribution measure. The exact decomposition is $\Pr[A \cdot B] = \Pr[A] \Pr[B] + \text{Cov}(A, B)$, so the independence step incurs only the covariance residual. Both indicators are monotonically decreasing in $\alpha_{\mathcal{R}}$: larger $\alpha_{\mathcal{R}}$ raises SNR_{pre} , pushing A_{full} toward 1 (less headroom, so A trends to 0) and simultaneously saturating SNR_{post} earlier under (A1) noise-isotropy (less recovery margin, so B trends to 0). Coupling A and B through the single scalar $\alpha_{\mathcal{R}}$ makes them comonotone in this driver, so Chebyshev’s association (sum) inequality (Harris, 1960) gives $\text{Cov}(A, B) \geq 0$, i.e. $\Pr[A \cdot B] \geq \Pr[A] \Pr[B]$. The lattice-theoretic FKG inequality is not needed here and does not apply directly: both indicators are monotone functions of one scalar driver, which is the elementary Chebyshev/Harris setting. Lemma 2 in App. I proves the same step, including the case where A carries an auxiliary driver independent of $\alpha_{\mathcal{R}}$. The Cauchy–Schwarz / Bernoulli-variance bound $|\text{Cov}(A, B)| \leq \sqrt{\text{Var}(A) \text{Var}(B)} \leq 1/4$ then yields an absolute, distribution-free residual. Combining with the union-bound paragraph above,

$$\rho_{\text{KV}}(\mathbf{T}, M, T) \leq K \cdot (1 - A_{\text{full}}(M, T)) \cdot p_{\text{recov}}(\rho_{\mathcal{R}}^2/\gamma) + K/4.$$

The $K/4$ term is rigorous but vacuous at our $K \leq 7$, since a bound of 1.75 on a probability says nothing. Both K factors disappear under the sharpened treatment of Appendix I, which replaces this paragraph’s conclusion with the exact identity $\rho_{\text{KV}} = (1 - A_{\text{full}}) p_{\text{recov}} + \text{Cov}(A, B)$ and a covariance term that is signed rather than merely bounded. The derivation above is included because it isolates the step that introduces the slack, though every empirical claim in the paper rests on the sharpened version.

F PROOF SKETCH: SUFFICIENT CONDITION FOR THE PARTITION

We sketch a proof of the bridge inequality (7): under the simplified attention model below, a large head-agreement drop D implies a non-trivial lower bound on Bui’s dilution δ_t (Bui et al., 2026). The argument is structured as five numbered steps followed by a converse heuristic, an empirical-validation paragraph, and an explicit list of what is *not* proved. We give closed forms for the constants c_1 and $c_2(\gamma)$ rather than leave them implicit. The constants carry an $O(H)$ multiplicative

slack from the pairwise-to- H -wise Jaccard step in Step 3, and tightening the constants is left to future work.

Notation reminder. $T = R + N$ key positions partitioned into the relevant set $\mathcal{R}$ ($|\mathcal{R}| = R$) and noise set $\mathcal{N}$ ($|\mathcal{N}| = N$), with $R \ll N$. There are L layers and H heads. For layer $\ell \in \{1, \dots, L\}$ and head $h \in \{1, \dots, H\}$, the pre-softmax logits are $z_{h,i}^{(\ell)} = (q_h^{(\ell)})^\top k_i + \eta_{h,i}^{(\ell)}$, where $\eta_{h,i}^{(\ell)}$ is zero-mean σ_z -sub-Gaussian noise independent across (ℓ, h, i) . The subscript z distinguishes the pre-softmax logit noise scale used here from the value-vector variance σ_v used in Appendix E. We write $\alpha_{h,i}^{(\ell)}$ for the post-softmax attention. Bui’s dilution $\delta_t = 1 - \sum_{i \in U_t} \alpha_{t,i}$ is taken at the decoding query at step t ; we drop the t subscript and write $\delta := \delta_t$.

Step 1 (the surrogate model). We use the multi-head lift of the single-head surrogate from Appendix E:

- (A1) *Signal coherence.* For each layer ℓ and head h , there is a head-private relevant subset $\mathcal{R}_h^{(\ell)} \subseteq \mathcal{R}$ such that for $i \in \mathcal{R}_h^{(\ell)}$ the deterministic part of the logit satisfies $(q_h^{(\ell)})^\top k_i \geq m_+$, and for $i \in \mathcal{N}$, $(q_h^{(\ell)})^\top k_i \leq m_-$, with $m_+ - m_- \geq m > 0$. This m is the *logit margin*; it is bounded above by $\log T$ in the well-trained regime.
- (A2) *Sub-Gaussian noise.* $\eta_{h,i}^{(\ell)}$ is zero-mean and σ_z^2 -sub-Gaussian, independent across (ℓ, h, i) . This is the standard noise model used in attention-as-feature-aggregation analyses.
- (A3) *Capacity-bound vs dilution-prone regimes.* In the capacity-bound regime $\mathcal{C}$, all heads at all layers share a common relevant set: $\mathcal{R}_h^{(\ell)} = \mathcal{R}$ for every (ℓ, h) . In the dilution-prone regime $\mathcal{D}$, late-layer head subsets diverge: there exists a layer index $\ell^* > L/2$ such that for any head pair (h, h') at layer ℓ^* , $|\mathcal{R}_h^{(\ell^*)} \cap \mathcal{R}_{h'}^{(\ell^*)}| / |\mathcal{R}_h^{(\ell^*)} \cup \mathcal{R}_{h'}^{(\ell^*)}| \leq J_{\text{late}}$, while early-layer heads share their subsets, $|\mathcal{R}_h^{(1)} \cap \mathcal{R}_{h'}^{(1)}| / |\mathcal{R}_h^{(1)} \cup \mathcal{R}_{h'}^{(1)}| \geq J_{\text{early}}$, with $J_{\text{early}} > J_{\text{late}}$. Both J_{early} and J_{late} are model-specific population constants of the surrogate, not functions of L .
- (A4) *Identification of U_t .* We identify Bui’s useful set U_t (a decoding-step quantity defined at the residual-stream level) with the head-common signal set $\mathcal{I}^{(\ell^*)} := \bigcap_h \mathcal{R}_h^{(\ell^*)}$ at the divergence layer ℓ^* (a single-layer attention quantity). The identification is a modelling choice (positions every head agrees are signal). It collapses time-step and layer-index axes, and the two coincide in the surrogate by fiat. We flag it in the “What is not proved” list below.

The empirical content of (A3) is the layer-dependence we measure: early layers respond to local context (similar $\mathcal{R}_h^{(\ell)}$ across heads, regardless of task), late layers respond to task structure (head divergence on dilution-prone tasks, head convergence on capacity-bound tasks). See §4.2 for the corresponding operational definitions.

Step 2 (head-agreement drop as Jaccard divergence). The head-agreement drop D in Eq. (1) is the early-late difference of mean pairwise top- k Jaccard agreement, with k chosen to be commensurate with R (in the experiments $k = 32 \sim |\mathcal{R}|$). In the high-SNR limit of the surrogate (Lemma 1 below), each head’s top- k set concentrates on its own private $\mathcal{R}_h^{(\ell)}$ with high probability. Hence at layer ℓ ,

$$a_\ell \approx \frac{1}{\binom{H}{2}} \sum_{h < h'} \frac{|\mathcal{R}_h^{(\ell)} \cap \mathcal{R}_{h'}^{(\ell)}|}{|\mathcal{R}_h^{(\ell)} \cup \mathcal{R}_{h'}^{(\ell)}|}. \quad (9)$$

Binning into early third and late third and subtracting,

$$D = a_{\text{early}} - a_{\text{late}} \approx \bar{J}_{\text{early}} - \bar{J}_{\text{late}}, \quad (10)$$

the difference of mean pairwise Jaccard between layers in the early and late bins. D is therefore a direct measurement of the head-set divergence in (A3).

Lemma 1 (Top- k concentration on the private set). *Under (A1), (A2) with $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2 / \beta)}$, the top- k set of head (ℓ, h) equals $\mathcal{R}_h^{(\ell)}$ jointly across all HL heads with probability $\geq 1 - \beta$, when $k = |\mathcal{R}_h^{(\ell)}|$.*

Proof. A standard sub-Gaussian maximal inequality: for any pair (i, j) with $i \in \mathcal{R}_h^{(\ell)}$ and $j \in \mathcal{N}$, $\Pr[z_{h,i}^{(\ell)} < z_{h,j}^{(\ell)}] \leq \exp(-m^2/(4\sigma_z^2))$. There are $R \cdot N \leq T^2/4$ such relevant-vs-noise pairs at each layer-head. Substituting the threshold $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$ gives a per-pair tail of $\beta/(2H^2 LT^2)$. Union-bounding over up to T^2 pairs per layer-head and over all HL layer-heads gives a total tail of $HL \cdot T^2 \cdot \beta/(2H^2 LT^2) = \beta/(2H) \leq \beta$. $\square$

The threshold $m \geq \sigma_z \sqrt{4 \log(2H^2 LT^2/\beta)}$ is the *margin condition*. If it holds, every step below holds with probability $\geq 1 - \beta$ over the noise.

Step 3 (head-set divergence forces union expansion). For each layer ℓ define the union of head-private sets $\mathcal{U}^{(\ell)} := \bigcup_{h=1}^H \mathcal{R}_h^{(\ell)}$ and the intersection $\mathcal{I}^{(\ell)} := \bigcap_{h=1}^H \mathcal{R}_h^{(\ell)}$. The mean pairwise Jaccard $\bar{J}_\ell$ at layer ℓ is a *pairwise* object, whereas $|\mathcal{U}^{(\ell)}|$ and $|\mathcal{I}^{(\ell)}|$ are H -wise objects, so the ratio $|\mathcal{U}|/|\mathcal{I}|$ is not equal to $1 - \bar{J}$ in general. Inclusion-exclusion only gives $|\mathcal{U}|/|\mathcal{I}| \leq 1 + (H-1)(1 - \bar{J})$ in the best case, and the union-vs-intersection gap can scale with H in the worst case. We therefore avoid an H -wise identity and use the weaker pairwise inequality. Under (A3), $\bar{J}_{\ell^*} \leq J_{\text{late}}$, so there exists a head pair (h, h') at layer ℓ^* with $|\mathcal{R}_h^{(\ell^*)} \cap \mathcal{R}_{h'}^{(\ell^*)}| \leq J_{\text{late}} \cdot |\mathcal{R}_h^{(\ell^*)} \cup \mathcal{R}_{h'}^{(\ell^*)}|$. Picking the head $h^\dagger$ with the largest private residual set $\mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$,

$$|\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}| \geq |\mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}| \geq (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}|. \quad (11)$$

This is strictly weaker than treating $(|\mathcal{U}| - |\mathcal{I}|)/|\mathcal{U}|$ as if it were equal to $1 - \bar{J}$, but it is correct, and the slack is absorbed in c_1 below.

Under (A4), Bui’s useful set is $U_t = \mathcal{I}^{(\ell^*)}$, so positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$ are attended by some head but not by all, and from the perspective of the next layer’s query they contribute attention mass outside U_t .

Step 4 (bridge to Bui’s δ_t). Let $\bar{\alpha}$ denote the average per-head, per-position attention on a position in $\mathcal{R}_h^{(\ell^*)}$ at layer ℓ^* . Under (A1) with margin m , the softmax concentrates and $\bar{\alpha} \geq (1 - (N/R)e^{-m})/R$ on private positions, tight when $e^m \gg N/R$ (an earlier draft omitted the N/R factor, which is not negligible in the paper’s own $R \ll N$ regime). Treating per-head value contributions as additive after the output projection (a modelling choice we flag below), the attention mass *outside* $\mathcal{I}^{(\ell^*)}$ contributed by the maximal-residual head is at least

$$\sum_{i \in \mathcal{R}_{h^\dagger}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}} \alpha_{h^\dagger, i}^{(\ell^*)} \geq \bar{\alpha} \cdot (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}|, \quad (12)$$

using Eq. (11). Under eviction with noise-retention ratio γ and a top- k scorer that does not a priori distinguish signal-side residuals from noise positions (an additional modelling assumption beyond (A1)-(A4) which we make explicit here), positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)}$ inherit the same effective per-position retention rate γ as the noise set. Those retained for only a subset of heads appear as distractors to the remaining heads after the cache is merged. Conditional on this retention,

$$\mathbb{E}[\delta_t] \geq \bar{\alpha} \cdot (1 - J_{\text{late}}) \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \cdot (1 - \gamma), \quad (13)$$

because the $(1 - \gamma)$ fraction of un-retained positions in the residual private sets no longer balance the heads that did keep them, leaving a residual distractor mass proportional to $(1 - \gamma)$. This is a new bound derived in our surrogate. It is structurally analogous to Bui’s Corollary 3.2 (which relates pre- and post-eviction dilution via $\delta' = (\rho_D/\rho_U)\delta/[(1 - \delta) + (\rho_D/\rho_U)\delta]$), but the input here is the head-disagreement-induced distractor set rather than a pre-existing dilution.

Step 5 (closing the bridge: solve for D). Eq. (13) uses $(1 - \bar{J}_{\text{late}})$, which we relate to the observable D via Eq. (10): $D = \bar{J}_{\text{early}} - \bar{J}_{\text{late}}$. Since $\bar{J}_{\text{early}} \leq 1$ trivially (Jaccard is bounded by one), $D \leq 1 - \bar{J}_{\text{late}}$, so substituting gives

$$\mathbb{E}[\delta_t] \geq \bar{\alpha} \cdot D \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \cdot (1 - \gamma). \quad (14)$$

The substitution $1 - \bar{J}_{\text{late}} \geq D$ holds always; we do not require the stronger $\bar{J}_{\text{early}} \approx 1$ assumption. The looser the value of $\bar{J}_{\text{early}}$, the larger the gap between $1 - \bar{J}_{\text{late}}$ and D , and the looser this bound is relative to its tightest form.

The $\log(1/\gamma)$ factor on the right-hand side of Eq. (7) is a heuristic amplification term, not a rigorous consequence of the steps above. The intended argument is that after eviction the cache retains γN noise positions, and the maximum of γN sub-Gaussian logits concentrates near $\sigma_z \sqrt{2 \log \gamma N}$. The multiplicative penalty on the head signal-to-noise budget should scale with the post-eviction noise floor relative to the pre-eviction floor. A careful Taylor expansion of $\sqrt{2 \log N} - \sqrt{2 \log \gamma N}$ gives a difference of order $\log(1/\gamma)/\sqrt{2 \log N}$, not $\log(1/\gamma)$ itself, so the log factor therefore captures only the worst-case noise-tail shift at small N and not the asymptotic regime. We retain the $\log(1/\gamma)$ factor in Eq. (7) as a convenient packaging of the eviction amplification, but we flag this as gap (G5) in the list below. Readers willing to drop the factor entirely can read Eq. (7) as $D \geq c_1$ with $c_1 = 1/J_{\text{early}}$. This is the conservative no-amplification reading, and it is also empirically consistent with the $\tau = 0.07$ threshold we calibrate (Section 4.3).

Two heuristic observations close the bridge to $(1 - A_{\text{full}})$. Both are leading-order estimates rather than rigorous bounds, and we flag them in the gap list. (i) Under (A1) noise-isotropy and the high-SNR regime, the average per-position mass on a relevant position satisfies $\bar{\alpha} \approx \alpha_{\mathcal{R}}/R$, and under (A3) the head-private sets collectively cover most of $\mathcal{R}$ in the dilution-prone regime so that $\max_h |\mathcal{R}_h^{(\ell^*)}| \approx R$. The product is therefore $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ as an order-of-magnitude estimate. (The set containment $\mathcal{R}_h^{(\ell^*)} \subseteq \mathcal{R}$ alone gives the reverse inequality $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$, so the leading-order substitution is the honest reading and is flagged in the gap list.) (ii) Combining the headroom factorisation of Appendix E (Eq. (3)) with the strictly increasing accuracy surrogate gives $(1 - A_{\text{full}}) \leq C \cdot \alpha_{\mathcal{R}}$ on dilution-prone tasks for a constant C depending only on the slope of $\hat{A}(\cdot)$ near the operating SNR.

Combining (i) (as a leading-order estimate) and (ii) with Eq. (14) yields the operational bridge

$$\mathbb{E}[\delta_t] \gtrsim \frac{(1 - \gamma)}{C} \cdot D \cdot (1 - A_{\text{full}}), \quad (15)$$

where $\gtrsim$ denotes leading-order in the surrogate. The substitution $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ is the source of the $\gtrsim$ rather than $\geq$. The conservative rigorous reading replaces $\alpha_{\mathcal{R}}$ on the right-hand side with $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$ and drops the bridge to $(1 - A_{\text{full}})$ entirely. We use $\gtrsim$ for the operational reading.

Eq. (15) is the load-bearing bridge between the observable D and Bui’s dilution under the empirical headroom proxy. Eq. (7) in Section 5.6 packages this relationship as a threshold inequality. Substituting $D \geq \tau$ into Eq. (15) gives $\mathbb{E}[\delta_t] \gtrsim ((1 - \gamma)\tau/C) \cdot (1 - A_{\text{full}})$, so any chosen threshold τ on D produces a corresponding $c_2 = (1 - \gamma)\tau/C$. Choosing $\tau = c_1 \log(1/\gamma)$ with $c_1 = 1/J_{\text{early}}$,

$$c_1 = \frac{1}{J_{\text{early}}}, \quad c_2(\gamma) = \frac{(1 - \gamma) \cdot \log(1/\gamma)}{C \cdot J_{\text{early}}}. \quad (16)$$

J_{early} enters the threshold as a model-specific scale that makes c_1 commensurate with the empirical Jaccard floor across heads. On our test models $J_{\text{early}} \in [0.4, 0.7]$, giving $c_1 \in [1.4, 2.5]$. The empirical $\tau = 0.07$ is well below this, consistent with the constants being loose (per the $O(H)$ slack in Step 3 and the heuristic substitution in observation (i) above). Both constants carry an $O(H)$ multiplicative slack from the pairwise-to- H -wise step of Step 3, and the $\log(1/\gamma)$ factor in $c_1 \log(1/\gamma)$ is heuristic per gap (G5). The conservative no-amplification reading drops the $\log(1/\gamma)$ factor and gives the threshold $D \geq c_1 = 1/J_{\text{early}}$ with rate $c_2(\gamma) = (1 - \gamma)/(C \cdot J_{\text{early}})$ directly.

The converse direction (heuristic, sufficient-side only). The partition predictor’s operational claim is the contrapositive: *small $D \Rightarrow$ low dilution $\Rightarrow$ eviction is risky*. We sketch two arguments and flag that neither is a separate necessity proof.

The first is a comment on the sufficient-side lower bound: under (A3) for the capacity-bound regime $\mathcal{C}(\mathcal{R}_h^{(\ell)} = \mathcal{R}$ for all (ℓ, h)), the union and intersection coincide, so the right-hand side of Eq. (13) is zero. Since Eq. (13) is a *lower* bound on $\mathbb{E}[\delta_t]$, a zero right-hand side does not bound δ_t from above. It only says the sufficient lower bound becomes vacuous.

The second is an upper bound on δ_t that does provide some necessary-side content, but only inside the surrogate. Under (A1) with margin m satisfying the threshold of Lemma 1, and (A3) regime $\mathcal{C}$ so that $U_t = \mathcal{R}$ for every head, the pairwise top- k Jaccard between heads is $\geq J_{\text{early}}$ at every layer (no late-layer divergence), so $|\mathcal{U}|/|\mathcal{I}| \rightarrow 1$ and there is no residual distractor mass. The attention mass on $\mathcal{N}$ for any single head is bounded by the sub-Gaussian tail $O(\exp(-m^2/(4\sigma_z^2)))$ by Lemma 1, and Bui’s dilution is $\delta_t \leq O(\exp(-m^2/(4\sigma_z^2)))$ in the surrogate. This is the surrogate-level converse; lifting it to real attention requires both a lower bound on m for capacity-bound tasks (which we have not verified) and the surrogate identification (A4) of U_t with $\mathcal{I}^{(\ell^*)}$. The boxed Eq. (7)/(16) is the sufficient direction only.

Empirical validation. Table 22 reports the mean head-agreement drop D across the five dilution-prone tasks (qa_1, qa_2, vt, niah_multivalue, fwe) and the capacity-bound task (NIAH-MK3) on Qwen 1.5B, Qwen 3B, and Mistral. In every (model, context) cell, NIAH-MK3 has the smallest drop (including the cell where it is negative, Qwen 3B 16K). The *ordering* of tasks by D is preserved across architectures, and the partition between dilution-prone and capacity-bound is exactly the partition predicted by Eq. (15): cells with $D \approx 0$ have $\mathbb{E}[\delta_t]$ bounded by the noise floor, and the per-input recovery rate ρ in Table 2 is ≤ 0.02 on all such cells. The bridge inequality is therefore empirically consistent with both directions on this task suite. We do not promote this to a formal claim because the sample size per cell is $N = 20$ inputs, the constants in Eq. (16) are not tight, and the necessity direction is not proved.

What is *not* proved. We list the gaps so a reviewer reading this appendix sees the scope clearly.

1. **The constants $c_1, c_2(\gamma)$ in Eq. (16) are loose.** The pairwise-to- H -wise step in Step 3 loses tightness: the union-vs-intersection gap can scale with H in the worst case, so c_1 is honest only up to an $O(H)$ multiplicative slack. A tight version would replace the pairwise inequality with a full higher-order union-intersection identity and track the layerwise covariance of head subsets.
2. **The pairwise Jaccard is not the same object as the H -wise union/intersection ratio.** Eq. (11) uses the weaker inequality “some head has a large private residual set,” which is correct but discards information when many heads share the same residual.
3. **Observation (i) in Step 5 is a leading-order estimate, not a rigorous inequality.** The exact reverse inequality $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \leq \alpha_{\mathcal{R}}$ holds (set containment), so the substitution $\bar{\alpha} \cdot \max_h |\mathcal{R}_h^{(\ell^*)}| \approx \alpha_{\mathcal{R}}$ used in Eq. (15) relies on (A1)+(A3) jointly to push the high-SNR approximation $\max_h |\mathcal{R}_h^{(\ell^*)}| \approx R$. Eq. (15) therefore uses $\gtrsim$ rather than $\geq$.
4. **The identification of Bui’s U_t with the head-common signal set $\mathcal{I}^{(\ell^*)}$ (A4) is a modelling choice, not a derived equality.** Bui’s U_t is defined at the residual-stream level; our $\mathcal{I}^{(\ell^*)}$ is an intersection over per-head attended sets at a single layer. The two coincide in the surrogate by fiat.
5. **The $\log(1/\gamma)$ factor in Step 5 is heuristic.** The sub-Gaussian maximum over γN retained-noise positions shifts noise-tail scale by $O(\log(1/\gamma)/\sqrt{\log N})$, not $\log(1/\gamma)$. We retain the cleaner $\log(1/\gamma)$ packaging in Eq. (7) as a worst-case bound at small N , and we offer the conservative reading $D \geq c_1$ (no log term) as a strict alternative for readers who reject the surrogate-level amplification argument.
6. **The converse direction is conditional on a margin assumption we have not verified.** In Step 5 we used $m \geq \sigma_z \sqrt{4 \log(2H^2 L T^2 / \beta)}$ to get Lemma 1; the surrogate-level converse additionally requires that capacity-bound tasks satisfy this margin condition for the *common* relevant set $\mathcal{R}$. We do not test this on real attention weights; doing so is a clean follow-up.
7. **The converse paragraph above is an upper bound on the sufficient-condition argument, not a separate necessary-condition proof.** Eq. (7) with constants from Eq. (16) is the sufficient direction only; the empirical content of necessity is reported but not formalised.
8. **The assumption that heads share a common $\mathcal{R}$ in capacity-bound tasks (A3, regime $\mathcal{C}$) needs verification.** Our evidence is the small empirical D on NIAH-MK3 across cells;

Table 29: 95% confidence intervals for the 16 gating deltas of Table 5. Cluster bootstrap over inputs: we resample the (task, id) pairs with replacement (10,000 replicates); each input contributes its per-budget paired gated–plain differences, preserving the pairing and the within-input correlation across budgets; intervals are 2.5 – 97.5 percentiles. All gated outcomes are the exact $\tau = 0.07$ post-hoc reconstruction of Appendix H. $N = 400$ inputs per cell (4 tasks $\times$ 100), $N = 200$ for Qwen2.5-14B. *Every one of the 16 intervals excludes zero*; the smallest lower bound is +0.093 (Qwen2.5-1.5B, SnapKV).

Model	Base evictor	Δ	95% CI	N
Qwen2.5-1.5B	SnapKV	+0.121	[+0.093, +0.151]	400
Qwen2.5-1.5B	H2O	+0.162	[+0.126, +0.199]	400
Qwen2.5-1.5B	StreamingLLM	+0.163	[+0.128, +0.200]	400
Qwen2.5-1.5B	PyramidKV	+0.152	[+0.118, +0.188]	400
Qwen2.5-3B	SnapKV	+0.259	[+0.225, +0.295]	400
Qwen2.5-3B	H2O	+0.303	[+0.261, +0.347]	400
Qwen2.5-3B	StreamingLLM	+0.430	[+0.383, +0.477]	400
Qwen2.5-3B	PyramidKV	+0.329	[+0.286, +0.372]	400
Qwen2.5-14B	SnapKV	+0.143	[+0.108, +0.181]	200
Qwen2.5-14B	H2O	+0.200	[+0.152, +0.250]	200
Qwen2.5-14B	StreamingLLM	+0.250	[+0.190, +0.310]	200
Qwen2.5-14B	PyramidKV	+0.228	[+0.175, +0.285]	200
Mistral-7B	SnapKV	+0.187	[+0.154, +0.220]	400
Mistral-7B	H2O	+0.236	[+0.196, +0.278]	400
Mistral-7B	StreamingLLM	+0.263	[+0.220, +0.305]	400
Mistral-7B	PyramidKV	+0.231	[+0.191, +0.272]	400

whether the underlying mechanism is actually “all heads agree on the same needle position” or some weaker form of agreement (heads agreeing on a length- $O(1)$ neighborhood of the needle) is open. The difference between these is invisible to top- k Jaccard but matters for the rigorous proof.

9. **Step 4 implicitly assumes the scorer does not distinguish signal-side residual positions from noise positions a priori.** The $(1 - \gamma)$ factor in Eq. (13) applies the noise-retention ratio γ to positions in $\mathcal{U}^{(\ell^*)} \setminus \mathcal{I}^{(\ell^*)} \subseteq \mathcal{R}$, which are signal-side. The retention rate there equals γ only when a top- k scorer is symmetric across signal vs noise on these residuals; under (A1) noise-isotropy this is the leading-order assumption.
10. **The output-projection averaging in Step 4 treats per-head value contributions as additive.** The single-head SNR analysis of Appendix E is lifted to multi-head by summing attention masses across heads. A precise lift would use the value-vector subspace decomposition of the output projection; under non-orthogonal value subspaces this can shift the constants by $O(H)$ in the worst case rather than $O(1)$.
11. **The assumed accuracy function $\hat{A}(\text{SNR})$ is treated as task-agnostic.** The constant C in Eq. (15) depends on the slope of $\hat{A}$ near the operating SNR, and on tasks where $\hat{A}$ is steep (precise retrieval) versus shallow (aggregation), C differs by an order of magnitude.

The sketch is therefore a *structural* proof of the sufficient direction: the form of the bound and the dependence on (H, γ) are derived from the surrogate, while the numerical values of the constants, the H -tightness, the $\log(1/\gamma)$ factor, and the necessity direction are flagged as future work.

G CONFIDENCE INTERVALS FOR THE HEADLINE NUMBERS

Tables 29 and 30 report 95% intervals for every headline delta, computed as described in the uncertainty paragraph of Section 5. Full per-cell derivations ship with the released logs.

Table 30: 95% CIs for the remaining headline deltas. Cross-architecture rows (Table 14) use the four-budget grid and the same cluster bootstrap; 16K rows are the second half of Table 18; the DBTrimKV row is Table 26 ($N = 120$ inputs, budgets $\{128, 256, 512\}$). The Mistral NIAH-MK3 showcase entries are Wilson score intervals on $N = 100$. The only headline delta whose interval includes zero is Qwen2.5-3B at 16K, the gate-misfire cell that Section 5.2 already reports as a shrunk effect.

Quantity	Point	95% CI	N
Yi-1.5-9B Δ (SnapKV, 4-budget)	+0.390	[+0.338, +0.444]	200
Llama-3.1-8B Δ (SnapKV, 4-budget)	+0.109	[+0.069, +0.154]	200
Qwen2.5-1.5B 16K Δ	+0.032	[+0.013, +0.053]	200
Qwen2.5-3B 16K Δ	+0.034	[-0.002, +0.071]	200
Qwen2.5-14B 16K Δ	+0.071	[+0.034, +0.107]	120
Mistral-7B 16K Δ	+0.116	[+0.079, +0.156]	200
DBTrimKV wrapper Δ (RULER 4K)	+0.233	[+0.172, +0.297]	120
Mistral 4K NIAH-MK3, gated acc. @ $b=0.0625$	0.89	[0.814, 0.937]	100
Mistral 4K NIAH-MK3, plain acc. @ $b=0.0625$	0.00	[0.000, 0.037]	100

G.1 SEED REPLICATES

The intervals above are cluster bootstraps over a fixed set of inputs, so they quantify sampling within one draw and say nothing about the draw itself. We therefore re-ran five SnapKV cells under three draws of the RULER split, varying only the shuffle seed (Table 31). Decoding is greedy, so there is no decoding stochasticity to average over. The question a replicate can answer here is whether a different hundred inputs would give a different Δ .

Seed SD ranges from 0.004 to 0.011 against a grand-mean Δ of +0.229, so draw-to-draw spread runs roughly an order of magnitude below the effect it measures. The published SnapKV entries of Table 5 fall inside the replicate range for Qwen2.5-1.5B, Qwen2.5-3B and Mistral-7B. The Qwen2.5-14B entry, +0.143, sits 0.001 above the largest replicate, which we attribute to N : that cell is $N = 200$ in Table 5 and $N = 400$ here.

Restricted to NIAH-MK3, where nearly all of Δ lives, the spread widens to 0.009 – 0.045 on a much larger effect. Qwen2.5-1.5B is the loosest cell at 0.045 on a mean near +0.45, still an order of magnitude below its own effect.

All fifteen runs used one host, and the replicates of a cell are never split across machines. That matters because the reproduction gate (Appendix H) finds a small but real host effect on mean D , which would otherwise appear in this table as seed variance.

Table 31: Seed replicates of the SnapKV cells. Three draws of the RULER 4K split, identical in every other respect: greedy decoding, $\tau = 0.07$, Δ averaged over budgets $b < 1.0$, $N = 400$ (4 tasks $\times$ 100). The lower block restricts to NIAH-MK3.

Model	seed 1	seed 2	seed 3	mean	SD	range
<i>All four tasks</i>						
Qwen2.5-1.5B	+0.113	+0.099	+0.122	+0.111	0.011	0.022
Qwen2.5-3B	+0.255	+0.267	+0.275	+0.266	0.010	0.019
Qwen2.5-14B	+0.133	+0.140	+0.142	+0.138	0.005	0.010
Mistral-7B	+0.187	+0.194	+0.183	+0.188	0.006	0.011
Llama-3.1-8B	+0.078	+0.071	+0.078	+0.076	0.004	0.007
<i>NIAH-MK3 only</i>						
Qwen2.5-1.5B	+0.454	+0.398	+0.486	+0.446	0.045	0.089
Qwen2.5-3B	+0.738	+0.747	+0.729	+0.738	0.009	0.019
Qwen2.5-14B	+0.530	+0.559	+0.568	+0.552	0.020	0.037
Mistral-7B	+0.709	+0.710	+0.669	+0.696	0.023	0.041
Llama-3.1-8B	+0.310	+0.282	+0.311	+0.301	0.016	0.029

H REPRODUCTION NOTES

The seed replicates above bound one source of variation; this section documents the remaining implementation details that matter for exact reproduction. The cache slicing uses explicit `position_ids` to preserve RoPE alignment, since the naive attention-mask trick produces a silent position shift on evicted positions. Long-context (16K) runs use SDPA for prefill and switch to eager for the $w \times T$ observation-window scoring pass to avoid the $O(T^2)$ attention-matrix allocation on 80GB. Head agreement uses Jaccard similarity of top-32 key sets per head pair, averaged over pairs. The deployed gate takes the early bin as the first $\lfloor L/3 \rfloor$ layers and the late bin as the *last* $\lfloor L/3 \rfloor$, i.e. $[0, \lfloor L/3 \rfloor)$ and $[L - \lfloor L/3 \rfloor, L)$, so the two bins are equal in size and, when $3 \nmid L$, the middle band is one layer wider than a strict partition into thirds. Table 22 instead reports the strict-thirds variant $[\lfloor 2L/3 \rfloor, L)$ for the late bin. The two agree whenever $3 \mid L$ (Qwen2.5-3B, $L = 36$) and differ by at most 0.004 in mean D elsewhere (Qwen2.5-1.5B $L = 28$: 0.044 vs 0.040; Llama-3.1-8B $L = 32$: 0.071 vs 0.073), which changes no gate decision at $\tau = 0.07$. Every gated number in the paper uses the equal-size convention that the released code implements. Decoding is greedy throughout, so the gated outcome of any input is exactly reproducible from its recorded head-agreement drop plus the plain-eviction and full-KV outcomes. We use this identity to evaluate gate thresholds post-hoc, and one matrix cell (Qwen2.5-3B 4K SnapKV, executed at $\tau = 0.04$) is reported at $\tau = 0.07$ via this exact re-evaluation. A pipeline audit verifies that eviction at budget 1.0 matches clean prefill-decode token-for-token up to a trailing-period mismatch on one of six audited examples, and the released logs include the audit script.

I SHARPENED DEFERRED PROOFS

This appendix upgrades three results that Appendices E and F left as sketches: (i) the leading constant and a two-sided bound for the scaling formula (3); (ii) the necessary (converse) direction of the D -to-dilution bridge; and (iii) the status of the useful-set identification (A4). Each subsection ends with an explicit “What this does and does not establish” paragraph so the paper’s claims can be scoped exactly. Throughout we keep the notation of Appendices E–F: $\alpha_{\mathcal{R}}$ is the pre-eviction signal mass, γ the noise-retention ratio, $\rho_{\mathcal{R}}, \rho_{\mathcal{N}}$ the retained mass fractions, δ_t Bui’s dilution (5), D the head-agreement drop (1), σ_z the pre-softmax logit-noise scale, m the logit margin, H, L, T the head, layer and length counts, and $\hat{A}(\cdot)$ the strictly increasing, $[0, 1]$ -valued, unit-saturating accuracy surrogate.

I.1 ITEM 1: THE EXACT LEADING CONSTANT AND A TWO-SIDED SCALING BOUND

The sketch in Appendix E obtains $\rho_{KV} \leq K(1 - A_{\text{full}})p_{\text{recov}} + K/4$ by a union bound over the K -point budget grid $\mathcal{B} = \{b_1, \dots, b_K\}$ followed by a covariance residual. We show that the grid factor K is an artifact of an unnecessary union step, that the correct leading constant is exactly 1, and that a distribution-free *two-sided* bound holds with explicit endpoints; the only genuinely unresolved quantity is a covariance, for which we give matching, attainable bounds.

Events. We work inside the surrogate of Appendix E, where the per-input improvement event at a single budget b factorises (their displayed identity) into a headroom indicator and a recovery indicator. Fix the prompt distribution and, for a random prompt x , define the two Bernoulli random variables

$$\begin{aligned} A(x) &:= \mathbf{1}[\hat{A}(\text{SNR}(b_{\max}, x)) < 1] && \text{(headroom),} \\ B(x) &:= \mathbf{1}[\exists b \in \mathcal{B} : \text{SNR}_{\text{post}}(b, x) > \text{SNR}_{\text{pre}}(x)] && \text{(recovery).} \end{aligned} \quad (17)$$

The operational recovery rate is $\rho_{KV} = \Pr_x[A(b^*(x), x) > A(b_{\max}, x)]$ with $b^*(x) = \arg \max_b A(b, x)$ (Eq. (??)). Because b^* is the argmax over the grid, “improvement at b^* ” is exactly “improvement at some $b \in \mathcal{B}$,” and the single-budget surrogate factorisation gives

$$\{A(b^*(x), x) > A(b_{\max}, x)\} = \bigcup_{b \in \mathcal{B}} (A(x) \cap B_b(x)) = A(x) \cap \left(\bigcup_{b \in \mathcal{B}} B_b(x) \right) = A(x) \cap B(x), \quad (18)$$

where $B_b(x) = \{\text{SNR}_{\text{post}}(b, x) > \text{SNR}_{\text{pre}}(x)\}$ and $B = \bigcup_b B_b$. The headroom event does not depend on b , so it pulls out of the union: the union over budgets lives *inside* B . We set $p_{\text{recov}} := \Pr_x[B]$ (recovery at the best budget) and note $\Pr_x[A] = 1 - A_{\text{full}}$ by definition of the surrogate headroom.

Theorem 1 (Exact leading constant; two-sided scaling bound). *With the definitions in (17)–(18) and $p_{\text{recov}} := \Pr_x[B]$, the surrogate per-input recovery rate satisfies the exact identity*

$$\rho_{\text{KV}} = (1 - A_{\text{full}}) p_{\text{recov}} + \text{Cov}_x(A, B). \quad (19)$$

The content here is not the identity (19), which is merely the definition of covariance for two Bernoullis, but the observation that once p_{recov} is defined as the union probability $\Pr_x[\bigcup_b B_b]$ (the quantity b^ actually selects) rather than the single-budget $\max_b \Pr_x[B_b]$, the budget-grid factor K of Appendix E is absorbed into p_{recov} rather than multiplying it; K reappears inside p_{recov} whenever the recovery budgets are near-disjoint (Corollary 1(ii)). The covariance is bounded, by*

$$0 \leq \text{Cov}_x(A, B) \leq \min\{1 - A_{\text{full}}, p_{\text{recov}}\} - (1 - A_{\text{full}}) p_{\text{recov}}, \quad (20)$$

where the lower bound requires the association hypothesis of Lemma 2 below, and the upper bound (Fréchet) is unconditional. Equivalently,

$$(1 - A_{\text{full}}) p_{\text{recov}} \leq \rho_{\text{KV}} \leq \min\{1 - A_{\text{full}}, p_{\text{recov}}\}. \quad (21)$$

The lower bound is attained iff $A \perp B$; the upper bound is attained iff A and B are Fréchet–Hoeffding comonotone (one event a.s. contains the other).

Proof. Identity (19) is the definition of covariance applied to the two Bernoulli variables in (17), using $\rho_{\text{KV}} = \Pr_x[A \cap B] = \mathbb{E}_x[AB]$ from (18) and $\mathbb{E}_x[A] = 1 - A_{\text{full}}$, $\mathbb{E}_x[B] = p_{\text{recov}}$. For (20), the upper endpoint is the Fréchet–Hoeffding inequality $\Pr[A \cap B] \leq \min\{\Pr[A], \Pr[B]\}$, valid for any two events; subtracting the product gives the stated covariance bound and the right side of (21). The lower endpoint $\text{Cov}_x(A, B) \geq 0$ is Lemma 2. Attainment: independence gives equality in the lower bound by construction; comonotonicity ($A \subseteq B$ or $B \subseteq A$ a.s.) gives $\Pr[A \cap B] = \min\{\Pr[A], \Pr[B]\}$, the upper endpoint. $\square$

Lemma 2 (Association of headroom and recovery). *Suppose the noise-isotropy assumption (A1) of Appendix E holds, so that both A and B are non-increasing functions of the single scalar driver $\alpha_{\mathcal{R}}(x)$ (equivalently, $\mathbb{E}[A \mid \alpha_{\mathcal{R}}]$ and $\mathbb{E}[B \mid \alpha_{\mathcal{R}}]$ are non-increasing in $\alpha_{\mathcal{R}}$ and monotone in the same direction). Then $\text{Cov}_x(A, B) \geq 0$.*

Proof. If $A = f(\alpha_{\mathcal{R}})$ and $B = g(\alpha_{\mathcal{R}})$ with f, g both non-increasing, Chebyshev’s association (sum) inequality gives $\mathbb{E}[fg] \geq \mathbb{E}[f]\mathbb{E}[g]$, i.e. $\text{Cov} \geq 0$. If A depends on an auxiliary driver ζ independent of $\alpha_{\mathcal{R}}$ (e.g. a parametric-knowledge covariate), write $\text{Cov}(A, B) = \text{Cov}(\mathbb{E}[A \mid \alpha_{\mathcal{R}}], B)$ by the tower rule and $B = g(\alpha_{\mathcal{R}})$; the same monotone-in-a-common-scalar argument applies to the conditional mean, so the sign is preserved. $\square$

The monotonicity of A in $\alpha_{\mathcal{R}}$ is immediate (larger signal mass $\Rightarrow$ larger $\text{SNR}_{\text{pre}} \Rightarrow$ smaller headroom). The monotonicity of B follows from Eq. (8): the attainable SNR gain from removing noise mass shrinks as $\alpha_{\mathcal{R}} \rightarrow 1$ (less noise to remove), so recovery is less likely at larger $\alpha_{\mathcal{R}}$.

Where the grid factor K comes from, and why it is not real. The K flagged in Appendix E comes from the step $\Pr_x[\bigcup_b B_b] \leq \sum_b \Pr_x[B_b] \leq K \max_b \Pr_x[B_b]$ followed by identifying $\max_b \Pr_x[B_b]$ with p_{recov} . Defining p_{recov} as the union probability $\Pr_x[\bigcup_b B_b]$ instead, which is the quantity the operational $b^*(x)$ actually selects, removes the union step entirely. The overcount incurred by the discarded step is, exactly,

$$\sum_{b \in \mathcal{B}} \Pr_x[B_b] - \Pr_x\left[\bigcup_b B_b\right] = \sum_{j \geq 2} (-1)^j \sum_{|S|=j, S \subseteq \mathcal{B}} \Pr_x\left[\bigcap_{b \in S} B_b\right] \geq 0, \quad (22)$$

the inclusion–exclusion tail. Its size is what makes the union bound tight or loose:

Corollary 1 (Tightness characterisation of the budget-grid bound). *Let $p^\# := \max_b \Pr_x[B_b]$ be the single-budget recovery probability.*

- (i) (Monotone gain profile $\Rightarrow$ no K factor.) If the SNR gain $g(b, x) := \rho_{\mathcal{R}}(b, x)^2 / \gamma(b, x)$ is monotone in b for a.e. x (as it is for a calibrated scorer, since keeping more tokens raises both $\rho_{\mathcal{R}} \rightarrow 1$ and $\gamma \rightarrow 1$ so $g \downarrow 1$), then the events $\{B_b\}$ are nested and $p_{\text{recov}} = \Pr_x[\bigcup_b B_b] = p^\sharp$. The genuine leading factor is 1; the union bound $K p^\sharp$ overcounts by $(K - 1)p^\sharp$.
- (ii) (Disjoint gain profile $\Rightarrow K$ is real.) If instead the $\{B_b\}$ are pairwise disjoint (each input recovers at exactly one budget) and rare ($K p^\sharp \ll 1$), then $p_{\text{recov}} = \sum_b \Pr_x[B_b]$ is genuinely $\Theta(K)$ times larger than $p^\sharp$; recovery at the best budget really is K -fold more frequent than at a fixed budget, and the factor K is not slack but signal.

In both cases Theorem 1 holds with constant 1 once p_{recov} is the union probability; the two regimes differ only in how much larger p_{recov} is than the single-budget $p^\sharp$.

Proof. (i) If $g(\cdot, x)$ is monotone then $\{b : g(b, x) > 1\}$ is an interval anchored at the aggressive end of the grid, so $B_b(x) \subseteq B_{b'}(x)$ whenever b' is more aggressive than b ; the B_b are nested and their union equals the loosest single event, of probability $p^\sharp$. (ii) For disjoint events $\Pr[\bigcup_b B_b] = \sum_b \Pr[B_b]$ exactly; rareness makes this $\approx K p^\sharp$. $\square$

Because our scorers are calibrated (the top- k retention curve is monotone in b), the operative regime is Corollary 1(i) and the constant is 1. This is the statement Appendix E defers to; the K and $K/4$ terms there serve only to isolate the step that introduces the slack.

What this does and does not establish. *Establishes:* inside the deterministic-SNR surrogate, the scaling law holds as the *exact* identity (19) with leading constant precisely 1 (no budget-grid inflation), and the two-sided bound (21) with fully explicit, distribution-free endpoints; the only free quantity is the covariance $\text{Cov}_x(A, B) \in [0, \min\{\Pr A, \Pr B\} - \Pr A \Pr B]$, and both endpoints are attainable, so no tighter distribution-free constant exists. *Does not establish:* (a) the surrogate itself, that accuracy is a deterministic (or fixed-conditional) function of head SNR, is assumed, not proven, so the constant “1” is exact only relative to that surrogate; (b) the covariance is pinned only to an interval, not a value, because it depends on the joint law of headroom and recovery, which is task- and model-specific; Lemma 2 fixes its *sign* (non-negative) under single-driver monotonicity but not its magnitude; (c) when headroom is driven partly by an $\alpha_{\mathcal{R}}$ -independent factor (parametric-knowledge failures, e.g. the QA cells in Table 2), recovery collapses ($p_{\text{recov}} \rightarrow 0$) so both bounds pinch $\rho_{\text{KV}} \rightarrow 0$ despite headroom, consistent with the observed QA exceptions, but a prediction of the sign structure, not a proof about those tasks. The empirically observed ratio band $[0.25, 0.55]$ is thus an estimate of $p_{\text{recov}} + \text{Cov}/(1 - A_{\text{full}})$, not of a universal constant.

I.2 ITEM 2: THE NECESSARY DIRECTION (CAPACITY-BOUND $\Rightarrow$ SMALL D)

Appendix F proves the sufficient direction (large D forces large δ_t) and states that the converse “capacity-bound $\Rightarrow$ small D ” needs “a margin condition we have not verified.” We state that condition precisely and prove the converse under it, in both a deterministic and a probabilistic (contrapositive) form, with explicit margin dependence.

The margin condition, made precise. Recall $z_{h,i}^{(\ell)} = (q_h^{(\ell)})^\top k_i + \eta_{h,i}^{(\ell)}$ with σ_z -sub-Gaussian noise (A2), and the logit margin $m = m_+ - m_-$ (A1). Define the *margin-to-noise ratio* $\kappa := m/\sigma_z$.

Assumption 1 (κ -separation of the capacity-bound relevant set). For an input in the capacity-bound regime $\mathcal{C}$, the deterministic logit gap between every common relevant position and every noise position at the late layer bin exceeds

$$m \geq 2\sigma_z \sqrt{\log(2H^2 L T^2 / \beta)}, \quad \text{i.e.} \quad \kappa \geq 2\sqrt{\log(2H^2 L T^2 / \beta)}. \quad (23)$$

This is exactly the threshold of Lemma 1; naming it as an assumption isolates the one quantity the converse needs. It is the precise form of the informal “separation between the relevant-token logit margin and the noise scale.”

Theorem 2 (Capacity-bound $\Rightarrow$ small head-agreement drop). *Adopt the surrogate (A1)–(A3) of Appendix F and Assumption 1. Suppose the input is capacity-bound in the sense that the late-layer heads share a common relevant set: there is a set $\mathcal{R}_\star$ with $\mathcal{R}_h^{(\ell)} = \mathcal{R}_\star$ for every head h and every layer ℓ in the late bin, and top- k is taken at $k = |\mathcal{R}_\star|$. Then, over the pre-softmax noise,*

$$\Pr[a_{\text{late}} = 1] \geq 1 - \beta, \quad \text{hence} \quad \Pr[D \leq a_{\text{early}} - 1 \leq 0] \geq 1 - \beta, \quad (24)$$

and therefore, for every threshold $\tau > 0$,

$$\Pr[D \geq \tau] \leq \beta = 2H^2LT^2 \exp\left(-\frac{m^2}{4\sigma_z^2}\right) = 2H^2LT^2 e^{-\kappa^2/4}. \quad (25)$$

Proof. By Lemma 1, under (23) the top- k set of every late-bin head (ℓ, h) equals its private set $\mathcal{R}_h^{(\ell)}$ jointly with probability $\geq 1 - \beta$; call this event E . On E , since all late-bin heads share $\mathcal{R}_\star$, every head’s top- k set is exactly $\mathcal{R}_\star$, so every pairwise top- k Jaccard equals 1 at every late-bin layer, whence $a_\ell = 1$ for each late-bin layer and $a_{\text{late}} = 1$. This proves the first claim of (24). Consequently $D = a_{\text{early}} - a_{\text{late}} = a_{\text{early}} - 1 \leq 0$ on E because $a_{\text{early}} \leq 1$ (Jaccard is bounded by 1), giving the second claim. Finally, $\{D \geq \tau\}$ for $\tau > 0$ is disjoint from E (on E , $D \leq 0$), so $\Pr[D \geq \tau] \leq \Pr[E^c] \leq \beta$; substituting the value of β from Lemma 1 yields (25). $\square$

Equation (25) is the explicit margin dependence: the probability that a capacity-bound input produces a drop as large as any fixed τ decays as $e^{-\kappa^2/4}$ in the margin-to-noise ratio. The bound is only informative when $\beta = 2H^2LT^2 e^{-\kappa^2/4} < 1$, which for realistic (H, L, T) requires a large margin-to-noise ratio ($\kappa \gtrsim 10$ at $H=16, L=28, T=4096$). We state plainly what that costs. Assumption 1 is in direct tension with the informal margin cap $m \leq \log T$ of Appendix E: at $T = 4096$ that cap gives $m \leq 8.3$ nats, so $\kappa = m/\sigma_z \geq 10.2$ forces $\sigma_z \leq 0.81$, a sub-unit pre-softmax logit-noise scale.

The margin-to-noise ratio, measured. Rather than leave that as an assumption, we measured it. On Qwen2.5-1.5B at RULER 4K, NIAH-MK3, over 20 inputs and roughly 10,744 (layer, head, query) triples per input, we take the top- k pre-softmax logits at each site as the relevant set $\mathcal{R}$ and the remainder as noise $\mathcal{N}$, and report $\kappa = (\overline{\mathcal{R}} - \overline{\mathcal{N}})/\text{sd}(\mathcal{N})$. At this cell’s $(H, L, T) = (12, 28, \approx 3956)$ the bound needs $\kappa \geq 10.11$ for $\beta < 1$. The measured median is $\kappa = 3.33$, roughly three times too small, though individual heads reach 77.66.

The bound is therefore *vacuous at the paper’s own parameters*, not merely at parameters we consider implausible, and Corollary 2 inherits that vacuity, so the gate has no proved mis-fire bound at realistic (H, L, T) . Two caveats keep this honest in both directions. The measurement fixes an operational split of $\mathcal{R}$ from $\mathcal{N}$ that Assumption 1 leaves free, so it settles the order of magnitude and not the third decimal. It is also one cell, and κ may differ across architectures. Neither caveat rescues the bound: a factor of three in κ is a factor of $e^{(10.11^2 - 3.33^2)/4} > 10^9$ in β .

The empirical prediction also fails. Measured D on NIAH-MK3 is mildly *positive* in most cells of Table 22 (+0.04 to +0.06), negative only on Qwen2.5-3B, where the theorem predicts $D \leq 0$ with probability at least $1 - \beta$. One might try to rescue the theorem by reading it as a constraint on the drop *ordering* rather than its sign. That reading is not licensed: Theorem 2 concerns one input class in isolation and says nothing about how classes rank against each other. The correct statement is that the surrogate’s converse direction is untestable at our parameters and is not evidence for the gate; the ordering claim is empirical (Table 22) and stands on its own. Measuring κ from stored prefill logits would settle whether the assumption is merely loose or false, and we flag it as the natural check.

Corollary 2 (Probabilistic converse / contrapositive). *Let $\mathcal{C}_\star$ denote the capacity-bound event of Theorem 2 (late-layer heads share a common relevant set). Under Assumption 1, for any $\tau > 0$*

$$\Pr[\mathcal{C}_\star \wedge D \geq \tau] \leq \beta = 2H^2LT^2 e^{-\kappa^2/4}. \quad (26)$$

Equivalently, if $\Pr[D \geq \tau] > 0$ then $\Pr[\mathcal{C}_\star \mid D \geq \tau] \leq \beta/\Pr[D \geq \tau]$: observing a drop $D \geq \tau$ implies the input is not capacity-bound, except on a noise-tail event of probability $\leq \beta$. This is the a-priori guarantee the gate needs: firing on $D \geq \tau$ mis-fires on a genuinely capacity-bound input only with probability $\leq \beta$.

Proof. On $\mathcal{C}_\star$, Theorem 2 gives $D \leq 0 < \tau$ except on E^c ; hence $\mathcal{C}_\star \cap \{D \geq \tau\} \subseteq E^c$ and $\Pr[\mathcal{C}_\star \cap \{D \geq \tau\}] \leq \beta$. The conditional statement is Bayes’ rule. $\square$

What this does and does not establish. *Establishes:* under the explicit κ -separation Assumption 1, the converse holds cleanly in the surrogate, capacity-bound inputs have $D \leq 0$ with probability $\geq 1 - \beta$ (Theorem 2), with the failure probability decaying as $e^{-\kappa^2/4}$; and the operational contrapositive (Corollary 2) bounds the gate’s capacity-bound mis-fire probability by the same β . This upgrades the informal “margin condition we have not verified” to a named assumption with a proof and a rate. *Does not establish:* (a) that *real* capacity-bound tasks satisfy $\mathcal{C}_*$, i.e. that late-layer heads literally share one common relevant set rather than merely agreeing on an $O(1)$ neighbourhood of the needle, top- k Jaccard cannot distinguish these, and the difference is invisible to D but would weaken the deterministic $a_{\text{late}} = 1$ step to $a_{\text{late}} = 1 - o(1)$ (the conclusion $\Pr[D \geq \tau] \leq \beta + o(1)$ survives, but $\mathcal{C}_*$ is then an idealisation); (b) the numerical value of κ on real attention weights, Assumption 1 is a hypothesis about the network, tested here only through its consequence (small D on NIAH-MK3 in every cell of Table 22), not measured directly; (c) a bound on a_{early} from below, so the theorem controls the sign and the upper tail of D but not how negative D becomes. The converse is thus proved *conditionally on the margin assumption and the common-set model of capacity-bound*; both are now explicit rather than hidden.

I.3 ITEM 3: THE A4 USEFUL-SET IDENTIFICATION

Assumption (A4) of Appendix F identifies Bui’s useful set U_t (a residual-stream, decoding-step object) with the head-common signal set $\mathcal{I}^{(\ell^*)} = \bigcap_h \mathcal{R}_h^{(\ell^*)}$ (a single-layer attention object), and the text flags it as “a modelling choice, not a derived equality.” We show that a clean exact equality cannot be proven, but a *quantitative approximate* identification can: under a named consensus–usefulness assumption plus ε -closeness of heads, the two sets induce dilution values that agree up to $O(\varepsilon)$, with an explicit constant. We therefore replace the “derived equality” language with a proved approximation plus a named residual assumption.

Set-up. At the divergence layer ℓ^* write the head-private top- k sets $\mathcal{R}_h := \mathcal{R}_h^{(\ell^*)}$, their intersection $\mathcal{I} := \bigcap_h \mathcal{R}_h$ and union $\mathcal{U} := \bigcup_h \mathcal{R}_h$. For a set $\mathcal{S}$ define its induced dilution $\delta(\mathcal{S}) := 1 - \sum_{i \in \mathcal{S}} \alpha_{t,i}$, so Bui’s is $\delta^{(U)} := \delta(U_t)$ and ours is $\delta^{(I)} := \delta(\mathcal{I})$. Let $\bar{\alpha}_{\max} := \max_i \alpha_{t,i}$.

Assumption 2 (ε -closeness of heads). The late-layer heads are ε -close in their top- k sets: $\max_{h,h'} |\mathcal{R}_h \Delta \mathcal{R}_{h'}| \leq \varepsilon k$, equivalently pairwise Jaccard $\geq (1 - \varepsilon/2)/(1 + \varepsilon/2) = 1 - O(\varepsilon)$.

Assumption 3 (Consensus–usefulness, CUA). The useful set is sandwiched between the head-consensus and head-union sets: $\mathcal{I} \subseteq U_t \subseteq \mathcal{U}$. The right inclusion ($U_t \subseteq \mathcal{U}$) holds whenever routing to the residual stream is through attention only, so a position no head attends to cannot affect the next-token margin. The left inclusion ($\mathcal{I} \subseteq U_t$) is the substantive content: a position every head attends to is margin-preserving.

Proposition 1 (Quantitative approximate identification). *Under Assumptions 2–3,*

$$|\delta_t^{(U)} - \delta_t^{(I)}| \leq \sum_{i \in \mathcal{U} \setminus \mathcal{I}} \alpha_{t,i} \leq H(H-1)\varepsilon k \bar{\alpha}_{\max} = O(H^2\varepsilon), \quad (27)$$

where the last step uses $\bar{\alpha}_{\max} = \Theta(1/k)$ in the high-SNR softmax. In particular, as $\varepsilon \rightarrow 0$ (heads perfectly agree) the two dilution values coincide: $\delta_t^{(U)} \rightarrow \delta_t^{(I)}$.

Proof. By Assumption 3, $U_t \Delta \mathcal{I} \subseteq \mathcal{U} \setminus \mathcal{I}$, because $\mathcal{I} \subseteq U_t \subseteq \mathcal{U}$ forces every element of $U_t \Delta \mathcal{I} = U_t \setminus \mathcal{I}$ to lie in $\mathcal{U} \setminus \mathcal{I}$. Since $\delta(\mathcal{S})$ is a signed sum of attention masses over $\mathcal{S}$, $|\delta_t^{(U)} - \delta_t^{(I)}| = |\sum_{i \in U_t} \alpha_{t,i} - \sum_{i \in \mathcal{I}} \alpha_{t,i}| \leq \sum_{i \in U_t \Delta \mathcal{I}} \alpha_{t,i} \leq \sum_{i \in \mathcal{U} \setminus \mathcal{I}} \alpha_{t,i}$, the first inequality of (27). For the cardinality bound, for each head h , $|\mathcal{R}_h \setminus \mathcal{I}| \leq \sum_{h' \neq h} |\mathcal{R}_h \setminus \mathcal{R}_{h'}| \leq \sum_{h' \neq h} |\mathcal{R}_h \Delta \mathcal{R}_{h'}| \leq (H-1)\varepsilon k$ by Assumption 2. Since $\mathcal{U} \setminus \mathcal{I} = \bigcup_h (\mathcal{R}_h \setminus \mathcal{I})$, $|\mathcal{U} \setminus \mathcal{I}| \leq \sum_h |\mathcal{R}_h \setminus \mathcal{I}| \leq H(H-1)\varepsilon k$. Bounding each mass by $\bar{\alpha}_{\max}$ gives the middle inequality; $\bar{\alpha}_{\max} = \Theta(1/k)$ gives the $O(H^2\varepsilon)$ order. $\square$

Remark 1 (When the identification fails). The left inclusion of Assumption 3 ($\mathcal{I} \subseteq U_t$) can fail, and this is exactly the case the equality cannot cover: if a *single* specialised head carries the decisive value for the next-token margin, then a position in $\mathcal{R}_{h_0} \setminus \mathcal{I}$ (attended by that one head, absent from the consensus) can be margin-critical, so $U_t \not\subseteq \mathcal{U} \setminus (\mathcal{R}_{h_0} \setminus \mathcal{I})$ and $U_t \neq \mathcal{I}$ even at $\varepsilon = 0$ of the *other* heads. This is the single-decisive-head regime, precisely a capacity-bound, single-needle

retrieval situation, where one retrieval head dominates. That regime is not hypothetical: it is the *retrieval-head* phenomenon documented by Wu et al. (2025), who show that a small, sparse and largely input-independent set of heads carries long-context factual recall, and that ablating those heads destroys retrieval while leaving other capabilities intact. Their account and ours agree on the mechanism and differ in what is done with it: they identify *which* heads matter, whereas we read a consensus statistic over all heads and treat its collapse as evidence that no subset of the cache is safe to drop. Their result also predicts our failure mode, since a single decisive head is invisible to a pairwise-agreement average. The approximation of Proposition 1 is therefore reliable in the dilution-prone regime (many heads, diffuse consensus, no single decisive head) and degrades exactly where the paper already declines to apply the bridge (capacity-bound).

What this does and does not establish. *Establishes:* the identification of U_t with $\mathcal{I}^{(\ell^*)}$ is not an exact derived equality, but under the named Assumptions 2–3 the two sets induce *dilution values* that agree up to $O(H^2\varepsilon)$ in attention mass (Proposition 1), with an explicit constant $H(H-1)k\bar{\alpha}_{\max}$; so wherever heads are ε -close, using $\mathcal{I}^{(\ell^*)}$ in place of U_t perturbs Bui’s δ_t by a controlled $O(\varepsilon)$ term rather than by an uncontrolled amount. *Does not establish:* (a) the substantive left inclusion $\mathcal{I} \subseteq U_t$ (CUA); it is named as Assumption 3, justified by the additive-routing picture, and shown in Remark 1 to fail precisely in the single-decisive-head (capacity-bound) regime, so it is an assumption about the model’s read-out, not a theorem; (b) any control of the H^2 prefactor, which is a worst-case union count and is loose when residuals overlap across heads (the same $O(H)$ -slack issue as Step 3 of Appendix F); (c) a bound when ε is $\Theta(1)$ (genuinely divergent heads, the dilution-prone late layers themselves), there the approximation is vacuous and $\mathcal{I}$ and U_t are related only through the CUA sandwich, not quantitatively. The honest scoping is thus: **A4 is a named modelling assumption (CUA), under which we prove an $O(H^2\varepsilon)$ approximate identification of the dilution values**, not a derived equality; the word “derived equality” in Appendix F should be read as this approximation.

J USE OF LARGE LANGUAGE MODELS

Large language models were used as general-purpose assistants during this project: for literature search and triage, for drafting and copy-editing prose, and as coding aids when writing and analyzing the experiment scripts. All research ideas, hypotheses, experimental designs, and claims are the authors’ own. Every reported number was produced by the released code on the stated models and data, and the authors verified the experiments and their reporting. LLMs were not used as a source of scientific facts or citations without verification against primary sources.